Winter Term 2025/26

Linear models and regression

Prof. Dr. Matthias Schulte[1]

January 7, 2026

[1]Institute of Mathematics, Hamburg University of Technology, matthias.schulte@tuhh.de

Contents

1 Introduction **1**

2 Preliminaries **3**
2.1 Important distributions of random variables 3
2.2 Random vectors and the multivariate normal distribution 4
2.3 The law of large numbers and the central limit theorem 6
2.4 Estimators, confidence intervals and tests 7
2.5 Maximum likelihood estimation 11

3 Linear regression **15**
3.1 Linear models 15
3.2 Gaussian linear model 23
3.3 Testing in the Gaussian linear model 24
3.4 Analysis of variance 34

4 Non-linear regression **43**
4.1 Non-linear regression and least squares estimation 43
4.2 Non-linear regression and maximum likelihood estimation 45

5 Maximum likelihood estimators and their asymptotic behaviour **49**
5.1 Maximum likelihood estimation for regression 49
5.2 Asymptotics of maximum likelihood estimators 52
5.3 Asymptotic confidence intervals and tests for maximum likelihood estimators 56
5.4 Asymptotic tests for non-linear regression 59

6 Logistic regression **63**
6.1 Binary regression 63
6.2 Maximum likelihood estimation for logistic regression 64
6.3 Asymptotics of the maximum likelihood estimator for logistic regression . . 66
6.4 Tests and confidence intervals for logistic regression 70

Appendix **73**

A Results on convergence in distribution **75**

Chapter 1

Introduction

Let observations of (real-valued) quantities Y and $x^{(1)}, \ldots, x^{(k)}$ be given. Our goal is to understand the relationship between Y and $x^{(1)}, \ldots, x^{(k)}$ and to explain Y by $x^{(1)}, \ldots, x^{(k)}$. We denote Y as *dependent variable* (or *response variable*) and $x^{(1)}, \ldots, x^{(k)}$ as *independent variables* (or *covariates*). In general, the assumption that Y is a deterministic function of $x^{(1)}, \ldots, x^{(k)}$ is too restrictive so that we allow Y to be random. Now we are interested in how the distribution of Y depends on $x^{(1)}, \ldots, x^{(k)}$. This type of problem is called *regression*.

Regression analysis is applied in many different areas. Examples are the following:

- How does the height of a child depend on its age?
- Is the satisfaction of a person with their job related with their education and gender?
- Can the failure of a component be explained by the environmental temperature?
- Can sepal length and width and petal length and width be used to distinct different species of flowers?

Understanding how the dependent variable Y is affected by the independent variables $x^{(1)}, \ldots, x^{(k)}$ can be used for two different purposes. In *causal inference*, one is interested how changing one of the independent variables changes the dependent variable, i.e., if some covariate has a positive or negative effect. For example, one can wonder whether a higher education level increases or decreases the job satisfaction. In *prediction*, for given $x^{(1)}, \ldots, x^{(k)}$ one provides a guess for Y. Here one could ask for the height of a child who is nine years and three months old. Prediction is strongly related to machine learning.

Throughout this class, we will consider several types of dependencies between the dependent variable and the independent variables. In *linear models*, Y is assumed to be an affine-linear function of $x^{(1)}, \ldots, x^{(k)}$ plus some random variable, while in *non-linear regression* the affine-linear function is replaced by a non-linear function. If the dependent variable Y takes only the values zero and one, the probability of $Y = 1$ determines the distribution of Y and can be represented as a function of the independent variables. This approach is denoted as *binary regression* and we will study in particular a model called

logistic regression. For $\mathbb{N}_0$-valued dependent variables we will consider *Poisson regression*. Certain of these regression models can be seen as special cases of *generalised linear models*.

Further statistical procedures we will study are *linear discriminant analysis* and *principal component analysis*.

In regression analysis, the first step is to estimate the model parameters from the observations. To this end, we will use in particular *maximum likelihood estimation*. In a second step, statistical tests are employed to draw conclusions from the data. For this it is often helpful to understand the asymptotic behaviour of maximum likelihood estimators, which we will study as well.

Chapter 2

Preliminaries

2.1 Important distributions of random variables

For a random variable X and a set $A \in \mathcal{B}(\mathbb{R})$ we denote by $\mathbf{P}(X \in A)$ the probability that X takes a value in A. For technical reasons we cannot assign a probability to every subset A of $\mathbb{R}$ but only to those from the so-called *Borel σ-algebra* $\mathcal{B}(\mathbb{R})$. We say that random variables $X_1, \ldots, X_n$ with $n \in \mathbb{N}$ are *independent* if

$$\mathbf{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \prod_{i=1}^{n} \mathbf{P}(X_i \in A_i) \tag{2.1}$$

for all $A_1, \ldots, A_n \in \mathcal{B}(\mathbb{R})$. A family of random variables $(X_i)_{i \in I}$, where I does not need to be finite, is called *independent* if for all finite subsets $J \subseteq I$, $(X_i)_{i \in J}$ are independent. We denote a collection of random variables $(X_i)_{i \in I}$ as *i.i.d.*, which stands for independent and identically distributed, if they are independent and all X_i with $i \in I$ follow the same distribution.

Recall that a random variable X has a *density* $f : \mathbb{R} \to [0, \infty)$ if

$$\mathbf{P}(X \leq t) = \int_{-\infty}^{t} f(u) \, \mathrm{d}u$$

for $t \in \mathbb{R}$. We say that a random variable X follows a *normal distribution* with parameters $\mu \in \mathbb{R}$ and $\sigma \in (0, \infty)$ if it has the density

$$\phi(u) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(- \frac{(u-\mu)^2}{2\sigma^2} \right)$$

for $u \in \mathbb{R}$ and write $X \sim \mathrm{Normal}(\mu, \sigma^2)$. The normal distribution with the parameters $\mu = 0$ and $\sigma = 1$ is called the *standard normal distribution*.

A random variable X follows a *chi-square distribution* with m degrees of freedom if it has the same distribution as $\sum_{i=1}^{m} N_i^2$ for independent standard normal random variables $N_1, \ldots, N_m$. In this case, we use the abbreviation $X \sim \chi_m^2$.

We say that a random variable X has a *t-distribution* with m degrees of freedom if it follows the same distribution as $\frac{N}{\sqrt{Z/m}}$ for independent $N \sim \text{Normal}(0,1)$ and $Z \sim \chi^2_m$. In this case, we write $X \sim t_m$.

For $m_1, m_2 \in \mathbb{N}$ let $Z_1 \sim \chi^2_{m_1}$ and $Z_2 \sim \chi^2_{m_2}$ be independent. If a random variable X has the same distribution as $\frac{Z_1/m_1}{Z_2/m_2}$, we call its distribution an *F-distribution* with m_1 and m_2 degrees of freedom and use the short hand notation $X \sim \mathcal{F}_{m_1,m_2}$.

For a random variable X and $p \in (0,1)$ we denote $q \in \mathbb{R}$ as *p-quantile* of the distribution of X if

$$\mathbf{P}(X \leq q) \geq p \quad \text{and} \quad \mathbf{P}(X \geq q) \geq 1-p.$$

In case that X has a continuous distribution function F, q is a p-quantile if and only if $F(q) = p$. Thus, the normal distribution, the chi-square distribution, the t-distribution and the F-distribution all have unique p-quantiles for $p \in (0,1)$, which we denote by z_p, $\chi^2_{n,p}$, $t_{n,p}$ and $\mathcal{F}_{n_1,n_2,p}$, respectively, where $n, n_1, n_2 \in \mathbb{N}$ are the degrees of freedom.

Random variables $X \sim \text{Normal}(0,1)$ and $Y \sim t_n$ with $n \in \mathbb{N}$ have symmetric distributions in the sense that

$$\mathbf{P}(X \leq -u) = \mathbf{P}(X \geq u) \quad \text{and} \quad \mathbf{P}(Y \leq -u) = \mathbf{P}(Y \geq u)$$

for $u \in \mathbb{R}$. For their quantiles this implies that

$$\mathbf{P}(X \leq -z_p) = \mathbf{P}(X \geq z_p) = 1 - \mathbf{P}(X \leq z_p) = 1 - p \tag{2.2}$$

and

$$\mathbf{P}(Y \leq -t_{n,p}) = \mathbf{P}(Y \geq t_{n,p}) = 1 - \mathbf{P}(Y \leq t_{n,p}) = 1 - p \tag{2.3}$$

for $p \in (0,1)$.

2.2 Random vectors and the multivariate normal distribution

A *random vector* $\mathbf{X}$ in $\mathbb{R}^n$ is a vector $\mathbf{X} = (X_1, \ldots, X_n)^T$ whose entries $X_1, \ldots, X_n$ are random variables. Similarly, a *random matrix* is a matrix whose entries are random variables. The *distribution function* $F_{\mathbf{X}} : \mathbb{R}^n \to [0,1]$ of the random vector $\mathbf{X}$ is given by

$$F_{\mathbf{X}}(\mathbf{t}) = \mathbf{P}(X_1 \leq t_1, \ldots, X_n \leq t_n)$$

for $\mathbf{t} = (t_1, \ldots, t_n)^T \in \mathbb{R}^n$. Analogously to the univariate case, $f : \mathbb{R}^n \to [0, \infty)$ is a *density* of $\mathbf{X}$ if

$$F_{\mathbf{X}}(\mathbf{t}) = \int_{-\infty}^{t_1} \cdots \int_{-\infty}^{t_n} f(u_1, \ldots, u_n) \, \mathrm{d}u_n \ldots \mathrm{d}u_1$$

for all $\mathbf{t} = (t_1, \ldots, t_n)^T \in \mathbb{R}^n$. By the *expectation* of a random vector or a random matrix we mean the component-wise expectations, i.e.,

$$\mathbf{E}[\mathbf{X}] = \begin{pmatrix} \mathbf{E}[X_1] \\ \vdots \\ \mathbf{E}[X_n] \end{pmatrix}$$

under the assumption $\mathbf{E}[\|\mathbf{X}\|] < \infty$. If $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$, the *covariance matrix* $\mathbf{Cov}(\mathbf{X})$ of $\mathbf{X}$ is given by

$$\mathbf{Cov}(\mathbf{X}) = \begin{pmatrix} \mathbf{Cov}(X_1, X_1) & \dots & \mathbf{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \mathbf{Cov}(X_n, X_1) & \dots & \mathbf{Cov}(X_n, X_n) \end{pmatrix},$$

where

$$\mathbf{Cov}(X_i, X_j) = \mathbf{E}[(X_i - \mathbf{E}[X_i])(X_j - \mathbf{E}[X_j])] = \mathbf{E}[X_i X_j] - \mathbf{E}[X_i]\mathbf{E}[X_j]$$

is the *covariance* of X_i and X_j for $i, j \in \{1, \dots, n\}$.

For collections of random vectors in $\mathbb{R}^n$ one can define independence as in (2.1) and thereafter (with $\mathcal{B}(\mathbb{R})$ replaced by the Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$ on $\mathbb{R}^n$). The notion of i.i.d. can be extended analogously.

Assume that the components $X_1, \dots, X_n$ of $\mathbf{X}$ all have densities $f_{X_1}, \dots, f_{X_n}$. Then, $X_1, \dots, X_n$ are independent if and only if $\mathbf{X}$ has the density

$$f(x_1, \dots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i)$$

for $(x_1, \dots, x_n)^T \in \mathbb{R}^n$.

We call the distribution of $\mathbf{N} = (N_1, \dots, N_n)^T$ with independent standard normal random variables $N_1, \dots, N_n$ the *multivariate standard normal distribution* on $\mathbb{R}^n$. We say that a random vector $\mathbf{X}$ in $\mathbb{R}^n$ follows a *multivariate normal distribution* with $\boldsymbol{\mu} \in \mathbb{R}^n$ and a positive semi-definite matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$ as parameters if $\mathbf{X}$ has the same distribution as $\boldsymbol{\Sigma}^{\frac{1}{2}}\mathbf{N} + \boldsymbol{\mu}$ with $\boldsymbol{\Sigma}^{\frac{1}{2}}$ the unique positive semi-definite matrix in $\mathbb{R}^{n \times n}$ such that $\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{\Sigma}^{\frac{1}{2}} = \boldsymbol{\Sigma}$. In this case, we write $\mathbf{X} \sim \mathrm{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

The following lemma is helpful for studying affine transformations of random vectors, which are again random vectors.

Lemma 2.1. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

a) Let $\mathbf{X}$ be a random vector in $\mathbb{R}^n$ such that $\mathbf{E}[\|\mathbf{X}\|] < \infty$. Then,

$$\mathbf{E}[\mathbf{AX} + \mathbf{b}] = \mathbf{A}\mathbf{E}[\mathbf{X}] + \mathbf{b}.$$

If, additionally, $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$,

$$\mathbf{Cov}(\mathbf{AX} + \mathbf{b}) = \mathbf{A}\mathbf{Cov}(\mathbf{X})\mathbf{A}^T.$$

b) If $\mathbf{X} \sim \mathrm{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} \in \mathbb{R}^n$ and a positive semi-definite matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$, then $\mathbf{AX} + \mathbf{b} \sim \mathrm{Normal}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T)$.

We call two random vectors $\mathbf{X} = (X_1, \dots, X_m)^T$ and $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ with $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$ and $\mathbf{E}[\|\mathbf{Y}\|^2] < \infty$ *uncorrelated* if $\mathbf{Cov}(X_i, Y_j) = 0$ for all $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$, i.e.,

$$\mathbf{Cov}(\mathbf{X}, \mathbf{Y}) := (\mathbf{Cov}(X_i, Y_j))_{i \in \{1,\dots,m\}, j \in \{1,\dots,n\}} = \mathbf{0}.$$

Note that independence is a stronger property than being uncorrelated. Two independent random vectors $\mathbf{X}$ and $\mathbf{Y}$ satisfying $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$ and $\mathbf{E}[\|\mathbf{Y}\|^2] < \infty$ are also uncorrelated, but uncorrelatedness does not imply independence. However, in the following situation this is the case.

Lemma 2.2. Let $\mathbf{X}$ and $\mathbf{Y}$ be random vectors in $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, such that $(\mathbf{X}, \mathbf{Y})$ is a normally distributed random vector in $\mathbb{R}^{m+n}$. If $\mathbf{X}$ and $\mathbf{Y}$ are uncorrelated, then $\mathbf{X}$ and $\mathbf{Y}$ are independent.

2.3 The law of large numbers and the central limit theorem

Let $(\mathbf{Y}_n)_{n \in \mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{Y}$ be a random vector in $\mathbb{R}^m$. We say that $(\mathbf{Y}_n)_{n \in \mathbb{N}}$ *converges in probability* to $\mathbf{Y}$ if

$$\lim_{n \to \infty} \mathbf{P}(\|\mathbf{Y}_n - \mathbf{Y}\| \geq \varepsilon) = 0$$

for all $\varepsilon \in (0, \infty)$. In this case, we use the short hand notation $\mathbf{Y}_n \xrightarrow{\mathbf{P}} \mathbf{Y}$ as $n \to \infty$. Moreover, we say that $(\mathbf{Y}_n)_{n \in \mathbb{N}}$ *converges almost surely* to $\mathbf{Y}$ if

$$\mathbf{P}\big(\lim_{n \to \infty} \mathbf{Y}_n = \mathbf{Y}\big) = 1$$

and write $\mathbf{Y}_n \xrightarrow{\text{a.s.}} \mathbf{Y}$ as $n \to \infty$. The most prominent result concerning convergence in probability and almost sure convergence is the law of large numbers.

Theorem 2.3 (Law of large numbers)**.** Let $(\mathbf{X}_i)_{i \in \mathbb{N}}$ be i.i.d. random vectors in $\mathbb{R}^m$ such that $\mathbf{E}[\|\mathbf{X}\|] < \infty$ and define $\mathbf{S}_n = \sum_{i=1}^n \mathbf{X}_i$ for $n \in \mathbb{N}$. Then,

$$\frac{1}{n}\mathbf{S}_n \xrightarrow{\mathbf{P}} \mathbf{E}[\mathbf{X}_1] \quad \text{as} \quad n \to \infty \tag{2.4}$$

and

$$\frac{1}{n}\mathbf{S}_n \xrightarrow{\text{a.s.}} \mathbf{E}[\mathbf{X}_1] \quad \text{as} \quad n \to \infty. \tag{2.5}$$

We call (2.4) the *weak law of large numbers*, while (2.5) is the *strong law of large numbers*. Note that almost sure convergence always implies convergence in probability. Thus, the weak law of large numbers is an immediate consequence of the strong law of large numbers. We say that an event A holds *almost surely* if $\mathbf{P}(A) = 1$.

Let $(\mathbf{Y}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{Y}$ be a random vector in $\mathbb{R}^m$, whose distribution functions we denote by $(F_{\mathbf{Y}_n})_{n\in\mathbb{N}}$ and $F_{\mathbf{Y}}$. We say that $(\mathbf{Y}_n)_{n\in\mathbb{N}}$ *converges in distribution* to $\mathbf{Y}$ and write $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$ as $n \to \infty$ if

$$\lim_{n\to\infty} F_{\mathbf{Y}_n}(\mathbf{t}) = F_{\mathbf{Y}}(\mathbf{t})$$

for all $\mathbf{t} \in \mathbb{R}^m$ that are continuity points of $F_{\mathbf{Y}}$ (i.e., $F_{\mathbf{Y}}$ is continuous in $\mathbf{t}$). If $(Y_n)_{n\in\mathbb{N}}$ and Y are random variables and Y has a continuous distribution function, then $Y_n \xrightarrow{d} Y$ as $n \to \infty$ implies

$$\lim_{n\to\infty} \mathbf{P}(Y_n \le t) = \mathbf{P}(Y \le t) \quad \text{and} \quad \lim_{n\to\infty} \mathbf{P}(Y_n < t) = \mathbf{P}(Y < t)$$

for all $t \in \mathbb{R}$ so that

$$\lim_{n\to\infty} \mathbf{P}(Y_n \in I) = \mathbf{P}(Y \in I) \tag{2.6}$$

for all intervals $I \in \{(a,b),(a,b],[a,b),[a,b]\}$ with $a,b \in \mathbb{R}$.

The following famous multivariate central limit theorem concerns the convergence of sums of i.i.d. random vectors to normal random vectors.

Theorem 2.4 (Multivariate central limit theorem)**.** Let $(\mathbf{X}_i)_{i\in\mathbb{N}}$ be i.i.d. random vectors in $\mathbb{R}^m$ such that $\mathbf{E}[\|\mathbf{X}_1\|^2] < \infty$ and define $\mathbf{S}_n = \sum_{i=1}^n \mathbf{X}_i$ for $n \in \mathbb{N}$. Let $\mathbf{\Sigma} = \mathbf{Cov}(\mathbf{X}_1)$ and $\mathbf{N} \sim \text{Normal}(\mathbf{0}, \mathbf{\Sigma})$. Then,

$$\frac{\mathbf{S}_n - n\mathbf{E}[\mathbf{X}_1]}{\sqrt{n}} \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty.$$

2.4 Estimators, confidence intervals and tests

In (parametric) statistics, one has some given data from a distribution depending on unknown parameters, which one aims to learn. More precisely, let the *sample space* $\mathcal{X}$ be the set of all possible observations, which is often $\mathbb{R}^n$. For technical reasons we equip $\mathcal{X}$ with a σ-algebra $\mathscr{F}_{\mathcal{X}}$. Let $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ be a family of probability distributions on $\mathcal{X}$, which are the possible underlying distributions of our data. Now we observe a random element $\mathbf{X}$ on $\mathcal{X}$ that is distributed according to $\mathbf{P}_\vartheta$ for some unknown $\vartheta \in \Theta$. Our goal is to reconstruct ϑ or retrieve at least some information about ϑ from the data $\mathbf{X}$.

In the following let $g : \Theta \to \mathbb{R}^m$ be measurable. An *estimator* U for $g(\vartheta)$ is now a guess for $g(\vartheta)$ based on $\mathbf{X}$. More formally, it is a measurable map $U : \mathcal{X} \to \mathbb{R}^m$, i.e., an estimator itself is a random vector in $\mathbb{R}^m$. For an estimator it is desirable that the estimates are close to the true value $g(\vartheta)$. We say that an estimator U for $g(\vartheta)$ is *unbiased* if

$$\mathbf{E}_\vartheta[U] = g(\vartheta) \quad \text{for all} \quad \vartheta \in \Theta,$$

where $\mathbf{E}_\vartheta$ stands for the expectation with respect to $\mathbf{P}_\vartheta$. As it is random, an estimator cannot always provide the true value in general, but an unbiased estimator has the correct

expectation, which can be interpreted in the way that its average is the desired value. It is important to note that an unbiased estimator needs to have $g(\vartheta)$ as expectation for all possible underlying parameters $\vartheta \in \Theta$. If Θ is an Euclidean space or a subset of it, g is often the identity, i.e., $g(\vartheta) = \vartheta$.

A sequence of estimators $(U_n)_{n\in\mathbb{N}}$ for $g(\vartheta)$ is called *strongly consistent* if

$$U_n \xrightarrow{\mathbf{P}_\vartheta\text{-a.s.}} g(\vartheta) \quad \text{as} \quad n \to \infty$$

for all $\vartheta \in \Theta$ and *weakly consistent* if

$$U_n \xrightarrow{\mathbf{P}_\vartheta} g(\vartheta) \quad \text{as} \quad n \to \infty$$

for all $\vartheta \in \Theta$, where $\xrightarrow{\mathbf{P}_\vartheta\text{-a.s.}}$ and $\xrightarrow{\mathbf{P}_\vartheta}$ denote almost sure convergence and convergence in probability with respect to $\mathbf{P}_\vartheta$. Typically, n is the number of observations used by the estimator. In this case, consistency means that the estimates tend to the true value as the sample size is increased. Note that strong consistency implies weak consistency.

Example 2.5 (Normal sample with unknown expectation and known variance)**.** Let $\mathcal{X} = \mathbb{R}^n$ for some fixed $n \in \mathbb{N}$, let $\mathscr{F}_\mathcal{X}$ be the Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$ and let $\mathbf{P}_\mu$ be the distribution of n i.i.d. random variables $X_1, \ldots, X_n$ that follow normal distributions with unknown expectation $\mu \in \mathbb{R}$ and known standard deviation $\sigma_0 \in (0, \infty)$. We call $\mathbf{X} = (X_1, \ldots, X_n)^T$ a *normal sample* with unknown expectation and known variance. Our goal is to interfere from the sample $\mathbf{X}$ to the underlying parameter μ. As an estimator for μ (i.e., the function g from above is the identity) we introduce the *sample mean*

$$\overline{\mathbf{X}} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

A straightforward calculation reveals that

$$\mathbf{E}_\mu[\overline{\mathbf{X}}] = \mathbf{E}_\mu\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbf{E}_\mu[X_i] = \mathbf{E}_\mu[X_1] = \mu$$

for all $\mu \in \mathbb{R}$ so that the sample mean is an unbiased estimator for μ. If $\mathbf{P}_\mu$ is the distribution of a sequence of i.i.d. random variables $(X_i)_{i\in\mathbb{N}}$ with $X_1 \sim \text{Normal}(\mu, \sigma_0^2)$, $\mathcal{X}$ and $\mathscr{F}_\mathcal{X}$ are changed accordingly and we denote by $\overline{\mathbf{X}}_n$ the sample mean of the first n observations, the sample means $(\overline{\mathbf{X}}_n)_{n\in\mathbb{N}}$ are strongly consistent estimators for μ by the strong law of large numbers.

Note that the sample mean is not only an unbiased and strongly consistent estimator for the expectation in case of normal samples, but that this holds more general for all i.i.d. samples where the underlying distributions have finite expectations.

The disadvantage of an estimator is that we do not know how close it is to the true value. This problem is addressed by confidence regions. We call a set $I_\mathbf{X} \subseteq \mathbb{R}^m$ depending on $\mathbf{X}$ a *confidence region* for $g(\vartheta)$ with error level $\alpha \in (0, 1)$ if

$$\mathbf{P}_\vartheta(g(\vartheta) \in I_\mathbf{X}) \geq 1 - \alpha \tag{2.7}$$

for all $\vartheta \in \Theta$. This means that for any choice of the underlying parameter ϑ, $g(\vartheta)$ is contained in $I_{\mathbf{X}}$ with probability at least $1-\alpha$, i.e., the probability of the error that $g(\vartheta)$ does not belong to $I_{\mathbf{X}}$ is at most α. While an estimator gives only a single value, a confidence region provides a whole set containing $g(\vartheta)$ with some certainty. If $m=1$ and $I_{\mathbf{X}}$ is an interval, we call $I_{\mathbf{X}}$ a *confidence interval*.

Example 2.6. For the normal sample from Example 2.5 and $\alpha \in (0,1)$ let

$$I_{\mathbf{X}} = \left(\overline{\mathbf{X}} - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \overline{\mathbf{X}} + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right),$$

where z_p denotes the p-quantile of the standard normal distribution for $p \in (0,1)$. Since $\overline{\mathbf{X}} \sim \mathrm{Normal}(\mu, \sigma_0^2/n)$ and, thus, $(\overline{\mathbf{X}}-\mu)/(\sigma_0/\sqrt{n}) \sim \mathrm{Normal}(0,1)$, we obtain

$$\begin{aligned}
\mathbf{P}_\mu(\mu \in I_{\mathbf{X}}) &= \mathbf{P}_\mu\left(\mu \in \left(\overline{\mathbf{X}} - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \overline{\mathbf{X}} + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right)\right) \\
&= \mathbf{P}_\mu\left(\overline{\mathbf{X}} - \mu \in \left(-\frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right)\right) \\
&= \mathbf{P}_\mu\left(\frac{\overline{\mathbf{X}}-\mu}{\sigma_0/\sqrt{n}} \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}})\right) = \mathbf{P}(N \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}})) \\
&= \Phi(z_{1-\frac{\alpha}{2}}) - \Phi(-z_{1-\frac{\alpha}{2}}),
\end{aligned}$$

where N is a standard normal random variable and Φ stands for the distribution function of the standard normal distribution. Using (2.2), we obtain

$$\mathbf{P}_\mu(\mu \in I_{\mathbf{X}}) = \Phi(z_{1-\frac{\alpha}{2}}) - (1-\Phi(z_{1-\frac{\alpha}{2}})) = 2\Phi(z_{1-\frac{\alpha}{2}}) - 1 = 2\cdot\left(1-\frac{\alpha}{2}\right) - 1 = 1-\alpha.$$

This shows that $I_{\mathbf{X}}$ is a confidence interval for μ with error level α.

Let $(I_n)_{n\in\mathbb{N}}$ be a sequence of random sets or random intervals and let $\alpha \in (0,1)$. We say that they are *asymptotic confidence regions* or *asymptotic confidence intervals* for $g(\vartheta)$ with error level α if

$$\liminf_{n\to\infty} \mathbf{P}_\vartheta(g(\vartheta) \in I_n) \geq 1-\alpha$$

for all $\vartheta \in \Theta$. This is a weaker condition than (2.7) as the inequality does not need to hold for fixed n but only in an asymptotic sense.

The third type of statistical procedures is hypothesis testing. For two sets $\Theta_0, \Theta_1 \subseteq \Theta$ such that $\Theta_0 \cap \Theta_1 = \varnothing$ and $\Theta_0 \cup \Theta_1 = \Theta$ we aim to decide which of the statements

$$H_0 : \vartheta \in \Theta_0 \quad \text{or} \quad H_1 : \vartheta \in \Theta_1$$

is true for the unknown underlying parameter ϑ of $\mathbf{X}$. Here we call H_0 the *null hypothesis* and H_1 the *alternative* and say that we *test* H_0 *against* H_1. A test is now a decision rule between H_0 and H_1. If we decide in favour of H_1, we say that we *reject* H_0, while

in case of H_0 we say that we *cannot reject* H_0. Formally a *test* is a map $\varphi : \mathcal{X} \to \{\text{cannot reject}, \text{reject}\}$. When applying a statistical test, we commit an error in two cases, a *type I error* if we reject H_0 although it is true (i.e., $\vartheta \in \Theta_0$) and a *type II error* if we do not reject H_0 although it is false (i.e., $\vartheta \in \Theta_1$). Since, in general, one cannot control both error types simultaneously, one focusses on the type I error. We say that a test φ has *level* $\alpha \in (0,1)$ (or is a *level-α test*) if

$$\mathbf{P}_\vartheta(\varphi(\mathbf{X}) = \text{reject}) \leq \alpha \tag{2.8}$$

for all $\vartheta \in \Theta_0$. This means that in situations where H_0 is true the probability of rejecting H_0 is at most α or, in other words, the probability of a type I error is at most α. Typical choices for α are 0.1, 0.05 and 0.01.

Note that the null hypothesis and the alternative are not symmetric since we control the type I error. In applications, the alternative is always the statement we aim to verify. The philosophy here is that the opposite (i.e., the null hypothesis) is true as long as the data do not convincingly contradict it. In the case that the data sufficiently disagree with the null hypothesis, we can show that the null hypothesis must be false and consequently the alternative is true.

Example 2.7. Consider the situation of a normal sample from Example 2.5. For some $\mu_0 \in \mathbb{R}$ we are interested in testing

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu \neq \mu_0. \tag{2.9}$$

For $\alpha \in (0,1)$ we define

$$\varphi : \mathbb{R}^n \ni \mathbf{x} \mapsto \begin{cases} \text{cannot reject}, & \mu_0 \in \big(\overline{\mathbf{x}} - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \overline{\mathbf{x}} + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\big), \\ \text{reject}, & \mu_0 \notin \big(\overline{\mathbf{x}} - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \overline{\mathbf{x}} + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\big). \end{cases}$$

Similarly as in Example 2.6 we obtain

$$\begin{aligned} \mathbf{P}_{\mu_0}(\varphi(\mathbf{X}) = \text{reject}) &= \mathbf{P}_{\mu_0}\left(\mu_0 \notin \left(\overline{\mathbf{X}} - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \overline{\mathbf{X}} + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right)\right) \\ &= 1 - \mathbf{P}_{\mu_0}\left(\mu_0 \in \left(\overline{\mathbf{X}} - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \overline{\mathbf{X}} + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right)\right) \\ &= 1 - \mathbf{P}_{\mu_0}\left(\frac{\overline{\mathbf{X}} - \mu_0}{\sigma_0/\sqrt{n}} \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}})\right) = 1 - (\Phi(z_{1-\frac{\alpha}{2}}) - \Phi(-z_{1-\frac{\alpha}{2}})) \\ &= 1 - (1-\alpha) = \alpha. \end{aligned}$$

This shows that φ is a level-α test for (2.9).

When using statistical software such as `R` for tests, one usually does check if the test statistic belongs to the rejection region for a fixed level $\alpha \in (0,1)$. Instead one compares the desired level α with the so-called *p-value*, which is the smallest level such that the

null hypothesis can be rejected. Thus, the null hypothesis can be rejected as long as the p-value does not exceed α.

We say that a sequence of tests $(\varphi_n)_{n\in\mathbb{N}}$ for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$ are *asymptotic tests* with level $\alpha \in (0,1)$ if

$$\limsup_{n\to\infty} \mathbf{P}_\vartheta(\varphi_n(\overline{\mathbf{X}}) = \text{reject}) \leq \alpha$$

for all $\vartheta \in \Theta_0$. Compared with (2.8) this means that not every test must have level α, but that the level α must be only reached in an asymptotic sense.

2.5 Maximum likelihood estimation

Maximum likelihood estimation is an approach to construct estimators. As in the previous section let $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ be a family of probability distributions on some space $\mathcal{X}$ and let $\mathbf{X}$ be a random element on $\mathcal{X}$ distributed according to $\mathbf{P}_\vartheta$ for some unknown $\vartheta \in \Theta$. Our goal is to derive an estimator for ϑ. We often denote estimators for parameters by adding a hat, i.e., we write $\widehat{\vartheta}$ for an estimator for ϑ.

First we consider the situation that $\mathbf{P}_\vartheta$ is discrete (i.e., it can take at most countably many values) for all $\vartheta \in \Theta$. In this case, we call an estimator $\widehat{\vartheta} : \mathcal{X} \to \Theta$ a *maximum likelihood estimator* for ϑ if

$$\mathbf{P}_{\widehat{\vartheta}(\mathbf{x})}(\{\mathbf{x}\}) = \sup_{\widetilde{\vartheta}\in\Theta} \mathbf{P}_{\widetilde{\vartheta}}(\{\mathbf{x}\})$$

for all $\mathbf{x} \in \mathcal{X}$. Thus, for given $\mathbf{x}$, $\widehat{\vartheta}(\mathbf{x})$ chooses $\vartheta \in \Theta$ such that the probability of $\mathbf{x}$ is maximised, which explains the name maximum likelihood estimator. To find the maximum likelihood estimator for our sample $\mathbf{X}$ one has to maximise the function $\vartheta \mapsto \mathbf{P}_\vartheta(\mathbf{X})$, which is called *likelihood function*. A common strategy is to maximise the *log likelihood function* $\vartheta \mapsto \log \mathbf{P}_\vartheta(\mathbf{X})$, which has the same maximisers as the likelihood function due to the monotonicity of the logarithm.

Example 2.8. For a fixed $n \in \mathbb{N}$ let $\mathcal{X} = \{0, 1, \ldots, n\}$, let $\mathscr{F}_\mathcal{X}$ be the power set of $\mathcal{X}$ (i.e., the set of all subsets of $\mathcal{X}$) and let $\mathbf{P}_p$ be a binomial distribution with parameters n and $p \in [0,1]$, i.e.,

$$\mathbf{P}_p(\{k\}) = \binom{n}{k} p^k (1-p)^{n-k}$$

for $k \in \{0, 1, \ldots, n\}$. Now let $Y \sim \mathbf{P}_p$ for some unknown $p \in [0,1]$, which we want to estimate. The likelihood function is given by

$$[0,1] \ni p \mapsto \mathbf{P}_p(\{Y\}) = \binom{n}{Y} p^Y (1-p)^{n-Y}$$

and the log likelihood function L is

$$L(p) = \log \binom{n}{Y} + Y \log p + (n-Y) \log(1-p)$$

for $p \in [0,1]$. For $Y = 0$ or $Y = n$, this function is maximised by $p = 0 = \frac{Y}{n}$ or $p = 1 = \frac{Y}{n}$, respectively. Assume that $Y \notin \{0, n\}$. Because of

$$L'(p) = \frac{Y}{p} - \frac{n-Y}{1-p},$$

we have

$$\begin{aligned} L'(p) = 0 \quad &\iff \quad \frac{Y}{p} - \frac{n-Y}{1-p} = 0 \quad \iff \quad \frac{Y}{p} = \frac{n-Y}{1-p} \quad \iff \quad Y - pY = np - pY \\ &\iff \quad Y = np \quad \iff \quad p = \frac{Y}{n}. \end{aligned}$$

Since we see that L is continuous and $L(p)$ becomes small as $p \to 0$ or $p \to 1$, its only stationary point must belong to a maximum. This means that $\widehat{p} = \frac{Y}{n}$ is a maximum likelihood estimator for p. As it is the unique maximum likelihood estimator, we call it the maximum likelihood estimator for p.

In the following we assume that $\mathcal{X} = \mathbb{R}^m$ and that $\mathbf{P}_\vartheta$ has a density f_ϑ for any $\vartheta \in \Theta$. Then the approach from above does not work since $\mathbf{P}_\vartheta(\{\mathbf{x}\}) = 0$ for all $\vartheta \in \Theta$ and $\mathbf{x} \in \mathbb{R}^m$. Thus, we replace the probability $\mathbf{P}_\vartheta$ by its density f_ϑ. We say that an estimator $\widehat{\vartheta} : \mathbb{R}^m \to \Theta$ is a *maximum likelihood estimator* for ϑ if

$$f_{\widehat{\vartheta}(\mathbf{x})}(\mathbf{x}) = \sup_{\widetilde{\vartheta} \in \Theta} f_{\widetilde{\vartheta}}(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{R}^m$. This means that for our sample $\mathbf{X}$ the maximum likelihood estimator chooses a $\vartheta \in \Theta$ that maximises $\vartheta \mapsto f_\vartheta(\mathbf{X})$. The latter function is called the *likelihood function* and as in the discrete case one often studies its logarithm, the *log likelihood function*, instead of the likelihood function itself.

Example 2.9. We consider the normal sample from Example 2.5. In order to derive a maximum likelihood estimator for μ, we note that the random vector $\mathbf{X} = (X_1, \ldots, X_n)^T$ has the density

$$f_\mu(x_1, \ldots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(- \frac{(x_i - \mu)^2}{2\sigma_0^2} \right)$$

for $\mu \in \mathbb{R}$. Thus, we obtain the likelihood function

$$\mathbb{R} \ni \mu \mapsto \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(- \frac{(X_i - \mu)^2}{2\sigma_0^2} \right)$$

and the log likelihood function

$$L : \mathbb{R} \ni \mu \mapsto -n \log(\sqrt{2\pi}\sigma_0) - \sum_{i=1}^n \frac{(X_i - \mu)^2}{2\sigma_0^2}.$$

This means that

$$L'(\mu) = \sum_{i=1}^{n} \frac{(X_i - \mu)}{\sigma_0^2}$$

for $\mu \in \mathbb{R}$ and

$$L'(\mu) = 0 \quad \Longleftrightarrow \quad \sum_{i=1}^{n} \frac{(X_i - \mu)}{\sigma_0^2} = 0 \quad \Longleftrightarrow \quad \sum_{i=1}^{n} (X_i - \mu) = 0 \quad \Longleftrightarrow \quad \mu = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Since L is continuous and $L(\mu) \to -\infty$ as $\mu \to \infty$ or $\mu \to -\infty$, the only stationary point of L must be a maximum. This shows that $\widehat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i$ is the unique maximum likelihood estimator for μ. Note that $\widehat{\mu}$ is the sample mean $\overline{\mathbf{X}}$ of $\mathbf{X}$.

Chapter 3

Linear regression

3.1 Linear models

In the following, we aim to explain a dependent variable Y in terms of independent variables $x^{(1)}, \ldots, x^{(k)}$, assuming that Y depends linearly on $x^{(1)}, \ldots, x^{(k)}$ and has a random error. More precisely, we suppose that

$$Y = \gamma_0 + \gamma_1 x^{(1)} + \ldots + \gamma_k x^{(k)} + \varepsilon = \gamma_0 + \sum_{j=1}^{k} \gamma_j x^{(j)} + \varepsilon \tag{3.1}$$

with $\gamma_0, \gamma_1, \ldots, \gamma_k \in \mathbb{R}$ and a random variable ε. This means that Y is an affine-linear function of $x^{(1)}, \ldots, x^{(k)}$ plus some random noise. We denote γ_0 as *intercept* and $\gamma_1, \ldots, \gamma_k$ as *coefficients*. For ε we require in the sequel that it has a finite second moment, i.e., $\mathbf{E}[\varepsilon^2] < \infty$, and that $\mathbf{E}[\varepsilon] = 0$. Here, $\mathbf{E}[\varepsilon] = 0$ is no loss of generality since, otherwise, we can subtract $\mathbf{E}[\varepsilon]$ from ε and add it to the intercept γ_0. The relation between Y and $x^{(1)}, \ldots, x^{(k)}$ in (3.1) is called *linear model*.

Example 3.1. In an experiment, guinea pigs were given different doses of vitamin C and the length of some cells responsible for the tooth growth was measured. Here, the cell length, abbreviated as **len**, is the dependent variable and the independent variable is the dose of vitamin C denoted by **dose**. Assuming a linear relation between **len** and **dose**, we consider the linear model

$$\mathbf{len} = \gamma_0 + \gamma_1 \mathbf{dose} + \varepsilon$$

with $\gamma_0, \gamma_1 \in \mathbb{R}$ and some random variable ε.

We say that a quantity z is *categorical* if it can take m different values that correspond to categories, which are not necessarily numbers. Now assume that z is also an independent variable for Y. Such an independent variable is called *factor* and we refer to its possible values as *levels*. Without loss of generality we can denote the m categories by $1, \ldots, m$. The naive idea is to add in (3.1) a further term $\tilde{\gamma} z$. However, this does not make sense as there is not necessarily an order between the categories. So if z is the factor colour with

levels blue, green and red, which we abbreviate by 1, 2 and 3, this would indicate that for $\tilde{\gamma} > 0$ red leads to the largest value of Y, while for $\tilde{\gamma} < 0$ blue maximises Y. The case that Y attains its maximum for green cannot occur. To overcome such problems, we introduce for the factor z with m levels $1, \ldots, m$ the *indicator* or *dummy variables*

$$\mathbf{1}\{z = \ell\} = \begin{cases} 1, & z = \ell, \\ 0, & z \neq \ell, \end{cases}$$

for $\ell \in \{1, \ldots, m\}$. In (3.1), we add the expression

$$\tilde{\gamma}_2 \mathbf{1}\{z = 2\} + \ldots + \tilde{\gamma}_m \mathbf{1}\{z = m\} = \sum_{\ell=2}^{m} \tilde{\gamma}_\ell \mathbf{1}\{z = \ell\}.$$

This means that we have $m - 1$ dummy variables for a factor with m levels. We can interpret $\tilde{\gamma}_\ell$ for $\ell \in \{2, \ldots, m\}$ as the effect on Y if we change z from 1 to ℓ. At a first glance, one might be tempted to include also the dummy variable $\mathbf{1}\{z = 1\}$ and to add

$$\tilde{\gamma}_1 \mathbf{1}\{z = 1\} + \ldots + \tilde{\gamma}_m \mathbf{1}\{z = m\}$$

in (3.1), but then one looses uniqueness since replacing the intercept γ_0 and the coefficients $\tilde{\gamma}_1, \ldots, \tilde{\gamma}_m$ by $\gamma_0 + a$ and $\tilde{\gamma}_1 - a, \ldots, \tilde{\gamma}_m - a$ leads to exactly the same values for Y.

Example 3.2. We consider the situation of Example 3.1. Beside different doses there are also two supplement types for vitamin C, namely orange juice (OJ) and ascorbic acid (VC). Thus we have a factor **supp** with the levels OJ and VC, which we incorporate into our linear model by introducing the dummy variable $\mathbf{1}\{\mathbf{supp} = \mathrm{VC}\}$. This leads to the linear model

$$\mathbf{len} = \gamma_0 + \gamma_1 \mathbf{dose} + \gamma_2 \mathbf{1}\{\mathbf{supp} = \mathrm{VC}\} + \varepsilon$$

with $\gamma_0, \gamma_1, \gamma_2 \in \mathbb{R}$ and some random error ε. Instead of $\mathbf{1}\{\mathbf{supp} = \mathrm{VC}\}$, one could also take $\mathbf{1}\{\mathbf{supp} = \mathrm{OJ}\}$.

Example 3.3. The job satisfaction score abbreviated as **score** of several people was measured. The aim is to explain it by the factors **gender** with the levels male and female and **education_level** with the levels school, college and university. Here, one can use the linear model

$$\begin{aligned} \mathbf{score} = \gamma_0 &+ \gamma_1 \mathbf{1}\{\mathbf{gender} = \mathrm{female}\} + \gamma_2 \mathbf{1}\{\mathbf{education_level} = \mathrm{college}\} \\ &+ \gamma_3 \mathbf{1}\{\mathbf{education_level} = \mathrm{university}\} + \varepsilon \end{aligned}$$

with $\gamma_0, \gamma_1, \gamma_2, \gamma_3 \in \mathbb{R}$ and some random error ε.

Sometimes one has interactions between independent variables, which means that the dependent variable is affected by the joint behaviour of several of the independent variables. Such effects can be incorporated into linear models by adding products of independent variables as additional independent variables.

Example 3.4. In the situation of Example 3.2, the effect of the dose could also depend on the supplement type. To include a potential interaction between the supplement type and the dose, one can modify the linear model to

$$\mathbf{len} = \gamma_0 + \gamma_1 \mathbf{dose} + \gamma_2 \mathbf{1}\{\mathbf{supp} = \mathrm{VC}\} + \gamma_3 \mathbf{1}\{\mathbf{supp} = \mathrm{VC}\}\mathbf{dose} + \varepsilon$$

with $\gamma_0, \gamma_1, \gamma_2, \gamma_3 \in \mathbb{R}$ and some random error ε. Note that adding also $\mathbf{1}\{\mathbf{supp} = \mathrm{OJ}\}\mathbf{dose}$ would lead to redundancies as before.

Similarly, one can take the dependence on some power of an independent variable into account by adding this power of the independent variable as a further independent variable. This, approach is called *polynomial regression*.

In the following, we consider the linear model in (3.1) and aim to estimate the parameters $\gamma_0, \gamma_1, \ldots, \gamma_k$ from observations for Y and $x^{(1)}, \ldots, x^{(k)}$. Let Y_i and $x_i^{(1)}, \ldots, x_i^{(k)}$ for $i \in \{1, \ldots, n\}$ be given such that

$$Y_i = \gamma_0 + \gamma_1 x_i^{(1)} + \ldots + \gamma_k x_i^{(k)} + \varepsilon_i = \gamma_0 + \sum_{j=1}^{k} \gamma_j x_i^{(j)} + \varepsilon_i \tag{3.2}$$

for $i \in \{1, \ldots, n\}$ with uncorrelated random variables $\varepsilon_1, \ldots, \varepsilon_n$ satisfying $\mathbf{E}[\varepsilon_i] = 0$ for $i \in \{1, \ldots, n\}$. Here n is called *sample size*.

We can rewrite (3.2) as

$$\mathbf{Y} = \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$$

with

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & x_1^{(1)} & \ldots & x_1^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n^{(1)} & \ldots & x_n^{(k)} \end{pmatrix}, \quad \boldsymbol{\gamma} = \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_k \end{pmatrix} \quad \text{and} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

The matrix $\mathbf{A}$ is called *design matrix*.

The *least squares estimator* $\widehat{\boldsymbol{\gamma}} = (\widehat{\gamma}_0, \widehat{\gamma}_1, \ldots, \widehat{\gamma}_k)^T$ for $\boldsymbol{\gamma}$ is the solution of the optimisation problem

$$\min_{\widetilde{\boldsymbol{\gamma}} \in \mathbb{R}^{k+1}} \|\mathbf{A}\widetilde{\boldsymbol{\gamma}} - \mathbf{Y}\|. \tag{3.3}$$

Since one can think of $\mathbf{A}\widetilde{\boldsymbol{\gamma}}$ as the prediction for $\mathbf{Y}$, the idea of the least squares estimator is to choose the prediction that fits best to the observed values. In the following we solve the optimisation problem in (3.3) by minimising the function

$$h : \mathbb{R}^{k+1} \to \mathbb{R}, \widetilde{\boldsymbol{\gamma}} \mapsto \|\mathbf{A}\widetilde{\boldsymbol{\gamma}} - \mathbf{Y}\|^2 = \widetilde{\boldsymbol{\gamma}}^T \mathbf{A}^T \mathbf{A} \widetilde{\boldsymbol{\gamma}} - 2\mathbf{Y}^T \mathbf{A} \widetilde{\boldsymbol{\gamma}} + \mathbf{Y}^T \mathbf{Y}.$$

The gradient is given by

$$\nabla h(\widetilde{\boldsymbol{\gamma}}) = 2\mathbf{A}^T \mathbf{A} \widetilde{\boldsymbol{\gamma}} - 2\mathbf{A}^T \mathbf{Y}$$

so that

$$\nabla h(\widetilde{\boldsymbol{\gamma}}) = \mathbf{0} \quad \Longleftrightarrow \quad 2\mathbf{A}^T\mathbf{A}\widetilde{\boldsymbol{\gamma}} - 2\mathbf{A}^T\mathbf{Y} = 0 \quad \Longleftrightarrow \quad \mathbf{A}^T\mathbf{A}\widetilde{\boldsymbol{\gamma}} = \mathbf{A}^T\mathbf{Y}.$$

The function h has the positive semi-definite Hessian matrix $2\mathbf{A}^T\mathbf{A}$ so that it is convex and any stationary point is a minimum. Thus, a solution of $\mathbf{A}^T\mathbf{A}\widetilde{\boldsymbol{\gamma}} = \mathbf{A}^T\mathbf{Y}$ is a minimiser of (3.3). If $\mathbf{A}^T\mathbf{A}$ is invertible, there is exactly one minimum of h and

$$\widehat{\boldsymbol{\gamma}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y}$$

is the least squares estimator for $\boldsymbol{\gamma}$.

Example 3.5. The data for the tooth growth of guinea pigs discussed in Example 3.1 is included in the statistical software `R` and can be loaded by the command `data("ToothGrowth")`. The intercepts and coefficients of linear models can be computed by using the command

`lm(`depend. variable $\sim$ independ. variable 1 `+ ... +` independ. variable k`,` data set`)`.

Note that the intercept is automatically included. For the linear models given in Examples 3.1, 3.2 and 3.4 one obtains the following estimates:

```
> lm(len ~ dose, ToothGrowth)

Call:
lm(formula = len ~ dose, data = ToothGrowth)

Coefficients:
(Intercept)       dose
7.422            9.764


> lm(len ~ dose+supp, ToothGrowth)

Call:
lm(formula = len ~ dose + supp, ToothGrowth)

Coefficients:
(Intercept)       dose       suppVC
9.272            9.764       -3.700

> lm(len ~ dose+supp+supp*dose, ToothGrowth)

Call:
lm(formula = len ~ dose + supp + supp * dose, data = ToothGrowth)
```

```
Coefficients:
(Intercept)       dose       suppVC       dose:suppVC
11.550             7.811      -8.255       3.904
```

From now on we always assume that $\mathbf{A}^T\mathbf{A}$ is invertible. This is equivalent to $\mathbf{A}$ having column rank $k+1$.

We define $\widehat{\mathbf{Y}} = \mathbf{A}\widehat{\boldsymbol{\gamma}}$. One can think of $\widehat{\mathbf{Y}}$ as the *predictions* for $\mathbf{Y}$ based on the estimates $\widehat{\boldsymbol{\gamma}}$ for $\boldsymbol{\gamma}$. The components of $\widehat{\mathbf{Y}}$ are also called *fitted values*. We denote the components of the vector

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \widehat{\mathbf{Y}}$$

as *residuals*. The random errors are given by $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{A}\boldsymbol{\gamma}$. Here, $\mathbf{A}\boldsymbol{\gamma}$ is unknown, but it can be approximated by $\mathbf{A}\widetilde{\boldsymbol{\gamma}}$. Thus, $\widehat{\boldsymbol{\varepsilon}}$ is an approximation for the unobserved $\boldsymbol{\varepsilon}$.

Let $\mathbb{U} \subseteq \mathbb{R}^d$ be a linear subspace. For any $\mathbf{z} \in \mathbb{R}^d$ there exists a unique solution $\mathbf{z}_0 \in \mathbb{U}$ of the optimisation problem $\min_{\mathbf{v}\in\mathbb{U}} \|\mathbf{z} - \mathbf{v}\|$, i.e., $\mathbf{z}_0$ is the point of $\mathbb{U}$ closest to $\mathbf{z}$. The function $\Pi_{\mathbb{U}} : \mathbb{R}^d \to \mathbb{U}$ that maps every $\mathbf{z}$ to its $\mathbf{z}_0$ is called the *orthogonal projection* onto $\mathbb{U}$. It is a linear map and satisfies

$$\Pi_{\mathbb{U}}^2 = \Pi_{\mathbb{U}}. \tag{3.4}$$

Note that $\mathbf{M} \in \mathbb{R}^{d\times d}$ is the matrix of $\Pi_{\mathbb{U}}$ if and only if

$$\mathbf{M}\mathbf{z} \in \mathbb{U} \quad \text{and} \quad \mathbf{z} - \mathbf{M}\mathbf{z} \perp \mathbb{U} \quad \text{for all} \quad \mathbf{z} \in \mathbb{R}^d. \tag{3.5}$$

For the matrix $\mathbf{M}$ of $\Pi_{\mathbb{U}}$ we obtain from (3.5) that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$\begin{aligned}\langle \mathbf{M}\mathbf{x}, \mathbf{y}\rangle - \langle \mathbf{x}, \mathbf{M}\mathbf{y}\rangle &= \langle \mathbf{M}\mathbf{x}, \mathbf{y}\rangle - \langle \mathbf{M}\mathbf{x}, \mathbf{M}\mathbf{y}\rangle + \langle \mathbf{M}\mathbf{x}, \mathbf{M}\mathbf{y}\rangle - \langle \mathbf{x}, \mathbf{M}\mathbf{y}\rangle \\ &= \langle \mathbf{M}\mathbf{x}, \mathbf{y} - \mathbf{M}\mathbf{y}\rangle + \langle \mathbf{M}\mathbf{x} - \mathbf{x}, \mathbf{M}\mathbf{y}\rangle = 0\end{aligned}$$

and, thus,

$$\mathbf{x}^T(\mathbf{M}^T - \mathbf{M})\mathbf{y} = 0$$

so that

$$\mathbf{M} = \mathbf{M}^T. \tag{3.6}$$

As usual we do not distinct between a matrix and the associated linear map. In particular, we denote by $\Pi_{\mathbb{U}}$ also the matrix of the projection onto $\mathbb{U}$.

We write $\mathbb{L}$ for the linear subspace generated by the columns of $\mathbf{A}$, i.e.,

$$\mathbb{L} = \{\mathbf{A}\mathbf{v} : \mathbf{v} \in \mathbb{R}^{k+1}\}.$$

Moreover, $\mathbf{I}_d \in \mathbb{R}^{d\times d}$ stands for the identity matrix (i.e., the diagonal matrix with ones on the diagonal) on $\mathbb{R}^d$.

Lemma 3.6.

a) The matrix $\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is the projection $\Pi_{\mathbb{L}}$ onto $\mathbb{L}$. In particular, this means that

$$\widehat{\mathbf{Y}} = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y} = \Pi_{\mathbb{L}}\mathbf{Y}$$

and

$$\widehat{\boldsymbol{\varepsilon}} = (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{Y} = (\mathbf{I}_n - \Pi_{\mathbb{L}})\mathbf{Y}.$$

b) The matrix $\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is symmetric and satisfies

$$(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^2 = (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T).$$

Proof. For $\mathbf{z} \in \mathbb{R}^n$ we have $\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z} = \mathbf{A}\mathbf{v}$ with $\mathbf{v} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z} \in \mathbb{R}^{k+1}$ so that $\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z} \in \mathbb{L}$. Since

$$\mathbf{A}^T(\mathbf{z} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z}) = \mathbf{A}^T\mathbf{z} - \mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z} = \mathbf{A}^T\mathbf{z} - \mathbf{A}^T\mathbf{z} = \mathbf{0},$$

we see that

$$\langle \mathbf{A}\mathbf{v}, \mathbf{z} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z}\rangle = \mathbf{v}^T\mathbf{A}^T(\mathbf{z} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z}) = 0$$

for all $\mathbf{v} \in \mathbb{R}^{k+1}$ and, thus, $\mathbf{z} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{z} \perp \mathbb{L}$. Now the assertion of a) follows from (3.5).

Because of $(\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T = (\mathbf{A}^T)^T((\mathbf{A}^T\mathbf{A})^{-1})^T\mathbf{A}^T = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$, where we used the symmetry of $\mathbf{A}^T\mathbf{A}$ and $(\mathbf{A}^T\mathbf{A})^{-1}$, the matrices $\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ and $\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ are both symmetric. From part a) and (3.4) we deduce

$$\begin{aligned}(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^2 &= (\mathbf{I}_n - \Pi_{\mathbb{L}})^2 = \mathbf{I}_n^2 - \mathbf{I}_n\Pi_{\mathbb{L}} - \Pi_{\mathbb{L}}\mathbf{I}_n + \Pi_{\mathbb{L}}^2 = \mathbf{I}_n^2 - 2\Pi_{\mathbb{L}} + \Pi_{\mathbb{L}}^2 \\ &= \mathbf{I}_n - 2\Pi_{\mathbb{L}} + \Pi_{\mathbb{L}} = \mathbf{I}_n - \Pi_{\mathbb{L}} = \mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T,\end{aligned}$$

which is b). □

The following theorem collects some important properties of the least squares estimator:

Theorem 3.7.

a) The least squares estimator $\widehat{\boldsymbol{\gamma}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y}$ is an unbiased estimator for $\boldsymbol{\gamma}$, i.e., $\mathbf{E}[\widehat{\boldsymbol{\gamma}}] = \boldsymbol{\gamma}$.

b) Let $\sigma \in (0, \infty)$ be such that $\mathbf{Var}[\varepsilon_i] = \sigma^2$ for all $i \in \{1, \ldots, n\}$. Then

$$\mathbf{Cov}(\widehat{\boldsymbol{\gamma}}) = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}$$

and, if $n > k + 1$,

$$\widehat{\sigma}^2 := \frac{1}{n-k-1}\|\widehat{\boldsymbol{\varepsilon}}\|^2$$

is an unbiased estimator for σ^2, i.e., $\mathbf{E}[\widehat{\sigma}^2] = \sigma^2$.

Before proving Theorem 3.7, we recall traces of matrices as well as a useful property of them. The *trace* of a symmetric matrix $\mathbf{M} = (M_{i,j})_{i,j\in\{1,\ldots,d\}} \in \mathbb{R}^{d\times d}$ is defined as

$$\mathrm{trace}(\mathbf{M}) = \sum_{i=1}^{d} M_{i,i},$$

i.e., the trace of $\mathbf{M}$ is the sum of its diagonal elements. For two given matrices $\mathbf{U} = (u_{i,j})_{i\in\{1,\dots,p\},j\in\{1,\dots,q\}} \in \mathbb{R}^{p\times q}$ and $\mathbf{V} = (v_{i,j})_{i\in\{1,\dots,q\},j\in\{1,\dots,p\}} \in \mathbb{R}^{q\times p}$ we have that

$$\operatorname{trace}(\mathbf{UV}) = \sum_{i=1}^{p}\sum_{j=1}^{q} u_{ij}v_{ji} = \sum_{j=1}^{q}\sum_{i=1}^{p} v_{ji}u_{ij} = \operatorname{trace}(\mathbf{VU}). \tag{3.7}$$

Proof of Theorem 3.7. First note that

$$\widehat{\boldsymbol{\gamma}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(\mathbf{A}\boldsymbol{\gamma}+\boldsymbol{\varepsilon}) = \boldsymbol{\gamma} + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\boldsymbol{\varepsilon}. \tag{3.8}$$

Since $\mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, it follows from Lemma 2.1 a) that

$$\mathbf{E}[\widehat{\boldsymbol{\gamma}}] = \boldsymbol{\gamma} + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\gamma} + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{0} = \boldsymbol{\gamma},$$

which is part a). Under the assumption of b), we have $\mathbf{E}[\varepsilon_i] = 0$ and $\mathbf{Var}[\varepsilon_i] = \sigma^2$ for all $i \in \{1,\dots,n\}$. Together with the fact that $\varepsilon_1,\dots,\varepsilon_n$ are uncorrelated, this yields

$$\mathbf{E}[\varepsilon_{i_1}\varepsilon_{i_2}] = \mathbf{Cov}(\varepsilon_{i_1},\varepsilon_{i_2}) = \begin{cases}\sigma^2, & i_1 = i_2,\\ 0, & i_1 \neq i_2,\end{cases} \tag{3.9}$$

for $i_1, i_2 \in \{1,\dots,n\}$ so that $\mathbf{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$. Thus, (3.8) and Lemma 2.1 a) lead to

$$\begin{aligned}\mathbf{Cov}(\widehat{\boldsymbol{\gamma}}) &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Cov}(\boldsymbol{\varepsilon})((\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T\\ &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\sigma^2\mathbf{I}_n\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\\ &= \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1},\end{aligned}$$

which is the first statement of part b). From (3.8) we derive

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \widehat{\mathbf{Y}} = \mathbf{Y} - \mathbf{A}\widehat{\boldsymbol{\gamma}} = \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} - \mathbf{A}(\boldsymbol{\gamma} + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\boldsymbol{\varepsilon}) = (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon}$$

so that

$$\mathbf{E}[\|\widehat{\boldsymbol{\varepsilon}}\|^2] = \mathbf{E}[\widehat{\boldsymbol{\varepsilon}}^T\widehat{\boldsymbol{\varepsilon}}] = \mathbf{E}[\boldsymbol{\varepsilon}^T(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon}].$$

By Lemma 3.6 b) this simplifies to

$$\mathbf{E}[\|\widehat{\boldsymbol{\varepsilon}}\|^2] = \mathbf{E}[\boldsymbol{\varepsilon}^T(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon}].$$

From (3.9) and (3.7) we deduce that

$$\begin{aligned}\mathbf{E}[\|\widehat{\boldsymbol{\varepsilon}}\|^2] &= \sigma^2\operatorname{trace}(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\\ &= \sigma^2\operatorname{trace}(\mathbf{I}_n) - \sigma^2\operatorname{trace}(\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\\ &= \sigma^2\operatorname{trace}(\mathbf{I}_n) - \sigma^2\operatorname{trace}(\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1})\\ &= \sigma^2\operatorname{trace}(\mathbf{I}_n) - \sigma^2\operatorname{trace}(\mathbf{I}_{k+1}) = \sigma^2(n-k-1).\end{aligned}$$

Consequently, we have

$$\mathbf{E}[\widehat{\sigma}^2] = \frac{1}{n-k-1}\mathbf{E}[\|\widehat{\boldsymbol{\varepsilon}}\|^2] = \sigma^2,$$

which completes the proof. □

The next result concerns in some sense the optimality of the least squares estimator.

Theorem 3.8. Assume that there exists some $\sigma \in (0, \infty)$ such that $\mathbf{Var}[\varepsilon^{(i)}] = \sigma^2$ for all $i \in \{1, \dots, n\}$ and let $\mathbf{a} \in \mathbb{R}^{k+1}$. Then, $\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle$ is the best linear unbiased estimator for $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle$, i.e., for all $\mathbf{b} \in \mathbb{R}^n$ satisfying $\mathbf{E}[\langle \mathbf{b}, \mathbf{Y} \rangle] = \langle \mathbf{a}, \boldsymbol{\gamma} \rangle$ for all $\boldsymbol{\gamma} \in \mathbb{R}^{k+1}$, one has

$$\mathbf{Var}[\langle \mathbf{b}, \mathbf{Y} \rangle] \geq \mathbf{Var}[\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle]$$

for all $\boldsymbol{\gamma} \in \mathbb{R}^{k+1}$.

Proof. We have $\langle \mathbf{b}, \mathbf{Y} \rangle = \langle \mathbf{b}, \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} \rangle = \mathbf{b}^T \mathbf{A}\boldsymbol{\gamma} + \mathbf{b}^T \boldsymbol{\varepsilon}$. Together with $\mathbf{E}[\boldsymbol{\varepsilon}] = 0$ and $\mathbf{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$, it follows from Lemma 2.1 a) that

$$\mathbf{E}[\langle \mathbf{b}, \mathbf{Y} \rangle] = \mathbf{b}^T \mathbf{A}\boldsymbol{\gamma} + \mathbf{b}^T \mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{b}^T \mathbf{A}\boldsymbol{\gamma}$$

and

$$\mathbf{Var}[\langle \mathbf{b}, \mathbf{Y} \rangle] = \mathbf{E}[(\mathbf{b}^T \mathbf{A}\boldsymbol{\gamma} + \mathbf{b}^T \boldsymbol{\varepsilon} - \mathbf{b}^T \mathbf{A}\boldsymbol{\gamma})^2] = \mathbf{E}[(\mathbf{b}^T \boldsymbol{\varepsilon})^2] = \mathbf{b}^T \sigma^2 \mathbf{I}_n \mathbf{b} = \sigma^2 \mathbf{b}^T \mathbf{b}. \tag{3.10}$$

The first identity and the assumption that $\langle \mathbf{b}, \mathbf{Y} \rangle$ is an unbiased estimator for $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle$ imply that $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle = \mathbf{b}^T \mathbf{A}\boldsymbol{\gamma}$ for all $\boldsymbol{\gamma} \in \mathbb{R}^{k+1}$ so that $\mathbf{a}^T = \mathbf{b}^T \mathbf{A}$.

From Theorem 3.7 b) and Lemma 2.1 a) we obtain

$$\mathbf{Var}(\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle) = \mathbf{a}^T \mathbf{Cov}(\widehat{\boldsymbol{\gamma}}) \mathbf{a} = \mathbf{a}^T \sigma^2 (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a} = \sigma^2 \mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}.$$

Together with (3.10) and $\mathbf{a}^T = \mathbf{b}^T \mathbf{A}$ this leads to

$$\mathbf{Var}[\langle \mathbf{b}, \mathbf{Y} \rangle] - \mathbf{Var}[\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle] = \sigma^2 (\mathbf{b}^T \mathbf{b} - \mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}) = \sigma^2 \mathbf{b}^T (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T) \mathbf{b}.$$

By Lemma 3.6 b) this can be rewritten as

$$\begin{aligned} \mathbf{Var}[\langle \mathbf{b}, \mathbf{Y} \rangle] - \mathbf{Var}[\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle] &= \sigma^2 \mathbf{b}^T (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T)^T (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T) \mathbf{b} \\ &= \sigma^2 \|(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T) \mathbf{b}\|^2 \geq 0, \end{aligned}$$

which completes the proof. □

Finally, we study the special case, where we do not have any independent variables.

Example 3.9. We consider the special case $k = 0$, i.e., we have that

$$Y_i = \gamma_0 + \varepsilon_i$$

for $i \in \{1, \dots, n\}$. This means that we have n uncorrelated random variables $Y_1, \dots, Y_n$ such that $\mathbf{E}[Y_i] = \gamma_0$ for $i \in \{1, \dots, n\}$. Then, $\mathbf{A} = (1, \dots, 1)^T \in \mathbb{R}^n$ so that

$$\mathbf{A}^T \mathbf{A} = n \quad \text{and} \quad \mathbf{A}^T \mathbf{Y} = \sum_{i=1}^n Y_i.$$

Thus, the least squares estimator for γ_0 is

$$\widehat{\gamma}_0 = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

The quantity on the right-hand side is the sample mean of $\mathbf{Y}$, which we denote by $\overline{\mathbf{Y}}$. The residuals are given by

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \begin{pmatrix} \overline{\mathbf{Y}} \\ \vdots \\ \overline{\mathbf{Y}} \end{pmatrix}$$

so that

$$\|\widehat{\boldsymbol{\varepsilon}}\|^2 = \sum_{i=1}^{n}(Y_i - \overline{\mathbf{Y}})^2$$

and

$$\widehat{\sigma}^2 = \frac{1}{n-1}\|\widehat{\boldsymbol{\varepsilon}}\|^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{\mathbf{Y}})^2.$$

This means that $\widehat{\sigma}^2$ is the well-known *sample variance* of $\mathbf{Y}$.

3.2 Gaussian linear model

In order to develop statistical tests, we need stronger assumptions on the distribution of the random error terms $\varepsilon_1, \ldots, \varepsilon_n$. From now on we assume that $\varepsilon_1, \ldots, \varepsilon_n$ are independent and are all normally distributed with expectation zero and the same unknown standard deviation $\sigma \in (0, \infty)$. This is equivalent to $\boldsymbol{\varepsilon} \sim \text{Normal}(\mathbf{0}, \sigma^2\mathbf{I}_n)$. We refer to the corresponding linear model as *Gaussian linear model*. As before we assume that $\mathbf{A}^T\mathbf{A}$ is invertible.

Theorem 3.10. Consider a Gaussian linear model and let $\widehat{\sigma}^2_{ML} = \frac{1}{n}\|\widehat{\boldsymbol{\varepsilon}}\|^2$. Then, $(\widehat{\boldsymbol{\gamma}}, \widehat{\sigma}^2_{ML})$ is the maximum likelihood estimator for $(\boldsymbol{\gamma}, \sigma^2)$.

Proof. Since $\mathbf{Y} = \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \text{Normal}(\mathbf{0}, \sigma^2\mathbf{I}_n)$, it follows from Lemma 2.1 b) that $\mathbf{Y} \sim \text{Normal}(\mathbf{A}\boldsymbol{\gamma}, \sigma^2\mathbf{I}_n)$. Consequently, $\mathbf{Y}$ has the density

$$f(\mathbf{y}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(- \frac{(y_i - (\mathbf{A}\boldsymbol{\gamma})_i)^2}{2\sigma^2} \right) = \frac{1}{\sqrt{2\pi}^n \sigma^n} \exp\left(- \frac{\|\mathbf{y} - \mathbf{A}\boldsymbol{\gamma}\|^2}{2\sigma^2} \right)$$

for $\mathbf{y} = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$. Thus, we obtain the likelihood function

$$(\boldsymbol{\gamma}, \sigma^2) \mapsto \frac{1}{\sqrt{2\pi}^n \sigma^n} \exp\left(- \frac{\|\mathbf{Y} - \mathbf{A}\boldsymbol{\gamma}\|^2}{2\sigma^2} \right)$$

and the log likelihood function

$$L(\boldsymbol{\gamma}, \sigma^2) = -n \log \sqrt{2\pi} - \frac{n}{2} \log \sigma^2 - \frac{\|\mathbf{Y} - \mathbf{A}\boldsymbol{\gamma}\|^2}{2\sigma^2},$$

which we have to maximise. For fixed σ^2 this function becomes maximal if $\|\mathbf{Y} - \mathbf{A}\boldsymbol{\gamma}\|^2$ is minimal. By definition of the least squares estimator, this is achieved for $\boldsymbol{\gamma} = \widehat{\boldsymbol{\gamma}}$. Thus, it remains to maximise the function $\sigma^2 \mapsto L(\widehat{\boldsymbol{\gamma}}, \sigma^2)$. Because of

$$\frac{\partial L(\widehat{\boldsymbol{\gamma}}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\|\mathbf{Y} - \mathbf{A}\widehat{\boldsymbol{\gamma}}\|^2}{2\sigma^4},$$

we have

$$\frac{\partial L(\widehat{\boldsymbol{\gamma}}, \sigma^2)}{\partial \sigma^2} = 0 \quad \Longleftrightarrow \quad -\frac{n}{2\sigma^2} + \frac{\|\mathbf{Y} - \mathbf{A}\widehat{\boldsymbol{\gamma}}\|^2}{2\sigma^4} = 0 \quad \Longleftrightarrow \quad \sigma^2 = \frac{1}{n}\|\mathbf{Y} - \mathbf{A}\widehat{\boldsymbol{\gamma}}\|^2 = \frac{1}{n}\|\widehat{\boldsymbol{\varepsilon}}\|^2.$$

By the form of $\sigma^2 \mapsto L(\widehat{\boldsymbol{\gamma}}, \sigma^2)$, this stationary point belongs to the unique maximum. Thus, $(\widehat{\boldsymbol{\gamma}}, \widehat{\sigma}^2_{ML})$ is the maximum likelihood estimator for $(\boldsymbol{\gamma}, \sigma^2)$. □

Note that the maximum likelihood estimator $\widehat{\sigma}^2_{ML}$ and the previously introduced estimator $\widehat{\sigma}^2$ for σ^2 are related by

$$\widehat{\sigma}^2_{ML} = \frac{n-k-1}{n}\widehat{\sigma}^2.$$

Since $\widehat{\sigma}^2$ is an unbiased estimator for σ^2 (see Theorem 3.7 b)), this implies that $\widehat{\sigma}^2_{ML}$ is not unbiased. However, since $\frac{n-k-1}{n} \to 1$ as $n \to \infty$, $\widehat{\sigma}^2_{ML}$ is asymptotically unbiased for $n \to \infty$.

3.3 Testing in the Gaussian linear model

In this section, we derive tests and confidence intervals for the Gaussian linear model introduced in the previous section. As before, we always assume tacitly that $\mathbf{A}^T\mathbf{A}$ is invertible.

Theorem 3.11. Consider a Gaussian linear model.

a) The random vectors $\widehat{\boldsymbol{\gamma}}$ and $\widehat{\boldsymbol{\varepsilon}}$ are independent and satisfy $\widehat{\boldsymbol{\gamma}} \sim \text{Normal}(\boldsymbol{\gamma}, \sigma^2(\mathbf{A}^T\mathbf{A})^{-1})$ and $\widehat{\boldsymbol{\varepsilon}} \sim \text{Normal}(\mathbf{0}, \sigma^2(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T))$.

b) If $n > k+1$, one has $\|\widehat{\boldsymbol{\varepsilon}}\|^2/\sigma^2 \sim \chi^2_{n-k-1}$.

Proof. Recall that

$$\widehat{\boldsymbol{\gamma}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(\mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}) = \boldsymbol{\gamma} + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\boldsymbol{\varepsilon}$$

and

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \widehat{\mathbf{Y}} = \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} - \mathbf{A}\widehat{\boldsymbol{\gamma}} = (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon}. \tag{3.11}$$

As affine transformations of the normal random vector $\boldsymbol{\varepsilon}$, $\widehat{\boldsymbol{\gamma}}$ and $\widehat{\boldsymbol{\varepsilon}}$ are also normal random vectors (see Lemma 2.1 b)). By Lemma 2.1 a) and $\boldsymbol{\varepsilon} \sim \text{Normal}(\mathbf{0}, \sigma^2\mathbf{I}_n)$, their expectations and covariance matrices are given by

$$\mathbf{E}[\widehat{\boldsymbol{\gamma}}] = \boldsymbol{\gamma} + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\gamma},$$

$$\mathbf{E}[\widehat{\boldsymbol{\varepsilon}}] = (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0},$$

$$\mathbf{Cov}(\widehat{\boldsymbol{\gamma}}) = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Cov}(\boldsymbol{\varepsilon})\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{I}_n\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}$$

and

$$\begin{aligned}\mathbf{Cov}(\widehat{\boldsymbol{\varepsilon}}) &= (\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{Cov}(\boldsymbol{\varepsilon})(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T \\ &= \sigma^2(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{I}_n(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T = \sigma^2(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^2 \\ &= \sigma^2(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T),\end{aligned}$$

where we used Lemma 3.6 b) in the last step. This shows that $\widehat{\boldsymbol{\gamma}}$ and $\widehat{\boldsymbol{\varepsilon}}$ have the stated distributions.

We obtain

$$\begin{aligned}\mathbf{Cov}(\widehat{\boldsymbol{\gamma}}, \widehat{\boldsymbol{\varepsilon}}) &= \mathbf{E}[(\widehat{\boldsymbol{\gamma}} - \mathbf{E}[\widehat{\boldsymbol{\gamma}}])(\widehat{\boldsymbol{\varepsilon}} - \mathbf{E}[\widehat{\boldsymbol{\varepsilon}}])^T] \\ &= \mathbf{E}[(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\boldsymbol{\varepsilon}((\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon})^T] \\ &= \mathbf{E}[(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)] \\ &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T](\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) \\ &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\sigma^2\mathbf{I}_n(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) \\ &= \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(\mathbf{I}_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) \\ &= \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T - (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) = \mathbf{0}\end{aligned}$$

so that $\widehat{\boldsymbol{\gamma}}$ and $\widehat{\boldsymbol{\varepsilon}}$ are uncorrelated. Since their joint distribution is also a multivariate normal distribution (as they are obtained as affine transformations of $\boldsymbol{\varepsilon}$), they are also independent by Lemma 2.2, which completes the proof of a).

From (3.11) and Lemma 3.6 a) it follows that

$$\|\widehat{\boldsymbol{\varepsilon}}\|^2 = \|(\mathbf{I}_n - \Pi_{\mathbb{L}})\boldsymbol{\varepsilon}\|^2.$$

Let $\mathbf{u}_1, \ldots, \mathbf{u}_n \in \mathbb{R}^n$ be an orthonormal base of $\mathbb{R}^n$ such that $\mathbf{u}_1, \ldots, \mathbf{u}_{k+1}$ generate the linear subspace $\mathbb{L}$. We denote by $\mathbf{O}$ the matrix with the column vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$. Obviously, we have $\mathbf{O}^T\mathbf{O} = \mathbf{I}_n$. This implies $\mathbf{O}\mathbf{O}^T\mathbf{O}\mathbf{O}^T = \mathbf{O}\mathbf{O}^T$ so that multiplication by $(\mathbf{O}\mathbf{O}^T)^{-1}$ leads to $\mathbf{O}\mathbf{O}^T = \mathbf{I}_n$. Note that $\mathbf{O}\boldsymbol{\varepsilon} \sim \text{Normal}(\mathbf{O}\cdot\mathbf{0}, \sigma^2\mathbf{O}\mathbf{I}_n\mathbf{O}^T) = \text{Normal}(\mathbf{0}, \sigma^2\mathbf{I}_n)$ has the same distribution as $\boldsymbol{\varepsilon}$ and that the multiplication by $\mathbf{O}^T$ does not change the norm of a vector. Thus, we obtain

$$\|\widehat{\boldsymbol{\varepsilon}}\|^2 \stackrel{d}{=} \|(\mathbf{I}_n - \Pi_{\mathbb{L}})\mathbf{O}\boldsymbol{\varepsilon}\|^2 = \|\mathbf{O}^T(\mathbf{I}_n - \Pi_{\mathbb{L}})\mathbf{O}\boldsymbol{\varepsilon}\|^2,$$

where $\stackrel{d}{=}$ stands for equality in distribution. It is easy to see that $\mathbf{O}^T\mathbf{I}_n\mathbf{O} = \mathbf{I}_n$. Because of

$$\Pi_{\mathbb{L}}\mathbf{u}_i = \begin{cases} \mathbf{u}_i, & i \in \{1, \ldots, k+1\}, \\ \mathbf{0}, & i \in \{k+2, \ldots, n\}, \end{cases}$$

$\mathbf{O}^T\Pi_{\mathbb{L}}\mathbf{O}$ is a diagonal matrix such that the first $k+1$ diagonal elements are one and the remaining diagonal elements are zero. Thus, $\mathbf{O}^T(\mathbf{I}_n - \Pi_{\mathbb{L}})\mathbf{O}$ is a diagonal matrix whose first $k+1$ diagonal elements are zero and the others are ones. This implies (see Subsection 2.1 for the definition of the chi-square distribution)

$$\|\widehat{\boldsymbol{\varepsilon}}\|^2 \stackrel{d}{=} \sum_{i=k+2}^{n} \varepsilon_i^2 \sim \chi^2_{n-k-1},$$

which is part b). □

From Theorem 3.11 we derive the following tests and confidence intervals. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$ with $\mathbf{u} = (u_1, \ldots, u_d)^T$ and $\mathbf{v} = (v_1, \ldots, v_d)^T$ we denote by $\langle \mathbf{u}, \mathbf{v} \rangle$ the scalar product of $\mathbf{u}$ and $\mathbf{v}$, i.e.,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{d} u_i v_i.$$

Corollary 3.12. Consider a Gaussian linear model with $n > k+1$ and let $\mathbf{a} \in \mathbb{R}^{k+1}$, $c_0 \in \mathbb{R}$ and $\alpha \in (0,1)$.

a) For $H_0 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle = c_0$ against $H_1 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle \neq c_0$,

$$\varphi_{\neq}(\mathbf{Y}) = \begin{cases} \text{cannot reject}, & c_0 \in I^{\neq}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}}, \\ \text{reject}, & c_0 \notin I^{\neq}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}}, \end{cases}$$

with

$$I^{\neq}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}} = \left(\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - t_{n-k-1,1-\alpha/2}\widehat{\sigma}\sqrt{\mathbf{a}^T(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{a}}, \langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle + t_{n-k-1,1-\alpha/2}\widehat{\sigma}\sqrt{\mathbf{a}^T(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{a}}\right)$$

is a level-α test. Moreover, $I^{\neq}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}}$ is a confidence interval for $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle$ with error level α.

b) For $H_0 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle \leq c_0$ against $H_1 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle > c_0$,

$$\varphi_{>}(\mathbf{Y}) = \begin{cases} \text{cannot reject}, & c_0 \in I^{>}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}}, \\ \text{reject}, & c_0 \notin I^{>}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}}, \end{cases}$$

with

$$I^{>}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}} = \left(\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - t_{n-k-1,1-\alpha}\widehat{\sigma}\sqrt{\mathbf{a}^T(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{a}}, \infty\right)$$

is a level-α test. Moreover, $I^{>}_{\mathbf{a},\widehat{\boldsymbol{\gamma}}}$ is a confidence interval for $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle$ with error level α.

Proof. Recall that $\widehat{\sigma}^2 = \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{n-k-1}$. From Theorem 3.11 and Lemma 2.1 b) it follows that

$$\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle \sim \text{Normal}(\langle \mathbf{a}, \boldsymbol{\gamma} \rangle, \sigma^2 \mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}) \quad \text{and} \quad (n-k-1)\widehat{\sigma}^2/\sigma^2 \sim \chi^2_{n-k-1}$$

and that both random variables are independent. Thus, by the definition of the t-distribution (see Subsection 2.1), we have

$$\frac{\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}} \widehat{\sigma}} = \frac{\frac{\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\sqrt{\sigma^2 \mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}}}{\sqrt{\frac{(n-k-1)\widehat{\sigma}^2}{(n-k-1)\sigma^2}}} \sim t_{n-k-1} \tag{3.12}$$

since the numerator of the second fraction follows a standard normal distribution.

From (3.12) we derive

$$\begin{aligned}
\mathbf{P}(\langle \mathbf{a}, \boldsymbol{\gamma} \rangle \in I^{\neq}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}) &= \mathbf{P}\left(\frac{\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}} \widehat{\sigma}} \in (-t_{n-k-1,1-\alpha/2}, t_{n-k-1,1-\alpha/2}) \right) \\
&= \mathbf{P}(Y \in (-t_{n-k-1,1-\alpha/2}, t_{n-k-1,1-\alpha/2})) \\
&= \mathbf{P}(Y \leq t_{n-k-1,1-\alpha/2}) - \mathbf{P}(Y \leq -t_{n-k-1,1-\alpha/2}) \\
&= 2\mathbf{P}(Y \leq t_{n-k-1,1-\alpha/2}) - 1 = 2 \cdot \left(1 - \frac{\alpha}{2}\right) - 1 = 1 - \alpha
\end{aligned}$$

with $Y \sim t_{n-k-1}$, where we used (2.3) in the second last step. Thus, $I^{\neq}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}$ is a confidence interval for $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle$ with error level α. Under $H_0 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle = c_0$ we have

$$\mathbf{P}(\varphi_{\neq} \text{ rejects } H_0) = 1 - \mathbf{P}(c_0 \in I^{\neq}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}) = 1 - \mathbf{P}(\langle \mathbf{a}, \boldsymbol{\gamma} \rangle \in I^{\neq}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}) = 1 - (1 - \alpha) = \alpha,$$

which shows that $\varphi_{\neq}$ is a level-α test.

Similarly as above with $Y \sim t_{n-k-1}$, we deduce

$$\begin{aligned}
\mathbf{P}(\langle \mathbf{a}, \boldsymbol{\gamma} \rangle \in I^{>}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}) &= \mathbf{P}\left(\frac{\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}} \widehat{\sigma}} \in (-\infty, t_{n-k-1,1-\alpha}) \right) \\
&= \mathbf{P}(Y \in (-t_{n-k-1,1-\alpha}, \infty)) = \mathbf{P}(Y \in (-\infty, t_{n-k-1,1-\alpha})) \\
&= \mathbf{P}(Y \leq t_{n-k-1,1-\alpha}) = 1 - \alpha
\end{aligned}$$

so that $I^{>}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}$ is a confidence interval for $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle$ with error level α. If $\langle \mathbf{a}, \boldsymbol{\gamma} \rangle \leq c_0$, we obtain

$$\begin{aligned}
\mathbf{P}(\varphi_{>} \text{ rejects } H_0) &= \mathbf{P}(c_0 \notin I^{>}_{\mathbf{a}, \widehat{\boldsymbol{\gamma}}}) \\
&= \mathbf{P}\left(\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - t_{n-k-1,1-\alpha} \widehat{\sigma} \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}} \geq c_0 \right) \\
&= \mathbf{P}\left(\frac{\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\widehat{\sigma} \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}} \geq \frac{c_0 - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\widehat{\sigma} \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}} + t_{n-k-1,1-\alpha} \right) \\
&\leq \mathbf{P}\left(\frac{\langle \mathbf{a}, \widehat{\boldsymbol{\gamma}} \rangle - \langle \mathbf{a}, \boldsymbol{\gamma} \rangle}{\widehat{\sigma} \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}} \geq t_{n-k-1,1-\alpha} \right) = \mathbf{P}(Y \geq t_{n-k-1,1-\alpha}) \\
&= 1 - \mathbf{P}(Y \leq t_{n-k-1,1-\alpha}) = 1 - (1 - \alpha) = \alpha,
\end{aligned}$$

which proves that $\varphi_{>}$ is a level-α test. □

In Corollary 3.12, two choices for $\mathbf{a}$ are of particular interest. If we are interested in predicting the dependent variable for values $\tilde{x}^{(1)}, \dots, \tilde{x}^{(k)}$ of the independent variables, we can choose $\mathbf{a} = (1, \tilde{x}^{(1)}, \dots, \tilde{x}^{(k)})^T$. On the other hand, for $\mathbf{a}$ being the vector in $\mathbb{R}^{k+1}$ such that the $(j+1)$-st component is one and all other components are zero, we can test $\gamma_j = 0$. We obtain that we can reject

$$H_0 : \gamma_j = 0 \quad \text{against} \quad H_1 : \gamma_j \neq 0$$

at level α if

$$0 \notin \big(\widehat{\gamma}_j - t_{n-k-1,1-\alpha/2}\widehat{\sigma}\sqrt{((\mathbf{A}^T\mathbf{A})^{-1})_{j+1,j+1}}, \widehat{\gamma}_j + t_{n-k-1,1-\alpha/2}\widehat{\sigma}\sqrt{((\mathbf{A}^T\mathbf{A})^{-1})_{j+1,j+1}}\big),$$

which is equivalent to

$$\frac{\widehat{\gamma}_j}{\widehat{\sigma}\sqrt{((\mathbf{A}^T\mathbf{A})^{-1})_{j+1,j+1}}} \notin (-t_{n-k-1,1-\alpha/2}, t_{n-k-1,1-\alpha/2}). \tag{3.13}$$

Corollary 3.12 allows to test if a single equation holds for the components of $\boldsymbol{\gamma}$. In order to test if several equations are satisfied, we need the following theorem.

Theorem 3.13. Consider a Gaussian linear model. Let $\mathbb{H} \subseteq \mathbb{R}^n$ be a linear subspace such that $\mathbb{H} \subseteq \mathbb{L}$, $r := \dim \mathbb{H} < k+1$ and $\mathbf{A}\boldsymbol{\gamma} \in \mathbb{H}$ and let $\Pi_{\mathbb{H}}$ be the orthogonal projection onto $\mathbb{H}$. Then,

a) $\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}$ and $\widehat{\boldsymbol{\varepsilon}}$ are independent,

b) $\frac{\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2}{\sigma^2} \sim \chi^2_{k+1-r}$,

c) if $n > k+1$, $\frac{(n-k-1)\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2}{(k+1-r)\|\widehat{\boldsymbol{\varepsilon}}\|^2} \sim \mathcal{F}_{k+1-r,n-k-1}$.

Proof. From Lemma 3.6 a) we obtain

$$\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y} = (\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\mathbf{Y} = (\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})(\mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}) = \mathbf{A}\boldsymbol{\gamma} - \mathbf{A}\boldsymbol{\gamma} + (\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\boldsymbol{\varepsilon} = (\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\boldsymbol{\varepsilon},$$

where we used $\mathbf{A}\boldsymbol{\gamma} \in \mathbb{H} \subseteq \mathbb{L}$, and

$$\widehat{\boldsymbol{\varepsilon}} = (\mathbf{I}_n - \Pi_{\mathbb{L}})\boldsymbol{\varepsilon}.$$

This leads to

$$\begin{aligned}\mathbf{E}[\widehat{\boldsymbol{\varepsilon}}(\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y})^T] &= \mathbf{E}[(\mathbf{I}_n - \Pi_{\mathbb{L}})\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})^T] = (\mathbf{I}_n - \Pi_{\mathbb{L}})\mathbf{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T](\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})^T \\ &= (\mathbf{I}_n - \Pi_{\mathbb{L}})\sigma^2\mathbf{I}_n(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})^T = \sigma^2(\mathbf{I}_n - \Pi_{\mathbb{L}})(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}}),\end{aligned}$$

where we employed in the last step that, by (3.6), $\Pi_{\mathbb{L}} = \Pi_{\mathbb{L}}^T$ and $\Pi_{\mathbb{H}} = \Pi_{\mathbb{H}}^T$. Since, by (3.4), $\Pi_{\mathbb{L}}^2 = \Pi_{\mathbb{L}}$, and $\Pi_{\mathbb{L}}\Pi_{\mathbb{H}} = \Pi_{\mathbb{H}}$ because $\mathbb{H} \subseteq \mathbb{L}$, we obtain

$$\mathbf{E}[\widehat{\boldsymbol{\varepsilon}}(\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y})^T] = \sigma^2(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}} - \Pi_{\mathbb{L}}^2 + \Pi_{\mathbb{L}}\Pi_{\mathbb{H}}) = \sigma^2(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}} - \Pi_{\mathbb{L}} + \Pi_{\mathbb{H}}) = \mathbf{0}.$$

This means that $\widehat{\boldsymbol{\varepsilon}}$ and $\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}$ are uncorrelated. Since their joint distribution is a multivariate normal distribution, by Lemma 2.2, they must be independent, which is part a).

Let $\mathbf{u}_1, \dots, \mathbf{u}_n \in \mathbb{R}^n$ be an orthonormal base of $\mathbb{R}^n$ such that $\mathbf{u}_1, \dots, \mathbf{u}_r$ generate the subspace $\mathbb{H}$ and $\mathbf{u}_1, \dots, \mathbf{u}_{k+1}$ generate the subspace $\mathbb{L}$. Let $\mathbf{O}$ be the matrix with the column vectors $\mathbf{u}_1, \dots, \mathbf{u}_n$. Since $\mathbf{O}\boldsymbol{\varepsilon}$ has the same distribution as $\boldsymbol{\varepsilon}$ and multiplication by $\mathbf{O}^T$ does not change the norm of a vector, we obtain

$$\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2 = \|(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\boldsymbol{\varepsilon}\|^2 \stackrel{d}{=} \|(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\mathbf{O}\boldsymbol{\varepsilon}\|^2 = \|\mathbf{O}^T(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\mathbf{O}\boldsymbol{\varepsilon}\|^2.$$

Because of

$$\Pi_{\mathbb{L}}\mathbf{u}_i = \begin{cases} \mathbf{u}_i, & i \in \{1, \dots, k+1\}, \\ \mathbf{0}, & i \in \{k+2, \dots, n\}, \end{cases} \quad \text{and} \quad \Pi_{\mathbb{H}}\mathbf{u}_i = \begin{cases} \mathbf{u}_i, & i \in \{1, \dots, r\}, \\ \mathbf{0}, & i \in \{r+1, \dots, n\}, \end{cases}$$

$\mathbf{O}^T(\Pi_{\mathbb{L}} - \Pi_{\mathbb{H}})\mathbf{O}$ is a diagonal matrix whose diagonal elements are zero except of the $(r+1)$-st to $(k+1)$-st diagonal elements which are one. This yields

$$\frac{1}{\sigma^2}\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2 \stackrel{d}{=} \frac{1}{\sigma^2}\sum_{i=r+1}^{k+1} \varepsilon_i^2 \sim \chi^2_{k+1-r}$$

and shows part b).

The random variables $\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2/\sigma^2$ and $\|\widehat{\boldsymbol{\varepsilon}}\|^2/\sigma^2$ are independent by part a) and they follow chi-square distributions with $k+1-r$ and $n-k-1$ degrees of freedom by Theorem 3.11 b) and part b). Thus,

$$\frac{(n-k-1)\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2}{(k+1-r)\|\widehat{\boldsymbol{\varepsilon}}\|^2} = \frac{\frac{\|\widehat{\mathbf{Y}} - \Pi_{\mathbb{H}}\mathbf{Y}\|^2}{(k+1-r)\sigma^2}}{\frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{(n-k-1)\sigma^2}}$$

is F-distributed with $k+1-r$ and $n-k-1$ degrees of freedom (see Section 2.1 for the definition of the F-distribution). □

Remark 3.14. One can also study the situation of a linear model as in (3.1) without the intercept γ_0. The intercept leads to ones in the first column of the design matrix $\mathbf{A}$. Since the proofs do not make use of this fact, all results continue to hold for linear regression without intercept if one replaces the number of parameters $k+1$ by k.

For $J \subseteq \{0, 1, \dots, k\}$ with $J \neq \varnothing$ one can consider the smaller linear model

$$Y = \mathbf{1}\{0 \in J\}\gamma_0 + \sum_{j=1}^{k} \mathbf{1}\{j \in J\}\gamma_j x^{(j)} + \varepsilon.$$

As discussed in Remark 3.14 our previous results still hold for the smaller linear model if it does not have an intercept. By $\mathbf{A}_J$ we denote the matrix that contains the columns of

the original design matrix $\mathbf{A}$ such that $j \in J$ for the corresponding γ_j. Then, the least squares estimator for $\boldsymbol{\gamma}_J = (\gamma_j)_{j\in J}$ is given by

$$\widehat{\boldsymbol{\gamma}}_J = (\mathbf{A}_J^T\mathbf{A}_J)^{-1}\mathbf{A}_J^T\mathbf{Y}.$$

Let $\mathbb{L}_J$ be the linear subspace spanned by the columns of $\mathbf{A}_J$. We also define

$$\widehat{\mathbf{Y}}_J = \mathbf{A}_J\widehat{\boldsymbol{\gamma}}_J = \mathbf{A}_J(\mathbf{A}_J^T\mathbf{A}_J)^{-1}\mathbf{A}_J^T\mathbf{Y} = \Pi_{\mathbb{L}_J}\mathbf{Y}$$

and

$$\widehat{\boldsymbol{\varepsilon}}_J = \mathbf{Y} - \widehat{\mathbf{Y}}_J = (\mathbf{I}_n - \Pi_{\mathbb{L}_J})\mathbf{Y},$$

where we used Lemma 3.6 a). The following result allows us to test the null hypothesis that some of the intercept and coefficients are zero, which means that one has in fact a smaller linear model.

Corollary 3.15. Consider a Gaussian linear model with $n > k+1$ and let $J \subseteq \{0, 1, \dots, k\}$ with $J \neq \varnothing$ and $\alpha \in (0, 1)$. For

$$H_0 : \gamma_j = 0 \quad \text{for all} \quad j \in \{0, 1, \dots, k\} \setminus J$$

against

$$H_1 : \gamma_j \neq 0 \quad \text{for some} \quad j \in \{0, 1, \dots, k\} \setminus J,$$

the test

$$\varphi(\mathbf{Y}) = \begin{cases} \text{cannot reject}, & \frac{(n-k-1)(\|\widehat{\boldsymbol{\varepsilon}}_J\|^2 - \|\widehat{\boldsymbol{\varepsilon}}\|^2)}{(k+1-|J|)\|\widehat{\boldsymbol{\varepsilon}}\|^2} < \mathcal{F}_{k+1-|J|,n-k-1,1-\alpha}, \\ \text{reject}, & \frac{(n-k-1)(\|\widehat{\boldsymbol{\varepsilon}}_J\|^2 - \|\widehat{\boldsymbol{\varepsilon}}\|^2)}{(k+1-|J|)\|\widehat{\boldsymbol{\varepsilon}}\|^2} \geq \mathcal{F}_{k+1-|J|,n-k-1,1-\alpha}, \end{cases}$$

has level α.

Proof. We have

$$\mathbf{Y} - \widehat{\mathbf{Y}}_J = \mathbf{Y} - \widehat{\mathbf{Y}} + \widehat{\mathbf{Y}} - \widehat{\mathbf{Y}}_J.$$

Since $\mathbf{Y} - \widehat{\mathbf{Y}} \perp \mathbb{L}$ and, because of $\widehat{\mathbf{Y}} \in \mathbb{L}$ and $\widehat{\mathbf{Y}}_J \in \mathbb{L}_J \subseteq \mathbb{L}$, $\widehat{\mathbf{Y}} - \widehat{\mathbf{Y}}_J \in \mathbb{L}$, the Pythagorean theorem yields

$$\|\mathbf{Y} - \widehat{\mathbf{Y}}_J\|^2 = \|\mathbf{Y} - \widehat{\mathbf{Y}}\|^2 + \|\widehat{\mathbf{Y}} - \widehat{\mathbf{Y}}_J\|^2.$$

Thus, we obtain

$$\|\widehat{\boldsymbol{\varepsilon}}_J\|^2 - \|\widehat{\boldsymbol{\varepsilon}}\|^2 = \|\mathbf{Y} - \widehat{\mathbf{Y}}_J\|^2 - \|\mathbf{Y} - \widehat{\mathbf{Y}}\|^2 = \|\widehat{\mathbf{Y}} - \widehat{\mathbf{Y}}_J\|^2.$$

Under H_0 we have $\mathbf{A}\boldsymbol{\gamma} = \mathbf{A}_J\boldsymbol{\gamma}_J \in \mathbb{L}_J$. Thus, from Theorem 3.13 with $\mathbb{H} = \mathbb{L}_J$ and $r = \dim \mathbb{L}_J = |J|$ it follows that

$$\frac{(n-k-1)(\|\widehat{\boldsymbol{\varepsilon}}_J\|^2 - \|\widehat{\boldsymbol{\varepsilon}}\|^2)}{(k+1-|J|)\|\widehat{\boldsymbol{\varepsilon}}\|^2} \sim \mathcal{F}_{k+1-|J|,n-k-1}$$

so that

$$\begin{aligned} \mathbf{P}(\varphi \text{ rejects } H_0) &= \mathbf{P}(F \geq \mathcal{F}_{k+1-|J|,n-k-1,1-\alpha}) \\ &= 1 - \mathbf{P}(F \leq \mathcal{F}_{k+1-|J|,n-k-1,1-\alpha}) = 1 - (1-\alpha) = \alpha \end{aligned}$$

with $F \sim \mathcal{F}_{k+1-|J|,n-k-1}$. This shows that φ is a level-α test. $\square$

The mean $\overline{\mathbf{Y}}$ of $\mathbf{Y}$ is given by

$$\overline{\mathbf{Y}} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

We call

$$\mathrm{TSS} = \sum_{i=1}^{n}(Y_i - \overline{\mathbf{Y}})^2$$

the *total sum of squares*. The *coefficient of determination* R^2 is defined as

$$\mathrm{R}^2 = 1 - \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{\sum_{i=1}^{n}(Y_i - \overline{\mathbf{Y}})^2}. \tag{3.14}$$

The total sum of squares is a measure of the variation within $\mathbf{Y}$, while $\|\widehat{\boldsymbol{\varepsilon}}\|^2$ measures the variation after the regression. Thus, we can think of R^2 as the ratio of the fluctuations within $\mathbf{Y}$ that are not explained by the linear model.

Corollary 3.16. Consider a Gaussian linear model with $n > k+1$ and let $\alpha \in (0,1)$. For

$$H_0 : \gamma_1 = \ldots = \gamma_k = 0 \quad \text{against} \quad H_1 : \gamma_j \neq 0 \quad \text{for some} \quad j \in \{1, \ldots, k\},$$

the test

$$\varphi(\mathbf{Y}) = \begin{cases} \text{cannot reject}, & \frac{(n-k-1)(\mathrm{TSS}-\|\widehat{\boldsymbol{\varepsilon}}\|^2)}{k\|\widehat{\boldsymbol{\varepsilon}}\|^2} < \mathcal{F}_{k,n-k-1,1-\alpha}, \\ \text{reject}, & \frac{(n-k-1)(\mathrm{TSS}-\|\widehat{\boldsymbol{\varepsilon}}\|^2)}{k\|\widehat{\boldsymbol{\varepsilon}}\|^2} \geq \mathcal{F}_{k,n-k-1,1-\alpha}, \end{cases}$$

has level α.

Proof. For $J = \{0\}$ it follows from Example 3.9 that $\|\widehat{\boldsymbol{\varepsilon}}\|^2 = \mathrm{TSS}$. Thus, the statement follows from Corollary 3.15 with $J = \{0\}$. □

Example 3.17. For the tooth growth of guinea pigs we consider the linear model with interaction from Example 3.4. Using the `summary` command, we obtain the following output in `R`:

```
> summary(lm(len ~ supp+dose+supp*dose, ToothGrowth))

Call:
lm(formula = len ~ supp + dose + supp * dose, data = ToothGrowth)

Residuals:
Min 1Q Median 3Q Max
-8.2264 -2.8462 0.0504 2.2893 7.9386
```

```
Coefficients:
             Estimate  Std.  Error  t value  Pr(>|t|)
(Intercept)    11.550        1.581    7.304  1.09e-09 ***
suppVC         -8.255        2.236   -3.691  0.000507 ***
dose            7.811        1.195    6.534  2.03e-08 ***
suppVC:dose     3.904        1.691    2.309  0.024631 *
Residual standard error:  4.083 on 56 degrees of freedom
Multiple R-squared:  0.7296, Adjusted R-squared:  0.7151
F-statistic:  50.36 on 3 and 56 DF, p-value:  6.521e-16
```

The estimated linear model is

$$\mathbf{len} = 11.550 - 8.255 \cdot \mathbf{1}\{\mathbf{supp} = \mathrm{VC}\} + 7.811 \cdot \mathbf{dose} + 3.904 \cdot \mathbf{dose} \cdot \mathbf{1}\{\mathbf{supp} = \mathrm{VC}\} + \varepsilon$$

with the normally distributed random error ε with $\mathbf{E}[\varepsilon] = 0$.

The residual standard error tells us that $\widehat{\sigma} = 4.083$, where $\widehat{\sigma}$ is the root of $\widehat{\sigma}^2 = \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{n-k-1}$, and the 56 degrees of freedom are $n - k - 1 = 56$ since we have $n = 60$ observations and $k = 3$. Moreover, we have for the coefficient of determination defined in (3.14) that $\mathrm{R}^2 = 0.7296$.

In the last line of the output, the test from Corollary 3.16 is used to test the hypothesis that all coefficients apart from the intercept are zero. As the p-value is very small, we can reject this null hypothesis at all reasonable levels. The F-statistic and the given degrees of freedom are those from Corollary 3.16.

In the table the estimates are the least squares estimators, while the standard errors are the roots of the diagonal elements of $(\mathbf{A}^T\mathbf{A})^{-1}$. The t-value is then the ratio of the estimate and the standard error, which is the test statistic from (3.13) for the null hypothesis that the coefficient is zero. It is compared with a t-distribution and the last column is the probability that the absolute value of a t-distributed random variable is at least as large. This is the p-value for the null hypothesis that the coefficient is zero.

If we test at level 0.05 (without correction for multiple testing), we can reject for the intercept and all coefficients the null hypotheses that they are zero. In this, case we can also conclude that they have the same sign as the estimates. Thus, we have shown that the intercept and the coefficients of **dose** and $\mathbf{dose} \cdot \mathbf{1}\{\mathbf{supp} = \mathrm{VC}\}$ are positive, while the coefficient of $\mathbf{1}\{\mathbf{supp} = \mathrm{VC}\}$ is negative. This leads to the interpretation that for both supplement types the cell length is predicted by a linear function of the dose. For orange juice (OJ) as supplement type the intercept is larger while the slope is smaller than for Ascorbic Acid (VC).

Example 3.18. We consider the job satisfaction score from Example 3.3. The underlying data are included as data set `jobsatisfaction` in the `R` package `datarium`. We obtain the following estimates:

```
> LM_jobsatisfaction<-lm(score ~ gender+education_level,jobsatisfaction)
> LM_jobsatisfaction
```

```
Call:
lm(formula = score ~ gender + education_level, data = jobsatisfaction)

Coefficients:
(Intercept)  genderfemale  education_levelcollege  education_leveluniversity
     5.6578       -0.1248                  0.7574                     3.2536
```

Thus, the estimated model is

$$\begin{aligned}\textbf{score} &= 5.6578 - 0.1248 \cdot \mathbf{1}\{\textbf{gender} = \text{female}\} + 0.7574 \cdot \mathbf{1}\{\textbf{education_level} = \text{college}\} \\ &\quad + 3.2536 \cdot \mathbf{1}\{\textbf{education_level} = \text{university}\} + \varepsilon\end{aligned}$$

with a random variable ε. Now our aim is to check if **gender** and **education_level** are relevant for the job satisfaction score. In case of **gender** we only have to check that the coefficient in front of $\mathbf{1}\{\textbf{gender} = \text{female}\}$ is non-zero, for which we can employ the t-test from Corollary 3.12. For **education_level** we have to test the null hypothesis that the coefficients for $\mathbf{1}\{\textbf{education_level} = \text{college}\}$ and $\mathbf{1}\{\textbf{education_level} = \text{university}\}$ are both zero, for which we can apply Corollary 3.15.

To obtain the results in `R`, we can use the command `Anova` from the package `car`.

```
> Anova(LM_jobsatisfaction, type="III")
Anova Table (Type III tests)

Response:  score
                Sum Sq  Df  F value Pr(>F)
(Intercept)     446.01   1       1194.0155  <2e-16  ***
gender            0.23   1          0.6031  0.4408
education_level 113.68   2        152.1718  <2e-16  ***
Residuals        20.17  54
---
Signif.  codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’  0.1 ‘ ’ 1
```

In the following let the test level be 0.01. We cannot reject the null hypothesis that the coefficient of $\mathbf{1}\{\textbf{gender} = \text{female}\}$ is zero, while we can reject the null hypothesis that the coefficients of $\mathbf{1}\{\textbf{education_level} = \text{college}\}$ and $\mathbf{1}\{\textbf{education_level} = \text{university}\}$ are both zero. This means that we can establish that the level of education is relevant for the job satisfaction, but we cannot show this for the gender.

Remark 3.19. So far we assumed that $x_i^{(1)}, \dots, x_i^{(k)}$ are deterministic for $i \in \{1, \dots, n\}$. However, all our results continue to hold if they are random as long as they are independent from $\varepsilon_1, \dots, \varepsilon_n$.

3.4 Analysis of variance

We consider the situation that a quantity Y is measured for m different groups $1, \ldots, m$. Our goal is to study if there are differences between the groups. We assume that there exist $\mu_1, \ldots, \mu_m \in \mathbb{R}$ and $\sigma \in (0, \infty)$ such that

$$Y = \mu_j + \varepsilon \tag{3.15}$$

with $\varepsilon \sim \text{Normal}(0, \sigma^2)$ for group $j \in \{1, \ldots, m\}$.

Example 3.20. In Example 3.3, the dependent variable **score**, which describes the job satisfaction, is explained by the factors **gender** with the levels male and female and **education_level** with the levels school, college and university. Here, it is reasonable to believe that there is an interaction between **gender** and **education_level**. Thus, we should distinguish all six different combinations of gender and education and assume that the job satisfaction of a person with gender $\mathrm{g} \in \{\text{male}, \text{female}\}$ and education $\mathrm{e} \in \{\text{school}, \text{college}, \text{university}\}$ is given by

$$\textbf{score} = \mu_{\mathrm{e,g}} + \varepsilon$$

with $\varepsilon \sim \text{Normal}(0, \sigma^2)$ and $\mu_{\text{school,male}}, \mu_{\text{college,male}}, \mu_{\text{university,male}}, \mu_{\text{school,female}}, \mu_{\text{college,female}}, \mu_{\text{university,female}} \in \mathbb{R}$ and $\sigma \in (0, \infty)$.

For group $j \in \{1, \ldots, m\}$ let the observations $Y_j^{(1)}, \ldots, Y_j^{(n_j)}$ be given. We assume that

$$Y_j^{(i)} = \mu_j + \varepsilon_j^{(i)}$$

for $i \in \{1, \ldots, n_j\}$ and $j \in \{1, \ldots, m\}$ with independent $\text{Normal}(0, \sigma^2)$-distributed random variables $(\varepsilon_j^{(i)})_{i \in \{1,\ldots,n_j\}, j \in \{1,\ldots,m\}}$. In the following, we use the abbreviation $\mathbf{Y}_j = (Y_j^{(1)}, \ldots, Y_j^{(n_j)})^T$ for $j \in \{1, \ldots, m\}$ and denote by $n = \sum_{j=1}^m n_j$ the total sample size. Apart from $n_j \in \mathbb{N}$ for $j \in \{1, \ldots, m\}$ we assume that $n > m$, i.e., we exclude $n_1 = \ldots = n_m = 1$. The problem to study the unknown $\mu_1, \ldots, \mu_m \in \mathbb{R}$ and $\sigma \in (0, \infty)$ for given $\mathbf{Y}_1, \ldots, \mathbf{Y}_m$ is called *analysis of variance*.

Using dummy variables, we can rephrase (3.15) as the Gaussian linear model

$$Y = \gamma_0 + \gamma_1 \mathbf{1}\{\text{group } 2\} + \ldots + \gamma_{m-1} \mathbf{1}\{\text{group } m\} + \varepsilon = \gamma_0 + \sum_{i=1}^{m-1} \gamma_i \mathbf{1}\{\text{group } i+1\} + \varepsilon \tag{3.16}$$

with $\gamma_0, \gamma_1, \ldots, \gamma_{m-1} \in \mathbb{R}$, where $\mathbf{1}\{\text{group } i\} = 1$ if and only if Y is from group i. This means that we have the relations

$$\gamma_0 = \mu_1 \quad \text{and} \quad \gamma_j = \mu_{j+1} - \mu_1 \quad \text{for} \quad j \in \{1, \ldots, m-1\} \tag{3.17}$$

between the parameters of (3.15) and (3.16). The underlying data for the linear model in (3.16) are

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_1 \\ \vdots \\ \mathbf{Y}_m \end{pmatrix} \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots \\ 1 & 0 & 0 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & 1 \end{pmatrix}.$$

Here, the first column of the design matrix $\mathbf{A}$ consists of ones, while there is a one in the j-th column for $j \in \{2, \dots, m\}$ if and only if the observation is from the j-th group. We obtain

$$\mathbf{A}^T\mathbf{A} = \begin{pmatrix} n & n_2 & n_3 & \dots & n_m \\ n_2 & n_2 & 0 & \dots & 0 \\ n_3 & 0 & n_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n_m & 0 & 0 & \dots & n_m \end{pmatrix} \quad \text{and} \quad \mathbf{A}^T\mathbf{Y} = \begin{pmatrix} \sum_{j=1}^m \sum_{i=1}^{n_j} Y_j^{(i)} \\ \sum_{i=1}^{n_2} Y_2^{(i)} \\ \vdots \\ \sum_{i=1}^{n_m} Y_m^{(i)} \end{pmatrix} = \begin{pmatrix} n\overline{\mathbf{Y}} \\ n_2\overline{\mathbf{Y}}_2 \\ \vdots \\ n_m\overline{\mathbf{Y}}_m \end{pmatrix},$$

where $\overline{\mathbf{Y}}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} Y_j^{(i)}$ is the sample mean of $\mathbf{Y}_j$ and $\overline{\mathbf{Y}}$ is the sample mean of $\mathbf{Y}$. Thus, the system $\mathbf{A}^T\mathbf{A}\widehat{\boldsymbol{\gamma}} = \mathbf{A}^T\mathbf{Y}$ is solved by $\widehat{\boldsymbol{\gamma}} = (\overline{\mathbf{Y}}_1, \overline{\mathbf{Y}}_2 - \overline{\mathbf{Y}}_1, \dots, \overline{\mathbf{Y}}_m - \overline{\mathbf{Y}}_1)^T$ since

$$n\overline{\mathbf{Y}}_1 + \sum_{j=2}^m n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}}_1) = \left(n - \sum_{j=2}^m n_j\right)\overline{\mathbf{Y}}_1 + \sum_{j=2}^m n_j\overline{\mathbf{Y}}_j = \sum_{j=1}^m n_j\overline{\mathbf{Y}}_j = \sum_{j=1}^m\sum_{i=1}^{n_j} Y_j^{(i)} = n\overline{\mathbf{Y}}$$

and

$$n_j\overline{\mathbf{Y}}_1 + n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}}_1) = n_j\overline{\mathbf{Y}}_j$$

for $j \in \{2, \dots, m\}$, which are the first and the remaining lines for the equation system. Together with (3.17) this provides the estimators

$$\widehat{\mu}_j = \overline{\mathbf{Y}}_j$$

for μ_j with $j \in \{1, \dots, m\}$. This shows that the least squares estimators are the sample means, which are the natural choice for the estimation of the group means. Since the prediction for $Y_j^{(i)}$ is $\widehat{\mu}_j = \overline{\mathbf{Y}}_j$, the residuals $\widehat{\boldsymbol{\varepsilon}}$ satisfy

$$\|\widehat{\boldsymbol{\varepsilon}}\|^2 = \sum_{j=1}^m\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2. \tag{3.18}$$

Thus, σ^2 can be estimated by

$$\widehat{\sigma}^2 = \frac{1}{n-(m-1)-1}\|\widehat{\boldsymbol{\varepsilon}}\|^2 = \frac{1}{n-m}\sum_{j=1}^{m}\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2. \tag{3.19}$$

Next we test if the group means are different.

Corollary 3.21. Consider analysis of variance and let $\alpha \in (0,1)$. For

$$H_0 : \mu_1 = \ldots = \mu_m \quad \text{against} \quad H_1 : \mu_u \neq \mu_v \quad \text{for some} \quad u, v \in \{1, \ldots, m\},$$

the test

$$\varphi(\mathbf{Y}_1, \ldots, \mathbf{Y}_m) = \begin{cases} \text{cannot reject}, & \frac{(n-m)\sum_{j=1}^m n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2}{(m-1)\sum_{j=1}^m \sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2} < \mathcal{F}_{m-1,n-m,1-\alpha}, \\ \text{reject}, & \frac{(n-m)\sum_{j=1}^m n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2}{(m-1)\sum_{j=1}^m \sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2} \geq \mathcal{F}_{m-1,n-m,1-\alpha}, \end{cases}$$

has level α.

Proof. By (3.17) we can rewrite the test problem as

$$H_0 : \gamma_1 = \ldots = \gamma_{m-1} = 0 \quad \text{against} \quad H_1 : \gamma_u \neq 0 \quad \text{for some} \quad u \in \{1, \ldots, m-1\},$$

which is exactly the situation considered in Corollary 3.16 with $k = m-1$. For the total sum of squares we have

$$\begin{aligned} \text{TSS} &= \sum_{j=1}^{m}\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}})^2 = \sum_{j=1}^{m}\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j + \overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2 \\ &= \sum_{j=1}^{m}\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2 + (\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2 + 2(Y_j^{(i)} - \overline{\mathbf{Y}}_j)(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}}) \\ &= \sum_{j=1}^{m}\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2 + \sum_{j=1}^{m} n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2 + 2\sum_{j=1}^{m}(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j). \end{aligned}$$

Because of

$$\sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j) = \sum_{i=1}^{n_j} Y_j^{(i)} - n_j\overline{\mathbf{Y}}_j = n_j\overline{\mathbf{Y}}_j - n_j\overline{\mathbf{Y}}_j = 0$$

for $j \in \{1, \ldots, m\}$, the last term on the right-hand side vanishes. Together with (3.18), we obtain

$$\text{TSS} - \|\widehat{\boldsymbol{\varepsilon}}\|^2 = \sum_{j=1}^{m} n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2.$$

Thus, we see that

$$\frac{(n-(m-1)-1)(\text{TSS} - \|\widehat{\boldsymbol{\varepsilon}}\|^2)}{(m-1)\|\widehat{\boldsymbol{\varepsilon}}\|^2} = \frac{(n-m)\sum_{j=1}^m n_j(\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}})^2}{(m-1)\sum_{j=1}^m \sum_{i=1}^{n_j}(Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2}$$

so that the statement follows from Corollary 3.16. □

If one knows that not all group means are identical, one is usually interested in deciding which of them are different. In order to test

$$H_0 : \mu_u = \mu_v \quad \text{against} \quad H_1 : \mu_u \neq \mu_v$$

for fixed $u, v \in \{1, \ldots, m\}$ with $u < v$, one can study by (3.17)

$$H_0 : \gamma_{u-1} = \gamma_{v-1} \quad \text{against} \quad H_1 : \gamma_{u-1} \neq \gamma_{v-1}$$

if $u \geq 2$ and

$$H_0 : \gamma_{v-1} = 0 \quad \text{against} \quad H_1 : \gamma_{v-1} \neq 0$$

if $u = 1$. Both test problems can be reformulated as

$$H_0 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle = 0 \quad \text{against} \quad H_1 : \langle \mathbf{a}, \boldsymbol{\gamma} \rangle \neq 0$$

with suitable vectors $\mathbf{a} \in \mathbb{R}^m$ so that we can apply the t-test from Corollary 3.12.

However, as we are not only interested in the comparison of a fixed pair of means but in all pairs, we would have to use $m(m-1)/2$ t-tests. Here we run into the problem of *multiple testing*. When applying a single test at level α, this means that we ensure that the probability of a type I error is at most α. When doing multiple tests at level α, the probabilities of type I errors accumulate so that the overall probability of falsely rejecting at least one of the null hypotheses can become much larger than α. In order to ensure that the overall probability of a type I error is at most α, one has to adjust the levels of the individual tests. The most straightforward approach is the so-called *Bonferroni correction* where one uses for each of ℓ tests the level $\frac{\alpha}{\ell}$. For the pairwise comparison of means the Bonferroni correction leads to the level $\frac{2\alpha}{m(m-1)}$. Unfortunately, this level is rather small even for small choices of m.

Instead of pairwise t-tests with Bonferroni correction, we will use the so-called Tukey or Tukey-Kramer tests, for which we have to introduce a new distribution.

Definition 3.22. For $k, \ell \in \mathbb{N}$ and independent random variables $N_1, \ldots, N_k \sim \text{Normal}(0, 1)$ and $Z \sim \chi^2_\ell$ let

$$Y = \frac{\max\{N_1, \ldots, N_k\} - \min\{N_1, \ldots, N_k\}}{\sqrt{Z/\ell}}.$$

We denote the distribution of Y as *studentised range distribution* with the parameters k and ℓ and write $Y \sim \mathcal{Q}_{k,\ell}$.

For $p \in (0, 1)$ and $k, \ell \in \mathbb{N}$ let $\mathcal{Q}_{k,\ell,p}$ be the p-quantile of the $\mathcal{Q}_{k,\ell}$-distribution.

Theorem 3.23. Consider analysis of variance, assume that $n_1 = \ldots = n_m$ and let $\alpha \in (0, 1)$. Then,

$$\mathbf{P}\big(\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D^{(\alpha)} \leq \mu_u - \mu_v \leq \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D^{(\alpha)} \text{ for all } u, v \in \{1, \ldots, m\} \text{ with } u \neq v\big) = 1 - \alpha$$

with

$$D^{(\alpha)} = \mathcal{Q}_{m,n-m,1-\alpha} \sqrt{\frac{1}{n_1(n-m)} \sum_{j=1}^{m} \sum_{i=1}^{n_j} (Y_j^{(i)} - \overline{\mathbf{Y}}_j)^2}.$$

Proof. Note that $D^{(\alpha)} = \mathcal{Q}_{m,n-m,1-\alpha}\sqrt{\widehat{\sigma}^2/n_1}$ by (3.19). We can rewrite the probability of interest as

$$\begin{aligned}
&\mathbf{P}\big(\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D^{(\alpha)} \le \mu_u - \mu_v \le \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D^{(\alpha)} \text{ for all } u,v \in \{1,\dots,m\} \text{ with } u \neq v\big)\\
&= \mathbf{P}\big(\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D^{(\alpha)} \le \mu_u - \mu_v \le \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D^{(\alpha)} \text{ for all } u,v \in \{1,\dots,m\}\big)\\
&= \mathbf{P}\big(-D^{(\alpha)} \le (\overline{\mathbf{Y}}_u - \mu_u) - (\overline{\mathbf{Y}}_v - \mu_v) \le D^{(\alpha)} \text{ for all } u,v \in \{1,\dots,m\}\big)\\
&= \mathbf{P}\big(|\overline{\mathbf{Y}}_u - \mu_u - (\overline{\mathbf{Y}}_v - \mu_v)| \le D^{(\alpha)} \text{ for all } u,v \in \{1,\dots,m\}\big)\\
&= \mathbf{P}\bigg(\frac{\sqrt{n_1}|\overline{\mathbf{Y}}_u - \mu_u - (\overline{\mathbf{Y}}_v - \mu_v)|}{\widehat{\sigma}} \le \mathcal{Q}_{m,n-m,1-\alpha} \text{ for all } u,v \in \{1,\dots,m\}\bigg)\\
&= \mathbf{P}\bigg(\frac{\sqrt{n_1}(\max_{u\in\{1,\dots,m\}} \overline{\mathbf{Y}}_u - \mu_u - \min_{v\in\{1,\dots,m\}} \overline{\mathbf{Y}}_v - \mu_v)}{\widehat{\sigma}} \le \mathcal{Q}_{m,n-m,1-\alpha}\bigg).
\end{aligned} \tag{3.20}$$

Since, for $j \in \{1,\dots,m\}$,

$$\overline{\mathbf{Y}}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} Y_j^{(i)} = \frac{1}{n_1}\sum_{i=1}^{n_1} \mu_j + \varepsilon_j^{(i)} = \mu_j + \frac{1}{n_1}\sum_{i=1}^{n_1} \varepsilon_j^{(i)},$$

we obtain that $\overline{\mathbf{Y}}_j \sim \text{Normal}(\mu_j, \sigma^2/n_1)$ for $j \in \{1,\dots,m\}$. Moreover, we have that $\overline{\mathbf{Y}}_1,\dots,\overline{\mathbf{Y}}_m$ are independent. This implies that

$$\frac{\sqrt{n_1}}{\sigma}(\overline{\mathbf{Y}}_1 - \mu_1,\dots,\overline{\mathbf{Y}}_m - \mu_m) \stackrel{d}{=} (N_1,\dots,N_m)$$

with independent $N_1,\dots,N_m \sim \text{Normal}(0,1)$. Thus, we have

$$\frac{\sqrt{n_1}}{\sigma}\big(\max_{u\in\{1,\dots,m\}} \overline{\mathbf{Y}}_u - \mu_u - \min_{v\in\{1,\dots,m\}} \overline{\mathbf{Y}}_v - \mu_v\big) \stackrel{d}{=} \max\{N_1,\dots,N_m\} - \min\{N_1,\dots,N_m\}.$$

Since $\overline{\mathbf{Y}}_1,\dots,\overline{\mathbf{Y}}_m$ are a function of $\widehat{\boldsymbol{\gamma}}$, it follows from Theorem 3.11 that $\overline{\mathbf{Y}}_1,\dots,\overline{\mathbf{Y}}_m$ are independent of $(n-m)\widehat{\sigma}^2/\sigma^2 \sim \chi^2_{n-m}$. Let Z denote a χ^2_{n-m}-distributed random variable independent of $N_1,\dots,N_m$. The previous observations lead to

$$\begin{aligned}
&\frac{\sqrt{n_1}(\max_{u\in\{1,\dots,m\}} \overline{\mathbf{Y}}_u - \mu_u - \min_{v\in\{1,\dots,m\}} \overline{\mathbf{Y}}_v - \mu_v)}{\widehat{\sigma}}\\
&= \frac{\frac{\sqrt{n_1}}{\sigma}(\max_{u\in\{1,\dots,m\}} \overline{\mathbf{Y}}_u - \mu_u - \min_{v\in\{1,\dots,m\}} \overline{\mathbf{Y}}_v - \mu_v)}{\sqrt{\frac{(n-m)\widehat{\sigma}^2}{(n-m)\sigma^2}}}\\
&\stackrel{d}{=} \frac{\max\{N_1,\dots,N_m\} - \min\{N_1,\dots,N_m\}}{\sqrt{Z/(n-m)}} \sim \mathcal{Q}_{m,n-m}.
\end{aligned}$$

Combining this with the calculation in (3.20) shows

$$\begin{aligned}
&\mathbf{P}(\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D^{(\alpha)} \le \mu_u - \mu_v \le \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D^{(\alpha)} \text{ for all } u,v \in \{1,\dots,m\} \text{ with } u \neq v)\\
&= \mathbf{P}(Y \le \mathcal{Q}_{m,n-m,1-\alpha}) = 1-\alpha
\end{aligned}$$

with $Y \sim \mathcal{Q}_{m,n-m}$ and completes the proof. □

The situation of Theorem 3.23 is called the *balanced* case as it is assumed that one has the same number of observations for all groups. For the *unbalanced* case, where the numbers of observations per group can be different, one has the following counterpart to Theorem 3.23.

Theorem 3.24. Consider analysis of variance and let $\alpha \in (0,1)$. Then,

$$\mathbf{P}\big(\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D_{u,v}^{(\alpha)} \leq \mu_u - \mu_v \leq \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D_{u,v}^{(\alpha)} \text{ for all } u,v \in \{1,\ldots,m\} \text{ with } u \neq v\big) \geq 1-\alpha$$

with

$$D_{u,v}^{(\alpha)} = \mathcal{Q}_{m,n-m,1-\alpha}\sqrt{\left(\frac{1}{2n_u}+\frac{1}{2n_v}\right)\frac{1}{(n-m)}\sum_{j=1}^{m}\sum_{i=1}^{n_j}(Y_j^{(i)}-\overline{\mathbf{Y}}_j)^2}$$

for $u,v \in \{1,\ldots,m\}$ with $u \neq v$.

We do not provide a proof of this statement. It was conjectured independently by Tukey and Kramer in the 50's but it remained unproven for more than 30 years (see [1]).

Theorems 3.23 and 3.24 provide simultaneous confidence regions for all differences $(\mu_u - \mu_v)_{u,v\in\{1,\ldots,m\},u\neq v}$. Here, the confidence region is a $m(m-1)$-dimensional set and with probability at least $1-\alpha$, $(\mu_u - \mu_v)_{u,v\in\{1,\ldots,m\},u\neq v}$ is contained in this set. As the set is a Cartesian product of intervals, we know that with probability at least $1-\alpha$ all differences belong to corresponding intervals. This is a much stronger statement than having a univariate confidence interval for every difference of means.

We are interested in testing the hypotheses

$$H_0^{u,v} : \mu_u = \mu_v \quad \text{against} \quad H_1^{u,v} : \mu_u \neq \mu_v$$

for all $u,v \in \{1,\ldots,m\}$ with $u < v$ simultaneously. Let $\alpha \in (0,1)$ be given. In the *Tukey-Kramer test*, one rejects $H_0^{(u,v)}$ for all u and v such that

$$0 \notin \left[\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D_{u,v}^{(\alpha)}, \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D_{u,v}^{(\alpha)}\right].$$

If we falsely reject $H_0^{u,v}$, we have

$$\mu_u - \mu_v \notin \left[\overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v - D_{u,v}^{(\alpha)}, \overline{\mathbf{Y}}_u - \overline{\mathbf{Y}}_v + D_{u,v}^{(\alpha)}\right]. \tag{3.21}$$

Since the probability that this is the case for any choice of $u,v \in \{1,\ldots,m\}$ with $u < v$ is at most α by Theorem 3.24, the probability of falsely rejecting at least one null hypothesis is bounded by α and the test has level α.

In case that one rejects $H_0^{u,v}$, one can also conclude that μ_u and μ_v satisfy the same inequality as $\overline{\mathbf{Y}}_u$ and $\overline{\mathbf{Y}}_v$. This is only false if (3.21) holds for some $u,v \in \{1,\ldots,m\}$ with $u < v$, an event whose probability is at most α.

The *Tukey test* is the Tukey-Kramer test for the balanced case, where one has the same number of observations for all groups.

Example 3.25. In order to study the model from Example 3.20, we first compute the linear model:

```
> LM<-lm(score ~ education_level*gender, jobsatisfaction)
> LM

Call:
lm(formula = score ~ education_level * gender, data = jobsatisfaction)

Coefficients:
                        (Intercept)                  education_levelcollege
                            5.42667                                 0.79667
          education_leveluniversity                            genderfemale
                            3.86533                                 0.31433
education_levelcollege:genderfemale  education_leveluniversity:genderfemale
                           -0.07467                                -1.20033
```

By summing up the intercept and the corresponding coefficients we can derive estimates for all group means. For example, we get

$$\widehat{\mu}_{\text{college,female}} = 5.42667 + 0.79667 + 0.31433 - 0.07467 = 6.463.$$

The null hypothesis of equal group means is equivalent to all coefficients in the linear model being zero apart from the intercept. We can find the p-value in the following output:

```
> summary(LM)

Call:
lm(formula = score ~ education_level * gender, data = jobsatisfaction)

Residuals:
    Min       1Q   Median      3Q     Max
-1.8860  -0.2325  -0.0040  0.3297  1.1640

Coefficients:
                                      Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)                            5.42667     0.18334   29.599   < 2e-16  ***
education_levelcollege                 0.79667     0.25928    3.073   0.00337  **
education_leveluniversity              3.86533     0.25272   15.295   < 2e-16  ***
genderfemale                           0.31433     0.25272    1.244   0.21915
education_levelcollege:genderfemale   -0.07467     0.35740   -0.209   0.83533
education_leveluniversity:genderfemale -1.20033    0.35266   -3.404   0.00129  **
---
Signif.  codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’  0.1 ‘ ’ 1

Residual standard error:  0.55 on 52 degrees of freedom
Multiple R-squared:  0.8829, Adjusted R-squared:  0.8717
```

```
F-statistic:  78.45 on 5 and 52 DF, p-value:  < 2.2e-16
```

In the last line, we see the value of the test statistic and the p-value for the null hypothesis that all coefficients are zero. Since the p-value is so small, we can reject the null hypothesis of equal means at all usual levels.

For pairwise comparisons we get the simultaneous confidence intervals and the p-values of the Tukey-Kramer test by using the command `TukeyHSD` in `R`:

```
> TukeyHSD(aov(LM), "education_level:gender")
Tukey multiple comparisons of means
95% family-wise confidence level

Fit:  aov(formula = LM)

$`education_level:gender`
                                        diff           lwr         upr      p adj
college:male-school:male           0.7966667   0.029551460   1.5637819  0.0374890
university:male-school:male        3.8653333   3.117641918   4.6130247  0.0000000
school:female-school:male          0.3143333  -0.433358082   1.0620247  0.8132166
college:female-school:male         1.0363333   0.288641918   1.7840247  0.0019203
university:female-school:male      2.9793333   2.231641918   3.7270247  0.0000000
university:male-college:male       3.0686667   2.320975251   3.8163581  0.0000000
school:female-college:male        -0.4823333  -1.230024749   0.2653581  0.4086560
college:female-college:male        0.2396667  -0.508024749   0.9873581  0.9317495
university:female-college:male     2.1826667   1.434975251   2.9303581  0.0000000
school:female-university:male     -3.5510000  -4.278749384  -2.8232506  0.0000000
college:female-university:male    -2.8290000  -3.556749384  -2.1012506  0.0000000
university:female-university:male -0.8860000  -1.613749384  -0.1582506  0.0087499
college:female-school:female       0.7220000  -0.005749384   1.4497494  0.0529764
university:female-school:female    2.6650000   1.937250616   3.3927494  0.0000000
university:female-college:female   1.9430000   1.215250616   2.6707494  0.0000000
```

Thus, we can reject at level 0.01 all null hypotheses whose p-value is at most 0.01. Moreover, we obtain that the differences have the signs of the observed differences. So we can conclude that

$\mu_{\text{university,male}} > \mu_{\text{school,male}}$, $\mu_{\text{college,female}} > \mu_{\text{school,male}}$, $\mu_{\text{university,female}} > \mu_{\text{school,male}}$, $\mu_{\text{university,male}} > \mu_{\text{college,male}}$, $\mu_{\text{university,female}} > \mu_{\text{college,male}}$, $\mu_{\text{school,female}} < \mu_{\text{university,male}}$, $\mu_{\text{college,female}} < \mu_{\text{university,male}}$, $\mu_{\text{university,female}} < \mu_{\text{university,male}}$, $\mu_{\text{university,female}} > \mu_{\text{school,female}}$, $\mu_{\text{university,female}} > \mu_{\text{college,female}}$.

Note that the table above provides family-wise confidence intervals with combined error level 0.05. Thus, 0 is not between the lower and upper bounds whenever the adjusted p-value is below 0.05.

Chapter 4

Non-linear regression

4.1 Non-linear regression and least squares estimation

As in the case of linear regression we are interested in explaining a dependent variable Y by independent variables $x^{(1)}, \dots, x^{(k)}$. Again we assume that Y is a deterministic function of $x^{(1)}, \dots, x^{(k)}$ plus some random noise, but we allow the deterministic function to be non-linear in $x^{(1)}, \dots, x^{(k)}$. To this end let $(g(\cdot;\beta))_{\beta \in B}$ be a family of functions such that $B \subseteq \mathbb{R}^m$ and $\mathbb{R}^k \ni \mathbf{x} \mapsto g(\mathbf{x};\beta)$ for any $\beta \in B$, where one can think of β as parameters. We assume that

$$Y = g(x^{(1)}, \dots, x^{(k)}; \beta) + \varepsilon \tag{4.1}$$

for some $\beta \in B$ and a random variable ε. Here we require ε to have a finite second moment, i.e., $\mathbf{E}[\varepsilon^2] < \infty$, and to satisfy $\mathbf{E}[\varepsilon] = 0$. The latter is no loss of generality because, otherwise, one could subtract $\mathbf{E}[\varepsilon]$ in the error term and incorporate it into the function g.

Example 4.1. In biochemistry, the Michaelis–Menten equation provides the relation

$$v = \frac{v_{\max} a}{K + a}$$

between the reaction rate v and the concentration a with some constants $v_{\max}, K \in (0, \infty)$. If the reaction rate v is measured in an experiment for given concentration a, it is reasonable to assume that

$$v = \frac{v_{\max} a}{K + a} + \varepsilon,$$

where ε is a random measurement error. This corresponds to the situation of (4.1) with

$$g(x;\beta) = \frac{\beta_1 x}{\beta_2 + x}$$

for $\beta = (\beta_1, \beta_2)^T \in (0, \infty)^2$.

In the following let observations $(Y_1, \mathbf{x}_1), \dots, (Y_n, \mathbf{x}_n) \in \mathbb{R} \times \mathbb{R}^k$ of $(Y, x^{(1)}, \dots, x^{(k)})$ be given. More precisely, we assume that

$$Y_i = g(\mathbf{x}_i; \beta) + \varepsilon_i \tag{4.2}$$

for $i \in \{1, \dots, n\}$ with some unknown $\beta \in B$ and uncorrelated random variables $\varepsilon_1, \dots, \varepsilon_n$ satisfying $\mathbf{E}[\varepsilon_i] = 0$ for $i \in \{1, \dots, n\}$.

Our goal is to estimate the unknown parameter β. We say that an estimator $\widehat{\beta}$ is a least squares estimator for β if it is a solution of the optimisation problem

$$\min_{\widetilde{\beta} \in B} \sum_{i=1}^{n} \big(Y_i - g(\mathbf{x}_i; \widetilde{\beta})\big)^2. \tag{4.3}$$

This means that the parameter $\widetilde{\beta}$ is chosen such that the $(g(\mathbf{x}_i; \widetilde{\beta}))_{i \in \{1,\dots,n\}}$ fit best to the observed values $(Y_i)_{i \in \{1,\dots,n\}}$.

From now on we assume that $g(\mathbf{x}; \widetilde{\beta})$ is differentiable with respect to $\widetilde{\beta}$ for all $\mathbf{x} \in \mathbb{R}^k$ and that B is open, which excludes extrema on the boundary of B. Then the objective function

$$S(\widetilde{\beta}) = \sum_{i=1}^{n} \big(Y_i - g(\mathbf{x}_i; \widetilde{\beta})\big)^2$$

is differentiable with respect to $\widetilde{\beta}$ and has the gradient

$$\nabla_{\widetilde{\beta}} S(\widetilde{\beta}) = -2 \sum_{i=1}^{n} \big(Y_i - g(\mathbf{x}_i; \widetilde{\beta})\big) \nabla_{\widetilde{\beta}} g(\mathbf{x}_i; \widetilde{\beta}).$$

If a least squares estimator $\widehat{\beta}$ exists, it must satisfy

$$-2 \sum_{i=1}^{n} \big(Y_i - g(\mathbf{x}_i; \widehat{\beta})\big) \nabla_{\widetilde{\beta}} g(\mathbf{x}_i; \widehat{\beta}) = 0. \tag{4.4}$$

Solving this equation and verifying that the solution is a global minimum of (4.3) allows us to estimate the unknown β.

In general, (4.4) does not have a closed form solution. Instead, one has to use numerical methods to solve it.

If we assume that the random variables $\varepsilon_1, \dots, \varepsilon_n$ in (4.2) all have the same variance $\sigma^2 \in [0, \infty)$, that $n > m$ and that we have a least squares estimator $\widehat{\beta}$, we can estimate σ^2 by

$$\widehat{\sigma}^2 = \frac{1}{n-m} \sum_{i=1}^{n} (Y_i - g(\mathbf{x}_i; \widehat{\beta}))^2.$$

Example 4.2. For the situation of Example 4.1 let observations $(v_1, a_1), \dots, (v_n, a_n)$ be given. Note that

$$\frac{\partial v}{\partial v_{\max}} = \frac{a}{K+a} \quad \text{and} \quad \frac{\partial v}{\partial K} = -\frac{v_{\max} a}{(K+a)^2}.$$

Thus, a least squares estimator $(\widehat{v}_{\max}, \widehat{K})$ for $(v_{\max}, K)$ has to satisfy

$$\sum_{i=1}^{n} \left(v_i - \frac{\widehat{v}_{\max} a_i}{\widehat{K} + a_i} \right) \begin{pmatrix} \frac{a_i}{\widehat{K}+a_i} \\ -\frac{\widehat{v}_{\max} a_i}{(\widehat{K}+a_i)^2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The data set `Puromycin` in `R` contains observations for the Michaelis-Menten model. In the following, we use the first 12 observations since the other measurements were for a different cell type. The command to compute a non-linear regression model in `R` is `nls`, whose arguments are the formula for the non-linear model, the data set and the start values used in the optimisation for the the unknown parameters. Note that one can obtain an error message if the start values are chosen badly.

```
> NLS_model<-nls(rate ~ vmax*conc/(K+conc),Puromycin[1:12,],
start=list(vmax=10,K=1))
> NLS_model
Nonlinear regression model
model:  rate ~ vmax * conc/(K + conc)
data:  Puromycin[1:12, ]
      vmax        K
 212.68377  0.06412
residual sum-of-squares:  1195

Number of iterations to convergence:  10
Achieved convergence tolerance:  1.671e-06
```

This provides the model

$$v = \frac{212.68377a}{0.06412 + a} + \varepsilon$$

with a random error ε for the relation between the reaction rate v and the concentration a.

The command `predict` can be used for predictions for new values of the independent variables:

```
> predict(NLS_model, data.frame(conc=c(0.4)))
[1] 183.3002
```

Without the second argument `predict` returns the fitted values for the original data of the non-linear regression model.

4.2 Non-linear regression and maximum likelihood estimation

In this section, we assume that the random errors $\varepsilon_1, \ldots, \varepsilon_n$ in (4.2) are independent normal random variables with expectation zero and the same standard deviation $\sigma \in (0, \infty)$.

Under this assumption it is possible to compute maximum likelihood estimators for β and σ^2.

Theorem 4.3. Assume that $\varepsilon_1, \dots, \varepsilon_n$ in the non-linear regression model are i.i.d. with $\varepsilon_1 \sim \text{Normal}(0, \sigma^2)$ for some $\sigma \in (0, \infty)$ and that $n > m$. Let $\widehat{\beta}$ be a least squares estimator for β and define $\widehat{\sigma}^2_{ML} = \frac{n-m}{n}\widehat{\sigma}^2$. Then, $(\widehat{\beta}, \widehat{\sigma}^2_{ML})$ is a maximum likelihood estimator for (β, σ^2).

Proof. Since $Y_i \sim \text{Normal}(g(\mathbf{x}_i; \beta), \sigma^2)$ for $i \in \{1, \dots, n\}$ and $Y_1, \dots, Y_n$ are independent, the random vector $(Y_1, \dots, Y_n)^T$ has the density

$$f(y_1, \dots, y_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(- \frac{(y_i - g(\mathbf{x}_i; \beta))^2}{2\sigma^2} \right).$$

Thus, the likelihood function is

$$B \times [0, \infty) \ni (\beta, \sigma^2) \mapsto \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(- \frac{(Y_i - g(\mathbf{x}_i; \beta))^2}{2\sigma^2} \right)$$

and the log likelihood function is

$$L(\beta, \sigma^2) = -n \log \sqrt{2\pi} - \frac{n}{2} \log \sigma^2 - \sum_{i=1}^n \frac{(Y_i - g(\mathbf{x}_i; \beta))^2}{2\sigma^2}.$$

In the following we maximise $L(\beta, \sigma^2)$. First we keep σ^2 fixed and maximise in β. For given σ^2, $L(\beta, \sigma^2)$ is maximised if

$$B \ni \beta \mapsto \sum_{i=1}^n (Y_i - g(\mathbf{x}_i; \beta))^2$$

is minimised. By definition the least squares estimator $\widehat{\beta}$ is a solution of this problem. Thus, it remains to maximise

$$[0, \infty) \ni \sigma^2 \mapsto L(\widehat{\beta}, \sigma^2) = -n \log \sqrt{2\pi} - \frac{n}{2} \log \sigma^2 - \sum_{i=1}^n \frac{(Y_i - g(\mathbf{x}_i; \widehat{\beta}))^2}{2\sigma^2}.$$

Whenever $\sum_{i=1}^n (Y_i - g(\mathbf{x}_i; \widehat{\beta}))^2 = 0$, the maximum is attained for $\sigma^2 = 0$, i.e.,

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - g(\mathbf{x}_i; \widehat{\beta}))^2 = \frac{n-k}{n} \widehat{\sigma}^2.$$

Now assume $\sum_{i=1}^n (Y_i - g(\mathbf{x}_i; \widehat{\beta}))^2 > 0$. Because of

$$\frac{\partial L(\widehat{\beta}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \sum_{i=1}^n \frac{(Y_i - g(\mathbf{x}_i; \widehat{\beta}))^2}{2\sigma^4},$$

we have

$$\begin{aligned}\frac{\partial L(\widehat{\beta},\sigma^2)}{\partial\sigma^2}=0 \quad &\Longleftrightarrow \quad -\frac{n}{2\sigma^2}+\sum_{i=1}^{n}\frac{(Y_i-g(\mathbf{x}_i;\widehat{\beta}))^2}{2\sigma^4}=0\\ &\Longleftrightarrow \quad \sigma^2=\frac{1}{n}\sum_{i=1}^{n}(Y_i-g(\mathbf{x}_i;\widehat{\beta}))^2=\frac{n-k}{n}\widehat{\sigma}^2.\end{aligned}$$

Since $\sigma^2 \mapsto L(\widehat{\beta},\sigma^2)$ is continuous and $L(\widehat{\beta},\sigma^2) \to -\infty$ as $\sigma^2 \to 0$ or $\sigma^2 \to \infty$, we have a maximum at $\sigma^2 = \frac{n-k}{n}\widehat{\sigma}^2$. Thus, the log likelihood function has a maximum at $(\beta,\sigma^2) = (\widehat{\beta}, \frac{n-k}{n}\widehat{\sigma}^2)$, which proves that $(\widehat{\beta},\widehat{\sigma}^2_{ML})$ is a maximum likelihood estimator for (β,σ^2). $\square$

Chapter 5

Maximum likelihood estimators and their asymptotic behaviour

5.1 Maximum likelihood estimation for regression

The aim of regression is to explain a dependent variable Y in terms of independent variables $x^{(1)}, \dots, x^{(k)}$. In case of linear and non-linear regression, we assumed that Y is a deterministic function of $x^{(1)}, \dots, x^{(k)}$ plus some random variable, but in order to deal with more general situations, we allow in the following that the distribution of Y depends on $x^{(1)}, \dots, x^{(k)}$. This dependence is given by some unknown parameters, which we are interested in. We will distinguish the cases that Y has a discrete or absolutely continuous distribution.

We denote the vector of the independent variables $x^{(1)}, \dots, x^{(k)}$ by $\mathbf{x}$ and let $D \subseteq \mathbb{R}^k$ be the set of possible values for $\mathbf{x}$.

In the discrete case, we consider a family of real-valued functions $(f(\cdot, \cdot; \vartheta))_{\vartheta \in \Theta}$ such that

$$\mathbb{N}_0 \ni y \mapsto f(y, \mathbf{x}; \vartheta)$$

is a probability mass function for all $\mathbf{x} \in D$ and $\vartheta \in \Theta$. We assume that there exists a $\vartheta_0 \in \Theta$ such that, for every given $\mathbf{x} \in D$, Y has the probability mass function $f(\cdot, \mathbf{x}; \vartheta_0)$.

For the absolutely continuous case let $(f(\cdot, \cdot; \vartheta))_{\vartheta \in \Theta}$ be a family of real-valued functions such that

$$\mathbb{R} \ni y \mapsto f(y, \mathbf{x}; \vartheta)$$

is a probability density function for all $\mathbf{x} \in D$ and $\vartheta \in \Theta$. Now we suppose that there exists a $\vartheta_0 \in \Theta$ such that for every given $\mathbf{x} \in D$, Y has the probability density function $f(\cdot, \mathbf{x}; \vartheta_0)$.

In the following, we consider the discrete and the absolutely continuous case simultaneously. Our goal is to estimate the unknown ϑ_0 from observations. For fixed sample size $n \in \mathbb{N}$ and $\vartheta_0 \in \Theta$ let $\mathbf{x}_1, \dots, \mathbf{x}_n \in D$ and independent random variables $Y_1, \dots, Y_n$ be given such that, for $i \in \{1, \dots, n\}$, Y_i has the probability mass function $f(\cdot, \mathbf{x}_i; \vartheta)$ in the

discrete case and the density $f(\cdot, \mathbf{x}_i; \vartheta)$ in the absolutely continuous case. We illustrate the described framework by two examples.

Example 5.1. Assume that no independent variables are given, i.e., $k = 0$. Then, we have an i.i.d. sample $Y_1, \dots, Y_n$. This is the standard situation of maximum likelihood estimation as considered in Section 2.5, where one also distinguishes the discrete and the absolutely continuous cases.

Example 5.2. Consider a non-linear regression problem as in Section 4.2. Let $(g(\cdot; \beta))_{\beta \in B}$ be a family of real-valued functions such that $B \subseteq \mathbb{R}^m$ is open and $\mathbb{R}^k \ni \mathbf{x} \mapsto g(\mathbf{x}; \beta)$ for each $\beta \in B$. For $n \in \mathbb{N}$ and $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^k$ let $Y_1, \dots, Y_n$ be given such that

$$Y_i = g(\mathbf{x}_i; \beta_0) + \varepsilon_i$$

for $i \in \{1, \dots, n\}$ with some unknown $\beta_0 \in B$ and i.i.d. random variables $\varepsilon_1, \dots, \varepsilon_n$ satisfying $\varepsilon_1 \sim \mathrm{Normal}(0, \sigma_0^2)$ with some unknown $\sigma_0 \in (0, \infty)$. Then, Y_i has the density

$$\mathbb{R} \ni y \mapsto \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(- \frac{(y - g(\mathbf{x}_i; \beta_0))^2}{2\sigma_0^2} \right)$$

for $i \in \{1, \dots, n\}$. This means that we have the situation described above with

$$f(y, \mathbf{x}; \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(- \frac{(y - g(\mathbf{x}; \beta))^2}{2\sigma^2} \right)$$

for $y \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^k$, $\beta \in B$ and $\sigma^2 \in (0, \infty)$.

From now on we assume that Θ is an open subset of $\mathbb{R}^m$ for some $m \in \mathbb{N}$. Let $M = \mathbb{N}_0$ in the discrete case and $M = \mathbb{R}$ in the absolutely continuous case. We assume that there exists a set $\overline{D} \subseteq M \times D$ such that

$$f(y, \mathbf{x}; \vartheta) > 0 \quad \text{for all} \quad (y, \mathbf{x}) \in \overline{D} \quad \text{and} \quad \vartheta \in \Theta$$

and

$$f(y, \mathbf{x}; \vartheta) = 0 \quad \text{for all} \quad (y, \mathbf{x}) \in (M \times D) \setminus \overline{D} \quad \text{and} \quad \vartheta \in \Theta.$$

We also suppose that $f(y, \mathbf{x}; \vartheta)$ is differentiable with respect to ϑ for all $(y, \mathbf{x}) \in \overline{D}$ and $\vartheta \in \Theta$.

Let $\widehat{\vartheta}_n$ be a maximum likelihood estimator for ϑ. It has to maximise the likelihood function

$$\Theta \ni \vartheta \mapsto \prod_{i=1}^n f(Y_i, \mathbf{x}_i; \vartheta)$$

and the log likelihood function

$$\Theta \ni \vartheta \mapsto L_n(\vartheta) = \sum_{i=1}^n \log f(Y_i, \mathbf{x}_i; \vartheta).$$

The latter has the gradient

$$\nabla_\vartheta L_n(\vartheta) = \sum_{i=1}^n \nabla_\vartheta \log f(Y_i, \mathbf{x}_i; \vartheta) = \sum_{i=1}^n \frac{1}{f(Y_i, \mathbf{x}_i; \vartheta)} \nabla_\vartheta f(Y_i, \mathbf{x}_i; \vartheta)$$

for $\vartheta \in \Theta$. Thus, the maximum likelihood estimator $\widehat{\vartheta}_n$ needs to be a solution of

$$\nabla_\vartheta L_n(\vartheta) = \sum_{i=1}^n \nabla_\vartheta \log f(Y_i, \mathbf{x}_i; \vartheta) = \sum_{i=1}^n \frac{1}{f(Y_i, \mathbf{x}_i; \vartheta)} \nabla_\vartheta f(Y_i, \mathbf{x}_i; \vartheta) = 0.$$

We say that $\widetilde{\vartheta} \in \Theta$ is a solution of the maximum likelihood equation if $\nabla_\vartheta L_n(\widetilde{\vartheta}) = 0$.

Example 5.3. Recall that a random variable follows a Weibull distribution with parameter $\tau \in (0, \infty)$ if it has the density

$$y \mapsto \begin{cases} \tau y^{\tau-1} \exp(-y^\tau), & y > 0, \\ 0, & y \leq 0. \end{cases}$$

For a fixed $n \in \mathbb{N}$ let i.i.d. random variables $Y_1, \ldots, Y_n$ be given such that Y_1 follows a Weibull distribution with unknown parameter τ, which we aim to estimate. This is the above situation with $k = 0$, $\overline{D} = (0, \infty)$ and

$$f(y; \tau) = \tau y^{\tau-1} \exp(-y^\tau)$$

for $y \in (0, \infty)$ and $\tau \in (0, \infty)$. Note that

$$\log f(y; \tau) = \log(\tau) + (\tau - 1) \log(y) - y^\tau$$

so that

$$\frac{\partial \log f(y; \tau)}{\partial \tau} = \frac{1}{\tau} + \log(y) - \log(y) y^\tau = \frac{1}{\tau} + \log(y)(1 - y^\tau). \tag{5.1}$$

Thus, a maximum likelihood estimator for τ must be a solution of

$$\sum_{i=1}^n \frac{\partial \log f(Y_i; \tau)}{\partial \tau} = \frac{n}{\tau} + \sum_{i=1}^n \log(Y_i)(1 - Y_i^\tau) = 0. \tag{5.2}$$

Because of

$$\frac{\partial^2 \log f(y; \tau)}{\partial^2 \tau} = -\frac{1}{\tau^2} - \log(y)^2 y^\tau < 0 \tag{5.3}$$

for $y \in (0, \infty)$, the left-hand side of (5.2) is strictly decreasing in τ. Together with

$$\lim_{\tau \to 0} \frac{n}{\tau} + \sum_{i=1}^n \log(Y_i)(1 - Y_i^\tau) = \infty \quad \text{and} \quad \limsup_{\tau \to \infty} \frac{n}{\tau} + \sum_{i=1}^n \log(Y_i)(1 - Y_i^\tau) < 0,$$

this means that (5.2) has a unique solution, which is the maximum likelihood estimator $\widehat{\tau}$ for τ.

In the following R code, we generate an i.i.d. sample for the Weibull distribution and compute the maximum likelihood estimate for the parameter. To this end, we solve the maximum likelihood equation by defining its left-hand side as function and using the command `nleqslv` from the package `nleqslv` in R.

```
> library(nleqslv)
> n<-100
> tau_0<-2.5
> set.seed(111)
> x<-rweibull(n,tau_0)
> DL<-function(tau){
+ y<-sum(1/tau+log(x)-log(x)*x^tau)
+ y
+ }
> MLE<-nleqslv(1, DL)
> MLE$x
[1] 2.474228
```

We see that the estimate is rather close to the true value 2.5.

5.2 Asymptotics of maximum likelihood estimators

In the following, we study the asymptotic behaviour of the maximum likelihood estimators from the previous section for increasing sample size, i.e., we let n go to infinity. Therefore, we assume in the sequel that $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are random as well. To this end let $\mathbf{X}$ be a random vector in D and let Y be a random variable. Let $(f(\cdot, \cdot; \vartheta))_{\vartheta \in \Theta}$ be as in the previous section. Assume that Y and $\mathbf{X}$ are such that, for a fixed $\vartheta_0 \in \Theta$, the conditional distribution of Y for given $\mathbf{X}$ has the probability mass function $f(\cdot, \mathbf{X}; \vartheta_0)$ in the discrete case and the density $f(\cdot, \mathbf{X}; \vartheta_0)$ in the absolutely continuous case.

Let $((Y_i, \mathbf{X}_i))_{i \in \mathbb{N}}$ be independent copies of $(Y, \mathbf{X})$. For $n \in \mathbb{N}$ we denote by L_n the log likelihood function

$$\Theta \ni \vartheta \mapsto L_n(\vartheta) = \sum_{i=1}^{n} \log f(Y_i, \mathbf{X}_i; \vartheta).$$

As the maximum likelihood estimator is a solution of $\nabla_\vartheta L_n(\vartheta) = 0$, we are interested in the solutions of this equation.

For $\vartheta \in \Theta$ such that $\mathbf{E}[\|\nabla_\vartheta \log f(Y, \mathbf{X}; \vartheta)\|^2] < \infty$ we call

$$I(\vartheta) = \mathbf{E}[\nabla_\vartheta \log f(Y, \mathbf{X}; \vartheta)(\nabla_\vartheta \log f(Y, \mathbf{X}; \vartheta))^T]$$

the *Fisher information matrix*.

The following result is a version of Theorem 18 in [2] and concerns the asymptotic behaviour of solutions of the maximum likelihood equation.

Theorem 5.4. Assume for the framework described above the following:

(A1) For all $i, j \in \{1, \dots, m\}$ and $(y, \mathbf{x}) \in \overline{D}$ the partial derivatives

$$\Theta \ni \vartheta \mapsto \frac{\partial^2 f(y, \mathbf{x}; \vartheta)}{\partial \vartheta_i \partial \vartheta_j}$$

exist and are continuous.

(A2) For all $i, j \in \{1, \dots, m\}$ and $\vartheta \in \Theta$ one has

$$\mathbf{E}\Big[\sum_{y=0}^{\infty} \Big|\frac{\partial^2 f(y, \mathbf{X}; \vartheta)}{\partial \vartheta_i \partial \vartheta_j}\Big|\Big] < \infty$$

and

$$\frac{\partial^2}{\partial \vartheta_i \partial \vartheta_j} \mathbf{E}\Big[\sum_{y=0}^{\infty} f(y, \mathbf{X}; \vartheta)\Big] = \mathbf{E}\Big[\sum_{y=0}^{\infty} \frac{\partial^2 f(y, \mathbf{X}; \vartheta)}{\partial \vartheta_i \partial \vartheta_j}\Big] \tag{5.4}$$

in the discrete case and

$$\mathbf{E}\Big[\int_{-\infty}^{\infty} \Big|\frac{\partial^2 f(y, \mathbf{X}; \vartheta)}{\partial \vartheta_i \partial \vartheta_j}\Big| \, \mathrm{d}y\Big] < \infty$$

and

$$\frac{\partial^2}{\partial \vartheta_i \partial \vartheta_j} \mathbf{E}\Big[\int_{-\infty}^{\infty} f(y, \mathbf{X}; \vartheta) \, \mathrm{d}y\Big] = \mathbf{E}\Big[\int_{-\infty}^{\infty} \frac{\partial^2 f(y, \mathbf{X}; \vartheta)}{\partial \vartheta_i \partial \vartheta_j} \, \mathrm{d}y\Big] \tag{5.5}$$

in the absolutely continuous case.

(A3) There exists a function $K : \overline{D} \to [0, \infty)$ with $\mathbf{E}[K(Y, \mathbf{X})] < \infty$ and a constant $\varrho \in (0, \infty)$ such that

$$\Big|\frac{\partial^2 \log f(y, \mathbf{x}; \vartheta)}{\partial \vartheta_i \partial \vartheta_j}\Big| \le K(y, \mathbf{x})$$

for all $i, j \in \{1, \dots, m\}$, $(y, \mathbf{x}) \in \overline{D}$ and $\vartheta \in \Theta$ with $\|\vartheta - \vartheta_0\| \le \varrho$.

(A4) One has $\mathbf{E}[\|\nabla_\vartheta \log f(Y, \mathbf{X}; \vartheta_0)\|^2] < \infty$ and the Fisher information matrix $I(\vartheta_0)$ is positive definite.

(A5) If $\overline{\vartheta} \in \Theta$ satisfies $f(y, \mathbf{x}; \overline{\vartheta}) = f(y, \mathbf{x}; \vartheta_0)$ for all $(y, \mathbf{x}) \in \overline{D}$, then $\overline{\vartheta} = \vartheta_0$.

Let $\mathbf{N} \sim \mathrm{Normal}(\mathbf{0}, I(\vartheta_0)^{-1})$. Then, there exists a sequence of random vectors $(\widetilde{\vartheta}_n)_{n \in \mathbb{N}}$ in Θ such that

$$\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } \nabla_\vartheta L_n(\widetilde{\vartheta}_n) = \mathbf{0} \text{ for all } n \ge n_0) = 1, \tag{5.6}$$

$$\widetilde{\vartheta}_n \xrightarrow{\text{a.s.}} \vartheta_0 \quad \text{as} \quad n \to \infty \tag{5.7}$$

and

$$\sqrt{n}(\widetilde{\vartheta}_n - \vartheta_0) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty. \tag{5.8}$$

Remark 5.5. Note that the expectations on the left-hand sides of (5.4) and (5.5) always equal one since $f(\cdot, \mathbf{X}; \vartheta)$ is a probability mass function or a probability density function, respectively. In order to check whether one can interchange limits and integrals, the following statement is helpful. Let I be an interval and let $(h(\cdot; u))_{u \in I}$ be real-valued functions such that $\mathbb{R}^\ell \ni \mathbf{z} \mapsto h(\mathbf{z}; u)$ for $u \in I$ and such that $h(\mathbf{z}; u)$ is differentiable with respect to u for all $\mathbf{z} \in \mathbb{R}^\ell$ and $u \in I$. Let $u_0 \in I$ and let $I_0 \subseteq I$ be an open interval satisfying $u_0 \in I_0$ and

$$\int_{\mathbb{R}^\ell} \sup_{u \in I_0} \left| \frac{\partial h(\mathbf{z}, u)}{\partial u} \right| \mathrm{d}\mathbf{z} < \infty.$$

Then, one has

$$\frac{\partial}{\partial u} \int_{\mathbb{R}^\ell} h(\mathbf{z}, u_0) \, \mathrm{d}\mathbf{z} = \int_{\mathbb{R}^\ell} \frac{\partial h(\mathbf{z}, u_0)}{\partial u} \, \mathrm{d}\mathbf{z}.$$

This can be shown by combining the definition of the derivative with the so-called dominated convergence theorem.

Note that Theorem 5.4 does not ensure consistency and asymptotic normality of maximum likelihood estimators, but only the existence of solutions of the maximum likelihood equation with the desired properties. Under additional assumptions, it can be used to derive the (strong) consistency and asymptotic normality of maximum likelihood estimators as in the next corollary.

Corollary 5.6. Let the assumptions of Theorem 5.4 be satisfied and let $(\widehat{\vartheta}_n)_{n \in \mathbb{N}}$ be a sequence of estimators such that

$$\begin{aligned} &\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } \widehat{\vartheta}_n \text{ is the} \\ &\quad \text{unique root of } \nabla_\vartheta L_n(\vartheta) = \mathbf{0} \text{ for all } n \geq n_0) = 1. \end{aligned} \tag{5.9}$$

Then,

$$\widehat{\vartheta}_n \xrightarrow{\text{a.s.}} \vartheta_0 \quad \text{as} \quad n \to \infty$$

and

$$\sqrt{n}(\widehat{\vartheta}_n - \vartheta_0) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty.$$

Proof. Let $(\widetilde{\vartheta}_n)_{n \in \mathbb{N}}$ be the sequence of random vectors from the conclusion of Theorem 5.4. From (5.6) and (5.9) it follows that

$$\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } \widehat{\vartheta}_n = \widetilde{\vartheta}_n \text{ for all } n \geq n_0) = 1.$$

This allows to replace $\widetilde{\vartheta}_n$ by $\widehat{\vartheta}_n$ in (5.7) and (5.8) and concludes the proof. □

Example 5.7. Consider the situation of Example 5.3, let $(Y_i)_{i \in \mathbb{N}}$ be i.i.d. Weibull distributed random variables and assume that the underlying parameter is $\tau_0 \in (0, \infty)$. In

order to apply Theorem 5.4, we need to check its assumptions. For $y \in (0,\infty)$ and $\tau \in (0,\infty)$ we have

$$\begin{aligned}\frac{\partial f(y;\tau)}{\partial \tau} &= y^{\tau-1}\exp(-y^\tau) + \tau\log(y)y^{\tau-1}\exp(-y^\tau) + \tau y^{\tau-1}\exp(-y^\tau)(-\log(y)y^\tau)\\ &= (y^{\tau-1} + \tau\log(y)y^{\tau-1} - \tau\log(y)y^{2\tau-1})\exp(-y^\tau)\end{aligned}$$

and

$$\begin{aligned}\frac{\partial^2 f(y;\tau)}{\partial^2 \tau} &= (\log(y)y^{\tau-1} + \log(y)y^{\tau-1} + \tau\log(y)^2y^{\tau-1} - \log(y)y^{2\tau-1} - 2\tau\log(y)^2y^{2\tau-1})\\ &\quad\times\exp(-y^\tau)\\ &\quad-(y^{\tau-1} + \tau\log(y)y^{\tau-1} - \tau\log(y)y^{2\tau-1})\log(y)y^\tau\exp(-y^\tau).\end{aligned}$$

Thus, we see that the second partial derivatives exist and are continuous in τ so that (A1) is satisfied. Using the argument with the integrable upper bound in Remark 5.5, we can verify (A2).

From (5.3) it follows that

$$\left|\frac{\partial^2 \log f(y;\tau)}{\partial^2\tau}\right| = \frac{1}{\tau^2} + \log(y)^2y^\tau.$$

Since one has

$$\frac{1}{\tau^2} + \log(y)^2y^\tau \le \frac{4}{\tau_0^2} + \log(y)^2(y^{\tau_0/2} + y^{3\tau_0/2})$$

for all $y \in (0,\infty)$ and $\tau \in (\tau_0/2, 3\tau_0/2)$ and

$$\int_0^\infty \left(\frac{4}{\tau_0^2} + \log(y)^2(y^{\tau_0/2} + y^{3\tau_0/2})\right)\tau_0 y^{\tau_0-1}\exp(-y^{\tau_0})\,\mathrm{d}y < \infty,$$

(A3) is satisfied. From (5.1) we obtain the Fisher information matrix

$$I(\tau_0) = \mathbf{E}[(1/\tau_0 + \log(Y_1)(1 - Y_1^{\tau_0}))^2],$$

where the expectation on the right-hand side is finite. Since the term in the expectation is non-negative due to the square as well as $\log(z)(1-z) \to -\infty$ as $z \to \infty$ and Y_1 can become arbitrarily large, we have that $I(\tau_0) > 0$ and (A4) is true. Note that (A5) is obviously satisfied. Thus, we have verified all assumptions of Theorem 5.4

Let $\widehat{\tau}_n$ be the maximum likelihood estimator with respect to $Y_1,\dots,Y_n$ derived in Example 5.3 for sample size n. Since it is the unique solution of the maximum likelihood equation, it follows from Corollary 5.6 that

$$\widehat{\tau}_n \xrightarrow{\text{a.s.}} \tau_0 \quad \text{as} \quad n \to \infty$$

and

$$\sqrt{n}(\widehat{\tau}_n - \tau_0) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty,$$

where $\mathbf{N}$ is a centred normal random variable with variance $1/I(\tau_0)$. In particular, this means that $(\widehat{\tau}_n)_{n\in\mathbb{N}}$ are strongly (and weakly) consistent estimators for the parameter of a Weibull distribution.

5.3 Asymptotic confidence intervals and tests for maximum likelihood estimators

In the previous section, we derived central limit theorems for maximum likelihood estimators. The asymptotic normality of a sequence of estimators allows us to construct asymptotic tests and confidence intervals as can be seen from the following theorem.

Theorem 5.8. Let $(\widehat{\vartheta}_n)_{n\in\mathbb{N}}$ be a sequence of estimators for a parameter ϑ, whose true value is $\vartheta_0 \in \mathbb{R}^m$, and let $\Gamma \in \mathbb{R}^{m\times m}$ be positive definite. Assume that

$$\sqrt{n}(\widehat{\vartheta}_n - \vartheta_0) \xrightarrow{d} \mathbf{N}_\Gamma \quad \text{as} \quad n \to \infty \tag{5.10}$$

with $\mathbf{N}_\Gamma \sim \text{Normal}(\mathbf{0}, \Gamma)$ and that there exists a sequence of random matrices $(\Gamma_n)_{n\in\mathbb{N}}$ in $\mathbb{R}^{m\times m}$ such that

$$\Gamma_n \xrightarrow{\mathbf{P}} \Gamma \quad \text{as} \quad n \to \infty. \tag{5.11}$$

Let $\mathbf{a} \in \mathbb{R}^m$ be fixed and let N be a standard normal random variable. Then,

$$\frac{\sqrt{n}(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - \langle \mathbf{a}, \vartheta_0\rangle)}{\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}}} \xrightarrow{d} N \quad \text{as} \quad n \to \infty \tag{5.12}$$

and

$$\lim_{n\to\infty} \mathbf{P}\big(\langle \mathbf{a}, \vartheta_0\rangle \in \big(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - z_{1-\frac{\alpha}{2}}\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}/n}, \langle \mathbf{a}, \widehat{\vartheta}_n\rangle + z_{1-\frac{\alpha}{2}}\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}/n}\big)\big) = 1-\alpha \tag{5.13}$$

for $\alpha \in (0,1)$. This means that $\big(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - z_{1-\frac{\alpha}{2}}\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}/n}, \langle \mathbf{a}, \widehat{\vartheta}_n\rangle + z_{1-\frac{\alpha}{2}}\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}/n}\big)$ is an asymptotic confidence interval for $\langle \mathbf{a}, \vartheta\rangle$ with error level α and that for

$$H_0 : \langle \mathbf{a}, \vartheta\rangle = c_0 \quad \text{against} \quad H_1 : \langle \mathbf{a}, \vartheta\rangle \neq c_0$$

with $c_0 \in \mathbb{R}$ the test that rejects H_0 if $c_0 \notin \big(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - z_{1-\frac{\alpha}{2}}\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}/n}, \langle \mathbf{a}, \widehat{\vartheta}_n\rangle + z_{1-\frac{\alpha}{2}}\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}/n}\big)$ is an asymptotic test with level α.

Note that the covariance matrix Γ in (5.10) usually depends on the true parameter ϑ_0 so that it is unknown. Thus, Γ is estimated by $(\Gamma_n)_{n\in\mathbb{N}}$. In (5.11), it is assumed that $(\Gamma_n)_{n\in\mathbb{N}}$ are weakly consistent estimators for Γ.

Since Γ_n is not required to be positive definite, $\mathbf{a}^T\Gamma_n\mathbf{a}$ does not need to be positive. However, since Γ is positive definite, $(\mathbf{a}^T\Gamma_n\mathbf{a})_{n\in\mathbb{N}}$ converges in probability to some positive constant, which ensures that the left-hand side of (5.12) is asymptotically well defined.

Proof of Theorem 5.8. From (5.10) and the Cramér-Wold device (see Theorem A.2) we obtain

$$\sqrt{n}(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - \langle \mathbf{a}, \vartheta_0\rangle) \xrightarrow{d} \langle \mathbf{a}, \mathbf{N}_\Gamma\rangle \quad \text{as} \quad n \to \infty.$$

Since $\langle \mathbf{a}, \mathbf{N}_\Gamma\rangle \sim \text{Normal}(0, \mathbf{a}^T\Gamma\mathbf{a})$ by Lemma 2.1, this implies

$$\frac{\sqrt{n}(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - \langle \mathbf{a}, \vartheta_0\rangle)}{\sqrt{\mathbf{a}^T\Gamma\mathbf{a}}} \xrightarrow{d} N \quad \text{as} \quad n \to \infty.$$

Note that

$$\frac{\sqrt{n}(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - \langle \mathbf{a}, \vartheta_0\rangle)}{\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}}} = \frac{\sqrt{\mathbf{a}^T\Gamma\mathbf{a}}}{\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}}}\frac{\sqrt{n}(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - \langle \mathbf{a}, \vartheta_0\rangle)}{\sqrt{\mathbf{a}^T\Gamma\mathbf{a}}}.$$

By (5.11) the first factor on the right-hand side converges in probability to 1 so that (5.12) follows from Slutsky's lemma (see Theorem A.1). Now (5.12) and (2.6) imply

$$\begin{aligned}
&\lim_{n\to\infty} \mathbf{P}\left(\frac{\sqrt{n}(\langle \mathbf{a}, \widehat{\vartheta}_n\rangle - \langle \mathbf{a}, \vartheta_0\rangle)}{\sqrt{\mathbf{a}^T\Gamma_n\mathbf{a}}} \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}})\right) = \mathbf{P}(N \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}))\\
&= \Phi(z_{1-\frac{\alpha}{2}}) - \Phi(-z_{1-\frac{\alpha}{2}}) = \Phi(z_{1-\frac{\alpha}{2}}) - (1 - \Phi(z_{1-\frac{\alpha}{2}}))\\
&= 2\Phi(z_{1-\frac{\alpha}{2}}) - 1 = 2\left(1 - \frac{\alpha}{2}\right) - 1 = 1 - \alpha,
\end{aligned}$$

which can be rewritten as in (5.13). From (5.13) we immediately see that the given interval is an asymptotic confidence interval with the required level. Under H_0 the probability that c_0 belongs to the confidence interval and that we do not reject tends to $1-\alpha$ as $n \to \infty$. Consequently, the probability of a Type I error converges to α, which shows that the asymptotic level of the test is α. □

Remark 5.9. Note that (5.12) in Theorem Theorem 5.8 also allows us to construct asymptotic tests for the one-sided null hypotheses $\langle \mathbf{a}, \vartheta\rangle \le c_0$ and $\langle \mathbf{a}, \vartheta\rangle \ge c_0$ as well as asymptotic one-sided confidence intervals.

In case that we reject the two-sided null hypothesis $\langle \mathbf{a}, \vartheta\rangle = c_0$, we can conclude that $\langle \mathbf{a}, \vartheta\rangle > c_0$ if $\langle \mathbf{a}, \widehat{\vartheta}\rangle > c_0$ and that $\langle \mathbf{a}, \vartheta\rangle < c_0$ if $\langle \mathbf{a}, \widehat{\vartheta}\rangle < c_0$.

Example 5.10. Let $(Y_i)_{i\in\mathbb{N}}$ be i.i.d. random variables that follow a Weibull distribution with parameter $\tau \in (0,\infty)$, which we aim to estimate. Assume that the true value for τ is $\tau_0 \in (0,\infty)$. For $n \in \mathbb{N}$ let $\widehat{\tau}_n$ be the maximum likelihood estimator with respect to $Y_1, \dots, Y_n$ derived in Example 5.3. From Example 5.7 we know that

$$\widehat{\tau}_n \xrightarrow{\text{a.s.}} \tau_0 \quad \text{as} \quad n \to \infty \tag{5.14}$$

and

$$\sqrt{n}(\widehat{\tau}_n - \tau_0) \xrightarrow{d} N \quad \text{as} \quad n \to \infty \tag{5.15}$$

with $N \sim \text{Normal}(0, 1/I(\tau_0))$, where

$$I(\tau_0) = \mathbf{E}[(1/\tau_0 + \log(Y_1)(1 - Y_1^{\tau_0}))^2].$$

Since τ_0 and, thus, also $I(\tau_0)$ are unknown, we have to estimate $I(\tau_0)$. To this end we define

$$\frac{1}{\widehat{\sigma}_n^2} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{\widehat{\tau}_n} + \log(Y_i)(1 - Y_i^{\widehat{\tau}_n})\right)^2$$

for $n \in \mathbb{N}$. Let $\varepsilon > 0$ be fixed such that $\tau_0 - \varepsilon > 0$. If $\widehat{\tau}_n \in (\tau_0 - \varepsilon, \tau_0 + \varepsilon)$, then

$$\begin{aligned}
R_n &:= \left| \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{\widehat{\tau}_n} + \log(Y_i)(1 - Y_i^{\widehat{\tau}_n}) \right)^2 - \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{\tau_0} + \log(Y_i)(1 - Y_i^{\tau_0}) \right)^2 \right| \\
&\le \frac{1}{n} \sum_{i=1}^n \left| \frac{1}{\widehat{\tau}_n} + \log(Y_i)(1 - Y_i^{\widehat{\tau}_n}) + \frac{1}{\tau_0} + \log(Y_i)(1 - Y_i^{\tau_0}) \right| \\
&\qquad \times \left| \frac{1}{\widehat{\tau}_n} + \log(Y_i)(1 - Y_i^{\widehat{\tau}_n}) - \frac{1}{\tau_0} - \log(Y_i)(1 - Y_i^{\tau_0}) \right| \\
&\le \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{\tau_0 - \varepsilon} + \frac{1}{\tau_0} + |\log(Y_i)|(2 + Y_i^{\tau_0} + Y_i^{\tau_0 - \varepsilon} + Y_i^{\tau_0 + \varepsilon}) \right) \\
&\qquad \times \left(\frac{1}{\tau_0 - \varepsilon} - \frac{1}{\tau_0 + \varepsilon} + |\log(Y_i)|\, |Y_i^{\tau_0 + \varepsilon} - Y_i^{\tau_0 - \varepsilon}| \right).
\end{aligned}$$

By the strong law of large numbers (see Theorem 2.3), the expression on the right-hand side converges almost surely to

$$\begin{aligned}
\mathbf{E}\Bigg[&\left(\frac{1}{\tau_0 - \varepsilon} + \frac{1}{\tau_0} + |\log(Y_1)|(2 + Y_1^{\tau_0} + Y_1^{\tau_0 - \varepsilon} + Y_1^{\tau_0 + \varepsilon}) \right) \\
&\times \left(\frac{1}{\tau_0 - \varepsilon} - \frac{1}{\tau_0 + \varepsilon} + |\log(Y_1)|\, |Y_1^{\tau_0 + \varepsilon} - Y_1^{\tau_0 - \varepsilon}| \right) \Bigg],
\end{aligned}$$

which we denote by $b(\varepsilon)$. Since, by the almost sure convergence of $(\widehat{\tau}_n)_{n \in \mathbb{N}}$ to τ_0 in (5.14),

$$\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } \widehat{\tau}_n \in (\tau_0 - \varepsilon, \tau_0 + \varepsilon) \text{ for all } n \ge n_0) = 1,$$

we see that

$$\mathbf{P}\left(\limsup_{n \to \infty} R_n \le 2b(\varepsilon) \right) = 1.$$

Because of $b(\varepsilon) \to 0$ as $\varepsilon \to 0$, we obtain $R_n \xrightarrow{\text{a.s.}} 0$ as $n \to \infty$. Together with

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{\tau_0} + \log(Y_i)(1 - Y_i^{\tau_0}) \right)^2 \xrightarrow{\text{a.s.}} \mathbf{E}\left[\left(\frac{1}{\tau_0} + \log(Y_1)(1 - Y_1^{\tau_0}) \right)^2 \right] \quad \text{as} \quad n \to \infty,$$

which follows from the strong law of large numbers, we derive that

$$\frac{1}{\widehat{\sigma}_n^2} \xrightarrow{\text{a.s.}} \mathbf{E}\left[\left(\frac{1}{\tau_0} + \log(Y_1)(1 - Y_1^{\tau_0}) \right)^2 \right] \quad \text{as} \quad n \to \infty$$

and, thus,

$$\widehat{\sigma}_n^2 \xrightarrow{\text{a.s.}} \mathbf{E}\left[\left(\frac{1}{\tau_0} + \log(Y_1)(1 - Y_1^{\tau_0}) \right)^2 \right]^{-1} = \frac{1}{I(\tau_0)} \quad \text{as} \quad n \to \infty. \tag{5.16}$$

For the test problem

$$H_0 : \tau = c_0 \quad \text{against} \quad H_1 : \tau \neq c_0$$

with $c_0 \in (0, \infty)$ and $\alpha \in (0, 1)$ we consider the test

$$\varphi(Y_1, \dots, Y_n) = \begin{cases} \text{cannot reject}, & c_0 \in \big(\widehat{\tau}_n - \frac{z_{1-\frac{\alpha}{2}}\widehat{\sigma}_n}{\sqrt{n}}, \widehat{\tau}_n + \frac{z_{1-\frac{\alpha}{2}}\widehat{\sigma}_n}{\sqrt{n}}\big), \\ \text{reject}, & c_0 \notin \big(\widehat{\tau}_n - \frac{z_{1-\frac{\alpha}{2}}\widehat{\sigma}_n}{\sqrt{n}}, \widehat{\tau}_n + \frac{z_{1-\frac{\alpha}{2}}\widehat{\sigma}_n}{\sqrt{n}}\big). \end{cases}$$

From (5.15), (5.16) and Theorem 5.8 with $\Gamma = 1/I(\tau_0)$ and $\Gamma_n = \widehat{\sigma}_n^2$ for $n \in \mathbb{N}$ it follows that this is an asymptotic test with level α and that $\big(\widehat{\tau}_n - \frac{z_{1-\frac{\alpha}{2}}\widehat{\sigma}_n}{\sqrt{n}}, \widehat{\tau}_n + \frac{z_{1-\frac{\alpha}{2}}\widehat{\sigma}_n}{\sqrt{n}}\big)$ is an asymptotic confidence interval for τ with error level α.

5.4 Asymptotic tests for non-linear regression

In the following, we consider non-linear regression, but assume that the independent variables are also random. Let $\mathbf{X}$ be a random vector in $\mathbb{R}^k$ and let $(g(\cdot;\beta))_{\beta\in B}$ be a family of real-valued functions such that $B \subseteq \mathbb{R}^m$ is open and $\mathbb{R}^k \ni \mathbf{x} \mapsto g(\mathbf{x};\beta)$ for $\beta \in B$. Let

$$Y = g(\mathbf{X};\beta_0) + \varepsilon$$

with $\beta_0 \in B$ and $\varepsilon \sim \text{Normal}(0, \sigma_0^2)$ independent from $\mathbf{X}$ with $\sigma_0 \in (0, \infty)$. Let $(Y_i, \mathbf{X}_i)_{i\in\mathbb{N}}$ be independent copies of $(Y, \mathbf{X})$.

Recall from Example 5.2 that

$$f(y, \mathbf{x}; \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\bigg(-\frac{(y - g(\mathbf{x};\beta))^2}{2\sigma^2}\bigg)$$

for $y \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^k$, $\beta \in B$ and $\sigma^2 \in (0, \infty)$. Next we compute the Fisher information matrix for this situation. We have

$$\log f(y, \mathbf{x}; \beta, \sigma^2) = -\log(\sqrt{2\pi}) - \frac{1}{2}\log(\sigma^2) - \frac{(y - g(\mathbf{x};\beta))^2}{2\sigma^2}$$

so that

$$\nabla_\beta \log f(y, \mathbf{x}; \beta, \sigma^2) = \frac{(y - g(\mathbf{x};\beta))}{\sigma^2}\nabla_\beta g(\mathbf{x};\beta)$$

and

$$\frac{\partial \log f(y, \mathbf{x}; \beta, \sigma^2)}{\partial\sigma^2} = -\frac{1}{2\sigma^2} + \frac{(y - g(\mathbf{x};\beta))^2}{2\sigma^4}.$$

Together with the independence of $\mathbf{X}$ and ε as well as $\mathbf{E}[\varepsilon] = \mathbf{E}[\varepsilon^3] = 0$, $\mathbf{E}[\varepsilon^2] = \sigma_0^2$ and $\mathbf{E}[\varepsilon^4] = 3\sigma_0^4$, this implies that

$$\begin{aligned}
&\mathbf{E}[\nabla_\beta \log f(Y, \mathbf{X}; \beta_0, \sigma_0^2)\nabla_\beta \log f(Y, \mathbf{X}; \beta_0, \sigma_0^2)^T] \\
&= \mathbf{E}\bigg[\frac{(Y - g(\mathbf{X};\beta_0))^2}{\sigma_0^4}\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T\bigg] = \mathbf{E}\bigg[\frac{\varepsilon^2}{\sigma_0^4}\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T\bigg] \\
&= \mathbf{E}\bigg[\frac{\varepsilon^2}{\sigma_0^4}\bigg]\mathbf{E}\bigg[\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T\bigg] = \frac{1}{\sigma_0^2}\mathbf{E}\bigg[\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T\bigg],
\end{aligned}$$

$$\begin{aligned}
&\mathbf{E}\Big[\nabla_\beta \log f(Y,\mathbf{X};\beta_0,\sigma_0^2)\frac{\partial \log f(Y,\mathbf{X};\beta_0,\sigma_0^2)}{\partial\sigma^2}\Big]\\
&= \mathbf{E}\Big[-\frac{(Y-g(\mathbf{X};\beta_0))}{2\sigma_0^4}\nabla_\beta g(\mathbf{X};\beta_0)+\frac{(Y-g(\mathbf{X};\beta_0))^3}{2\sigma_0^6}\nabla_\beta g(\mathbf{X};\beta_0)\Big]\\
&= \mathbf{E}\Big[-\frac{\varepsilon}{2\sigma_0^4}\nabla_\beta g(\mathbf{X};\beta_0)+\frac{\varepsilon^3}{2\sigma_0^6}\nabla_\beta g(\mathbf{X};\beta_0)\Big] = \mathbf{E}\Big[-\frac{\varepsilon}{2\sigma_0^4}+\frac{\varepsilon^3}{2\sigma_0^6}\Big]\mathbf{E}\Big[\nabla_\beta g(\mathbf{X};\beta_0)\Big] = 0
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{E}\Big[\Big(\frac{\partial \log f(Y,\mathbf{X};\beta_0,\sigma_0^2)}{\partial\sigma^2}\Big)^2\Big] &= \mathbf{E}\Big[\frac{1}{4\sigma_0^4}-\frac{(Y-g(\mathbf{X};\beta_0))^2}{2\sigma_0^6}+\frac{(Y-g(\mathbf{X};\beta_0))^4}{4\sigma_0^8}\Big]\\
&= \mathbf{E}\Big[\frac{1}{4\sigma_0^4}-\frac{\varepsilon^2}{2\sigma_0^6}+\frac{\varepsilon^4}{4\sigma_0^8}\Big] = \frac{1}{4\sigma_0^4}-\frac{\sigma_0^2}{2\sigma_0^6}+\frac{3\sigma_0^4}{4\sigma_0^8} = \frac{1}{2\sigma_0^4}.
\end{aligned}$$

Thus, we obtain the Fisher information matrix

$$I(\beta_0,\sigma_0^2) = \frac{1}{\sigma_0^2}\begin{pmatrix}\mathbf{E}[\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T] & \mathbf{0}\\ \mathbf{0}^T & \frac{1}{2\sigma_0^2}\end{pmatrix}$$

with $\mathbf{0}=(0,\dots,0)^T\in\mathbb{R}^m$.

For $n\in\mathbb{N}$ let $\widehat{\beta}_n$ be a least squares estimator for β with respect to $(Y_1,\mathbf{X}_1),\dots,(Y_n,\mathbf{X}_n)$ and let

$$\widehat{\sigma}_n^2 = \frac{1}{n-m}\sum_{i=1}^n (Y_i-g(\mathbf{X}_i;\widehat{\beta}_n))^2.$$

Recall from Theorem 4.3 that $(\widehat{\beta}_n,\frac{n-m}{n}\widehat{\sigma}_n^2)$ is a maximum likelihood estimator for (β,σ^2). Thus, our previous results on asymptotic normality of maximum likelihood estimators suggest that, under suitable assumptions,

$$\sqrt{n}\begin{pmatrix}\widehat{\beta}_n-\beta_0\\ \widehat{\sigma}_n^2-\sigma_0^2\end{pmatrix}\xrightarrow{d}\mathbf{N}_{\beta_0,\sigma_0^2}\quad\text{as}\quad n\to\infty$$

with $\mathbf{N}_{\beta_0,\sigma_0^2}\sim\text{Normal}(\mathbf{0},I(\beta_0,\sigma_0^2)^{-1})$. This implies that

$$\sqrt{n}(\widehat{\beta}_n-\beta_0)\xrightarrow{d}\mathbf{N}_{\beta_0}\quad\text{as}\quad n\to\infty$$

with $\mathbf{N}_{\beta_0}\sim\text{Normal}(\mathbf{0},\sigma_0^2\mathbf{E}[\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T]^{-1})$. Since β_0 and σ_0^2 are usually unknown, we cannot compute the covariance matrix. Instead we approximate it by using the observations together with the estimates for β and σ^2. Thus, we define

$$\widehat{M}_n = \frac{1}{n}\sum_{i=1}^n \nabla_\beta g(\mathbf{X}_i;\widehat{\beta}_n)\nabla_\beta g(\mathbf{X}_i;\widehat{\beta}_n)^T$$

for $n\in\mathbb{N}$.

Theorem 5.11. Assume that $M := \mathbf{E}[\nabla_\beta g(\mathbf{X};\beta_0)\nabla_\beta g(\mathbf{X};\beta_0)^T]$ is invertible, that

$$\sqrt{n}(\widehat{\beta}_n - \beta_0) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty$$

with $\mathbf{N} \sim \text{Normal}(\mathbf{0}, \sigma_0^2 M^{-1})$ and that

$$\widehat{\sigma}_n^2 \xrightarrow{\mathbf{P}} \sigma_0^2 \quad \text{and} \quad \widehat{M}_n \xrightarrow{\mathbf{P}} M \quad \text{as} \quad n \to \infty. \tag{5.17}$$

Consider the test problem

$$H_0 : \langle \mathbf{a}, \beta \rangle = c_0 \quad \text{against} \quad H_1 : \langle \mathbf{a}, \beta \rangle \neq c_0$$

for $\mathbf{a} \in \mathbb{R}^m$ and $c_0 \in \mathbb{R}$ and let $\alpha \in (0,1)$. Then,

$$\varphi((Y_1, \mathbf{X}_1), \ldots, (Y_n, \mathbf{X}_n)) = \begin{cases} \text{cannot reject}, & c_0 \in I_n, \\ \text{reject}, & c_0 \notin I_n, \end{cases}$$

with

$$I_n = \left(\langle \mathbf{a}, \widehat{\beta}_n \rangle - \frac{z_{1-\frac{\alpha}{2}} \widehat{\sigma}_n \sqrt{\mathbf{a}^T \widehat{M}_n^{-1} \mathbf{a}}}{\sqrt{n}}, \langle \mathbf{a}, \widehat{\beta}_n \rangle + \frac{z_{1-\frac{\alpha}{2}} \widehat{\sigma}_n \sqrt{\mathbf{a}^T \widehat{M}_n^{-1} \mathbf{a}}}{\sqrt{n}} \right)$$

is an asymptotic test for H_0 with level α. Moreover, I_n is an asymptotic confidence interval for $\langle \mathbf{a}, \beta \rangle$ with error level α.

Proof. From (5.17) it follows that

$$\frac{1}{\widehat{\sigma}_n^2} \widehat{M}_n \xrightarrow{\mathbf{P}} \frac{1}{\sigma_0^2} M \quad \text{as} \quad n \to \infty.$$

Since M is, by assumption, invertible, this implies that

$$\widehat{\sigma}_n^2 \widehat{M}_n^{-1} \xrightarrow{\mathbf{P}} \sigma_0^2 M^{-1} \quad \text{as} \quad n \to \infty.$$

Thus, all assumptions of Theorem 5.8 are satisfied with $\Gamma_n = \widehat{\sigma}_n^2 \widehat{M}_n^{-1}$ for $n \in \mathbb{N}$ and $\Gamma = \sigma_0^2 M^{-1}$, which yields the desired conclusions. □

The test in Theorem 5.11 rejects the null hypothesis if and only if

$$\frac{\langle \mathbf{a}, \widehat{\beta}_n \rangle - c_0}{\widehat{\sigma}_n \sqrt{\mathbf{a}^T \widehat{M}_n^{-1} \mathbf{a}}} \notin (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}) \tag{5.18}$$

Thus, the asymptotic p-value is given by

$$2\left(1 - \Phi\left(\frac{|\langle \mathbf{a}, \widehat{\beta}_n \rangle - c_0|}{\widehat{\sigma}_n \sqrt{\mathbf{a}^T \widehat{M}_n^{-1} \mathbf{a}}}\right)\right),$$

where Φ is the distribution function of the standard normal distribution. In the context of asymptotic tests, we use the term asymptotic p-value as it is only an approximation of the true p-value.

Note that Theorem 5.11 provides a test for the parameters of a non-linear regression model. However, one must be careful that it requires the asymptotic normality of the underlying maximum likelihood estimators as well as the weak consistency of the estimators for the Fisher information matrix.

Example 5.12. As in Examples 4.1 and 4.2, we consider the data set `Puromycin` in `R`, which can be described by the Michaelis-Menten equation. For the previously computed non-linear regression model we use the command `summary` in `R`:

```
> summary(NLS_model)

Formula:  rate ~ vmax * conc/(K + conc)

Parameters:
       Estimate  Std. Error  t value  Pr(>|t|)
vmax  2.127e+02   6.947e+00   30.615  3.24e-11  ***
K     6.412e-02   8.281e-03    7.743  1.57e-05  ***
---
Signif.  codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’  0.1 ‘ ’ 1

Residual standard error:  10.93 on 10 degrees of freedom

Number of iterations to convergence:  10
Achieved convergence tolerance:  1.671e-06
```

Here the residual standard error is the value of the estimator $\widehat{\sigma}_n^2$ for σ^2. The table with the estimates also provides the asymptotic p-values for the null hypotheses that the parameters are zero. In the column `Std. Error`, the roots of the diagonal elements of $\frac{\widehat{\sigma}_n^2}{n}\widehat{M}_n^{-1}$ in the notation of Theorem 5.11 are given, while the `t value` is the estimate divided by the standard error, which is the statistic in (5.18) for $c_0 = 0$. Theorem 5.11 suggests to compare this quantity with the quantiles of a standard normal distribution. In `R`, a t-distribution with $n - m$ degrees of freedom is used instead. The corresponding asymptotic p-values are given in the second last column. Since the quantiles of the t-distribution converge to those of the standard normal distribution as the number of degrees of freedom goes to infinity, asymptotically it does not make a difference whether the quantiles of the t-distribution or of the standard normal distribution are employed. Note that in general neither the standard normal distribution nor a t-distribution is the exact underlying distribution of the test statistics.

Chapter 6

Logistic regression

6.1 Binary regression

As in the cases of linear and non-linear regression we are interested in explaining a random quantity Y by deterministic quantities $x^{(1)}, \ldots, x^{(k)}$, but this time we assume that Y can take only the values zero and one. This situation, where one can think of 1 as success and 0 as failure, is called *binary regression*.

Example 6.1. On January 28, 1986 the Space Shuttle Challenger exploded directly after the start, killing all seven crew members aboard. The disaster was caused by the failure of so-called O-rings. At low temperatures such as on the day of the accident, their rubber becomes stiff and they seal less effectively. So our goal is to understand how the temperature affects whether there is a damage at an O-ring or not.

Since Y can only take the values zero and one, we cannot think of it as a deterministic function of $x^{(1)}, \ldots, x^{(k)}$ plus some random noise. Instead we aim to express the probability of $Y = 1$ as a function of $x^{(1)}, \ldots, x^{(k)}$. A first idea is to assume that

$$\mathbf{P}(Y = 1) = \beta^{(0)} + \beta^{(1)}x^{(1)} + \ldots + \beta^{(k)}x^{(k)}$$

for some $\beta^{(0)}, \beta^{(1)}, \ldots, \beta^{(k)} \in \mathbb{R}$. However, this does not work since there are always choices of $x^{(1)}, \ldots, x^{(k)}$ such that the right-hand side becomes negative or larger than one, which are contradictions as probabilities need to be between zero and one. To resolve this problem, we choose a strictly monotonically increasing function $h : \mathbb{R} \to (0, 1)$ and assume that

$$\mathbf{P}(Y = 1) = h(\beta^{(0)} + \beta^{(1)}x^{(1)} + \ldots + \beta^{(k)}x^{(k)}) \tag{6.1}$$

for $\beta^{(0)}, \beta^{(1)}, \ldots, \beta^{(k)} \in \mathbb{R}$. The function h is called *inverse link function*, while its inverse h^{-1} is denoted as *link function*.

The most common choice for h is

$$h(u) = \frac{\exp(u)}{1 + \exp(u)},$$

whose inverse

$$h^{-1} : (0,1) \to \mathbb{R}, \quad p \mapsto \log\left(\frac{p}{1-p}\right)$$

is named *logit function*. The corresponding type of binary regression is called *logistic regression*.

Another inverse link function is $h = \Phi$, where Φ stands for the distribution function of the standard normal distribution. This approach is referred to as *probit regression*.

In the following, we assume that n independent observations of Y and the corresponding values of $x^{(1)}, \dots, x^{(k)}$ are given. More precisely, let $Y_1, \dots, Y_n$ be independent Bernoulli-distributed random variables and let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^k$ be such that

$$\mathbf{P}(Y_i = 1) = h(\beta^T \mathbf{x}_i) \tag{6.2}$$

for $i \in \{1, \dots, n\}$ with an unknown $\beta \in \mathbb{R}^k$, which we aim to estimate. In contrast to (6.1), we do not have an intercept $\beta^{(0)}$ in (6.2). However, we can include the case of an intercept into this situation by letting the first entry of $\mathbf{x}_1, \dots, \mathbf{x}_n$ always being one. This could be also done for linear regression.

6.2 Maximum likelihood estimation for logistic regression

In the following, we consider the framework of the previous section for logistic regression, i.e., we take as inverse link function h the function

$$\ell : \mathbb{R} \to (0, \infty), \quad \ell(u) = \frac{\exp(u)}{1 + \exp(u)}.$$

For a Bernoulli distributed random variable Z with success probability $p \in [0,1]$ we have

$$\mathbf{P}(Z = z) = p^z (1-p)^{1-z}$$

for $z \in \{0, 1\}$. Thus, from our observations $(Y_1, \mathbf{x}_1), \dots, (Y_n, \mathbf{x}_n)$ we obtain the likelihood function

$$\mathbb{R}^k \ni \beta \mapsto \prod_{i=1}^n \ell(\beta^T \mathbf{x}_i)^{Y_i} (1 - \ell(\beta^T \mathbf{x}_i))^{1-Y_i}.$$

The corresponding log likelihood function is

$$\begin{aligned} L(\beta) &= \sum_{i=1}^n Y_i \log \ell(\beta^T \mathbf{x}_i) + (1 - Y_i) \log(1 - \ell(\beta^T \mathbf{x}_i)) \\ &= \sum_{i=1}^n Y_i \log \frac{\ell(\beta^T \mathbf{x}_i)}{1 - \ell(\beta^T \mathbf{x}_i)} + \log(1 - \ell(\beta^T \mathbf{x}_i)). \end{aligned}$$

Since

$$\log \frac{\ell(u)}{1-\ell(u)} = \log \frac{\frac{\exp(u)}{1+\exp(u)}}{1-\frac{\exp(u)}{1+\exp(u)}} = \log \exp(u) = u \tag{6.3}$$

and

$$\log(1-\ell(u)) = \log\left(1-\frac{\exp(u)}{1+\exp(u)}\right) = \log\left(\frac{1}{1+\exp(u)}\right) = -\log(1+\exp(u)) \tag{6.4}$$

for $u \in \mathbb{R}$, this can be rewritten as

$$L(\beta) = \sum_{i=1}^{n} Y_i \beta^T \mathbf{x}_i - \log(1+\exp(\beta^T \mathbf{x}_i)).$$

Because of

$$(\log(1+\exp(u)))' = \frac{\exp(u)}{1+\exp(u)} = \ell(u), \tag{6.5}$$

we obtain the gradient

$$\nabla L(\beta) = \sum_{i=1}^{n} (Y_i - \ell(\beta^T \mathbf{x}_i))\mathbf{x}_i$$

and, consequently, the Hessian matrix

$$\operatorname{Hess} L(\beta) = -\sum_{i=1}^{n} \ell'(\beta^T \mathbf{x}_i)\mathbf{x}_i \mathbf{x}_i^T. \tag{6.6}$$

Note that

$$\begin{aligned} \ell'(u) &= \left(\frac{\exp(u)}{1+\exp(u)}\right)' = \frac{\exp(u)(1+\exp(u)) - \exp(u)\exp(u)}{(1+\exp(u))^2} = \frac{\exp(u)}{(1+\exp(u))^2} \\ &= \frac{1}{1+\exp(u)} \frac{\exp(u)}{1+\exp(u)} = \left(1-\frac{\exp(u)}{1+\exp(u)}\right)\frac{\exp(u)}{1+\exp(u)} = (1-\ell(u))\ell(u). \end{aligned} \tag{6.7}$$

Because of $0 < \ell(u) < 1$ for $u \in \mathbb{R}$, we see that ℓ' is always positive. Thus, we have

$$\mathbf{a}^T \operatorname{Hess} L(\beta)\mathbf{a} = -\sum_{i=1}^{n} \ell'(\beta^T \mathbf{x}_i)(\mathbf{a}^T \mathbf{x}_i)^2 \leq 0$$

for $\mathbf{a} \in \mathbb{R}^k$ so that the Hessian matrix is negative semi-definite, which implies that the log likelihood function L is concave. This means that a root of

$$\nabla L(\beta) = \sum_{i=1}^{n} (Y_i - \ell(\beta^T \mathbf{x}_i))\mathbf{x}_i = \mathbf{0} \tag{6.8}$$

is a global maximum of L so that an estimator $\widehat{\beta}$ is a maximum likelihood estimator for the underlying parameter β if $\nabla L(\widehat{\beta}) = \mathbf{0}$.

Note that the equation in (6.8) does not need to have a solution. For instance, if $n \leq k$ and $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are linearly independent, this equation can only hold if $Y_i = \ell(\beta^T \mathbf{x}_i)$ for $i \in \{1, \ldots, n\}$, which is not possible since $Y_i \in \{0, 1\}$ but $\ell(\beta^T \mathbf{x}_i) \in (0, 1)$. This example can carry over to $n > k$ if several of the observations $(Y_i, \mathbf{x}_i)$, $i \in \{1, \ldots, n\}$, are identical.

Example 6.2. The observations of damaged O-rings and temperatures from Example 6.1 are given in the data set `orings` of the `R` package `faraway`. For each of 23 launches of Space Shuttles the temperature `temp` and the number `damage` of how many of the six O-rings were damaged are given. We assume that the different O-rings at a launch behave independently. The logistic regression model is computed by using the command `glm`, which includes an intercept by default:

```
> Model<-glm(cbind(damage,6-damage)~temp, family=binomial, orings)
> Model

Call:  glm(formula = cbind(damage, 6 - damage) ~ temp, family = binomial,
data = orings)

Coefficients:
(Intercept)     temp
    11.6630  -0.2162

Degrees of Freedom:  22 Total (i.e. Null); 21 Residual
Null Deviance:  38.9
Residual Deviance:  16.91   AIC: 33.67
```

This provides the following model:

$$\mathbf{P}(\text{O-ring is damaged}) = \ell(11.6630 - 0.2162 \cdot \texttt{temp})$$

The data set `orings` provides the numbers of damages for six O-rings. Alternatively, one can rewrite the data to a new data frame `orings_new` with a variable `damage` with values zero or one and a variable `temp` with the corresponding temperatures. Then

```
glm(damage ~ temp, family=binomial, orings_new)
```

leads to the same model as above.

6.3 Asymptotics of the maximum likelihood estimator for logistic regression

In the following, we study the asymptotic behaviour of the maximum likelihood estimators from the previous section for increasing sample size, i.e., we let the number n of $\mathbf{x}_1, \ldots, \mathbf{x}_n$ go to infinity. Therefore, we assume in the sequel that $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are random as well. To

this end let $\mathbf{X}$ be a random vector in $\mathbb{R}^k$. Moreover, let $\beta_0 \in \mathbb{R}^k$ and let Y be a random variable such that

$$\mathbf{P}(Y = 1|\mathbf{X}) = \ell(\beta_0^T\mathbf{X}) \quad \text{and} \quad \mathbf{P}(Y = 0|\mathbf{X}) = 1 - \ell(\beta_0^T\mathbf{X}), \tag{6.9}$$

i.e., for given $\mathbf{X}$, Y follows a Bernoulli distribution with parameter $\ell(\beta_0^T\mathbf{X})$. Here $\mathbf{P}(\cdot|\mathbf{X})$ and $\mathbf{E}[\cdot|\mathbf{X}]$ stand for the conditional probability and the conditional expectation for given $\mathbf{X}$, for which one has similar rules as for usual probabilities and expectations. Let $((Y_i, \mathbf{X}_i))_{i\in\mathbb{N}}$ be independent copies of $(Y, \mathbf{X})$. By L_n we denote the log likelihood function with respect to $(Y_1, \mathbf{X}_1), \ldots, (Y_n, \mathbf{X}_n)$.

Theorem 6.3. Consider the described situation and assume that $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$ and that $\mathbf{E}[\mathbf{X}\mathbf{X}^T]$ is positive definite.

a) One has

$$\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } L_n \text{ has a unique maximiser for all } n \geq n_0) = 1.$$

b) For $n \in \mathbb{N}$ let $\widehat{\beta}_n$ be the unique maximiser of L_n if it exists and $\widehat{\beta}_n = \mathbf{0}$, otherwise. Then,

$$\widehat{\beta}_n \xrightarrow{\text{a.s.}} \beta_0 \quad \text{as} \quad n \to \infty \tag{6.10}$$

and

$$\sqrt{n}(\widehat{\beta}_n - \beta_0) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty \tag{6.11}$$

with $\mathbf{N} \sim \text{Normal}(\mathbf{0}, \mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T]^{-1})$.

For $\mathbf{a} \in \mathbb{R}^k$ we have $\mathbf{a}^T\mathbf{X}\mathbf{X}^T\mathbf{a} > 0$ if and only if $\ell'(\beta_0^T\mathbf{X})\mathbf{a}^T\mathbf{X}\mathbf{X}^T\mathbf{a} > 0$ since ℓ' is always positive by (6.7). Thus, the positive definiteness of $\mathbf{E}[\mathbf{X}\mathbf{X}^T]$ implies that of $\mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T]$, which ensures that $\mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T]$ is invertible.

The almost sure convergence of $(\widehat{\beta}_n)_{n\in\mathbb{N}}$ to the true parameter β_0 in (6.10) means that $(\widehat{\beta}_n)_{n\in\mathbb{N}}$ are strongly consistent estimators for β. From (6.11) we see that they are approximately normally distributed if n is large.

Note that we obtain the same asymptotic behaviour as in (6.10) and (6.11) for estimators that are one of the maximisers of L_n instead of $\mathbf{0}$ in case that the maximiser of L_n is not unique.

We prepare the proof of Theorem 6.3 with the following lemma.

Lemma 6.4. Let the assumptions of Theorem 6.3 prevail. Then,

$$\mathbf{P}(\text{there exists an } n_0' \in \mathbb{N} \text{ such that } L_n \text{ is strictly concave for all } n \geq n_0') = 1.$$

Proof. Recall from (6.6) that

$$\operatorname{Hess} L_n(\beta) = -\sum_{i=1}^{n} \ell'(\beta^T\mathbf{X}_i)\mathbf{X}_i\mathbf{X}_i^T.$$

Since $\ell'(\beta^T\mathbf{X}_i) > 0$ by (6.7) and $\mathbf{X}_i\mathbf{X}_i^T$ is positive semi-definite for $i \in \{1,\dots,n\}$, $\operatorname{Hess} L_n(\beta)$ is negative definite if and only if for any $\mathbf{a} \in \mathbb{R}^k$ there exists an $i_0 \in \{1,\dots,n\}$ such that $\mathbf{a}^T\mathbf{X}_{i_0}\mathbf{X}_{i_0}^T\mathbf{a} > 0$. The latter condition is equivalent to $\sum_{i=1}^n \mathbf{X}_i\mathbf{X}_i^T$ being positive definite. As a function whose Hessian is always negative definite is strictly concave, this implies that L_n is strictly concave if $\frac{1}{n}\sum_{i=1}^n \mathbf{X}_i\mathbf{X}_i^T$ is positive definite. From the strong law of large numbers (see Theorem 2.3) it follows that

$$\frac{1}{n}\sum_{i=1}^n \mathbf{X}_i\mathbf{X}_i^T \xrightarrow{\text{a.s.}} \mathbf{E}[\mathbf{X}\mathbf{X}^T] \quad \text{as} \quad n \to \infty.$$

Since the matrix on the right-hand side is positive definite by assumption, the matrices on the left-hand side must be also positive definite for n sufficiently large. Thus, we see that

$$\mathbf{P}\left(\text{there exists an } n_0' \in \mathbb{N} \text{ such that } \frac{1}{n}\sum_{i=1}^n \mathbf{X}_i\mathbf{X}_i^T \text{ is positive definite for all } n \geq n_0'\right) = 1,$$

which completes the proof. □

Proof of Theorem 6.3. Note that we are exactly in the situation of Section 5.2 with

$$f(y,\mathbf{x};\beta) = \ell(\beta^T\mathbf{x})^y(1-\ell(\beta^T\mathbf{x}))^{1-y}$$

for $y \in \{0,1\}$, $\mathbf{x} \in \mathbb{R}^k$ and $\beta \in \mathbb{R}^k$. Thus, we check the assumptions of Theorem 5.4 in the sequel. For $y \in \{0,1\}$, $\mathbf{x} \in \mathbb{R}^k$, $\beta \in \mathbb{R}^k$ and $i,j \in \{1,\dots,k\}$ we have

$$\frac{\partial f(y,\mathbf{x};\beta)}{\partial \beta_i} = y\ell'(\beta^T\mathbf{x})\mathbf{x}^{(i)} - (1-y)\ell'(\beta^T\mathbf{x})\mathbf{x}^{(i)} = (2y-1)\ell'(\beta^T\mathbf{x})\mathbf{x}^{(i)}$$

and

$$\frac{\partial^2 f(y,\mathbf{x};\beta)}{\partial \beta_i \partial \beta_j} = (2y-1)\ell''(\beta^T\mathbf{x})\mathbf{x}^{(i)}\mathbf{x}^{(j)},$$

where we denote by $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$ the i-th and j-th entries of $\mathbf{x}$. From $\ell'(u) = \ell(u)(1-\ell(u))$ (see (6.7)) and

$$\begin{aligned}\ell''(u) &= (\ell(u)(1-\ell(u)))' = (\ell(u)-\ell(u)^2)' = \ell'(u) - 2\ell(u)\ell'(u)\\ &= (1-2\ell(u))\ell'(u) = (1-2\ell(u))(1-\ell(u))\ell(u)\end{aligned} \tag{6.12}$$

for $u \in \mathbb{R}$ it follows that

$$\frac{\partial^2 f(y,\mathbf{x};\beta)}{\partial \beta_i \partial \beta_j} = (2y-1)(1-2\ell(\beta^T\mathbf{x}))(1-\ell(\beta^T\mathbf{x}))\ell(\beta^T\mathbf{x})\mathbf{x}^{(i)}\mathbf{x}^{(j)}. \tag{6.13}$$

Thus, (A1) is satisfied.

Let $i, j \in \{1, \dots, k\}$. By (6.13), $|\ell(u)| \le 1$ for $u \in \mathbb{R}$ and $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$ we have

$$\mathbf{E}\left[\left|\frac{\partial^2 f(0, \mathbf{X}; \beta)}{\partial\beta_i \partial\beta_j}\right| + \left|\frac{\partial^2 f(1, \mathbf{X}; \beta)}{\partial\beta_i \partial\beta_j}\right|\right] \le 2\mathbf{E}[|\mathbf{X}^{(i)}\mathbf{X}^{(j)}|] \le 2\mathbf{E}[\|\mathbf{X}\|^2] < \infty.$$

Since the first factor on the right-hand side of (6.13) is ± 1, we see that

$$\mathbf{E}\left[\frac{\partial^2 f(0, \mathbf{X}; \beta)}{\partial\beta_i \partial\beta_j} + \frac{\partial^2 f(1, \mathbf{X}; \beta)}{\partial\beta_i \partial\beta_j}\right] = \mathbf{E}\left[\frac{\partial^2 f(0, \mathbf{X}; \beta)}{\partial\beta_i \partial\beta_j} - \frac{\partial^2 f(0, \mathbf{X}; \beta)}{\partial\beta_i \partial\beta_j}\right] = 0.$$

Together with Remark 5.5 this means that (A2) is true.

For $y \in \{0, 1\}$, $\mathbf{x} \in \mathbb{R}^k$ and $\beta \in \mathbb{R}^k$ one has

$$\log f(y, \mathbf{x}; \beta) = y \log \ell(\beta^T \mathbf{x}) + (1 - y) \log(1 - \ell(\beta^T \mathbf{x})) = y\beta^T \mathbf{x} - \log(1 + \exp(\beta^T \mathbf{x})),$$

where we used (6.3) and (6.4). Now (6.5) yields

$$\nabla_\beta \log f(y, \mathbf{x}; \beta) = (y - \ell(\beta^T \mathbf{x}))\mathbf{x}$$

so that

$$\mathrm{Hess}_\beta \log f(y, \mathbf{x}; \beta) = -\ell'(\beta^T \mathbf{x})\mathbf{x}\mathbf{x}^T.$$

Since $|\ell'(u)| \le 1$ for $u \in \mathbb{R}$, this means that (A3) holds with $K(y, \mathbf{x}) = \|\mathbf{x}\|^2$ due to $\mathbf{E}[\|\mathbf{X}\|^2] < \infty$. Moreover, the Fisher information matrix for β_0 is given by

$$\begin{aligned} I(\beta_0) &= \mathbf{E}[(Y - \ell(\beta_0^T \mathbf{x}))^2 \mathbf{X}\mathbf{X}^T] \\ &= \mathbf{E}[(\ell(\beta_0^T \mathbf{X})(1 - \ell(\beta_0^T \mathbf{X}))^2 + (1 - \ell(\beta_0^T \mathbf{X}))(0 - \ell(\beta_0^T \mathbf{X}))^2)\mathbf{X}\mathbf{X}^T] \\ &= \mathbf{E}[\ell(\beta_0^T \mathbf{X})(1 - \ell(\beta_0^T \mathbf{X}))(1 - \ell(\beta_0^T \mathbf{X}) + \ell(\beta_0^T \mathbf{X}))\mathbf{X}\mathbf{X}^T] \\ &= \mathbf{E}[\ell(\beta_0^T \mathbf{X})(1 - \ell(\beta_0^T \mathbf{X}))\mathbf{X}\mathbf{X}^T] = \mathbf{E}[\ell'(\beta_0^T \mathbf{X})\mathbf{X}\mathbf{X}^T], \end{aligned}$$

where we used that Y follows a Bernoulli distribution with parameter $\ell(\beta_0^T \mathbf{X})$ for given $\mathbf{X}$ for the second equality and (6.7) for the last equality. The right-most expression is positive definite as discussed next to the statement of Theorem 6.3 so that (A4) is satisfied. Finally, (A5) is obviously true.

Now an application of Theorem 5.4 yields the existence of a sequence of random vectors $(\widetilde{\beta}_n)_{n\in\mathbb{N}}$ in $\mathbb{R}^k$ such that

$$\mathbf{P}(\text{there exists an } n_0'' \in \mathbb{N} \text{ such that } \nabla_\beta L_n(\widetilde{\beta}_n) = \mathbf{0} \text{ for all } n \ge n_0'') = 0,$$

$$\widetilde{\beta}_n \xrightarrow{\text{a.s.}} \beta_0 \quad \text{as} \quad n \to \infty \tag{6.14}$$

and

$$\sqrt{n}(\widetilde{\beta}_n - \beta_0) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty \tag{6.15}$$

with $\mathbf{N} \sim \mathrm{Normal}(\mathbf{0}, I(\beta_0)^{-1})$. If $\widetilde{\beta}_n$ solves $\nabla_\beta L_n(\widetilde{\beta}_n) = \mathbf{0}$ and L_n is strictly concave, $\widetilde{\beta}_n$ is the unique maximiser of L_n. Since, by Lemma 6.4, there exists almost surely an $n_0' \in \mathbb{N}$ such that L_n is strictly concave for all $n \ge n_0'$, choosing $n_0 = \max\{n_0', n_0''\}$ leads to

$$\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } L_n \text{ has a unique maximiser for all } n \ge n_0) = 1,$$

which proves part a).

If $\widetilde{\beta}_n$ is the unique maximiser of L_n, we have $\widetilde{\beta}_n = \widehat{\beta}_n$ so that

$$\mathbf{P}(\text{there exists an } n_0 \in \mathbb{N} \text{ such that } \widetilde{\beta}_n = \widehat{\beta}_n \text{ for all } n \geq n_0) = 1.$$

Thus, we may replace $\widetilde{\beta}_n$ by $\widehat{\beta}_n$ in (6.14) and (6.15), which yields part b). □

6.4 Tests and confidence intervals for logistic regression

In the following, we employ Theorem 6.3 to derive asymptotic tests and confidence intervals for logistic regression. We consider the same framework as in the previous section. The matrix $\mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T]$ depends on β_0 and the distribution of $\mathbf{X}$, which are usually not given, and is, thus, unknown. We estimate it by

$$\widehat{M}_n = \frac{1}{n}\sum_{i=1}^{n} \ell'(\widehat{\beta}_n^T\mathbf{X}_i)\mathbf{X}_i\mathbf{X}_i^T$$

for $n \in \mathbb{N}$. The following lemma establishes that $(\widehat{M}_n)_{n\in\mathbb{N}}$ are strongly consistent estimators for $\mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T]$.

Lemma 6.5. Assume the situation of Theorem 6.3. Then,

$$\widehat{M}_n \xrightarrow{\text{a.s.}} \mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T] \quad \text{as} \quad n \to \infty.$$

Proof. For $n \in \mathbb{N}$ we can rewrite $\widehat{M}_n$ as

$$\widehat{M}_n = \frac{1}{n}\sum_{i=1}^{n}(\ell'(\widehat{\beta}_n^T\mathbf{X}_i) - \ell'(\beta_0^T\mathbf{X}_i))\mathbf{X}_i\mathbf{X}_i^T + \frac{1}{n}\sum_{i=1}^{n}\ell'(\beta_0^T\mathbf{X}_i)\mathbf{X}_i\mathbf{X}_i^T,$$

where we denote the first term on the right-hand side by $\mathbf{R}_n$. Combining $\ell' = (1-\ell)\ell$, (6.12) and $0 < \ell < 1$ with the mean value theorem and using the Cauchy-Schwarz inequality, we obtain

$$|\ell'(\widehat{\beta}_n^T\mathbf{X}_i) - \ell'(\beta_0^T\mathbf{X}_i)| \leq |\widehat{\beta}_n^T\mathbf{X}_i - \beta_0^T\mathbf{X}_i| \leq \|\widehat{\beta}_n - \beta_0\|\|\mathbf{X}_i\|$$

and

$$|\ell'(\widehat{\beta}_n^T\mathbf{X}_i) - \ell'(\beta_0^T\mathbf{X}_i)| \leq 1$$

for $i \in \{1, \ldots, n\}$ so that, for $j_1, j_2 \in \{1, \ldots, k\}$,

$$|(\mathbf{R}_n)_{j_1,j_2}| \leq \frac{1}{n}\sum_{i=1}^{n}\min\{\|\widehat{\beta}_n - \beta_0\|\|\mathbf{X}_i\|, 1\}\,|\mathbf{X}_i^{(j_1)}|\,|\mathbf{X}_i^{(j_2)}|.$$

Here $(\mathbf{R}_n)_{j_1,j_2}$ denotes the entry in the j_1-th row and j_2-th column of $\mathbf{R}_n$, while $\mathbf{X}_i^{(j_1)}$ and $\mathbf{X}_i^{(j_2)}$ are the j_1-th and the j_2-th entries of $\mathbf{X}_i$. Since, by Theorem 6.3,

$$\|\widehat{\beta}_n - \beta_0\| \xrightarrow{\text{a.s.}} 0 \quad \text{as} \quad n \to \infty,$$

for every $\varepsilon > 0$ one almost surely has

$$\limsup_{n\to\infty} |(\mathbf{R}_n)_{j_1,j_2}| \le \limsup_{n\to\infty} \frac{1}{n} \sum_{i=1}^n \min\{\varepsilon\|\mathbf{X}_i\|, 1\} \, |\mathbf{X}_i^{(j_1)}| \, |\mathbf{X}_i^{(j_2)}|.$$

Since, by the strong law of large numbers (see Theorem 2.3),

$$\frac{1}{n} \sum_{i=1}^n \min\{\varepsilon\|\mathbf{X}_i\|, 1\} \, |\mathbf{X}_i^{(j_1)}| \, |\mathbf{X}_i^{(j_2)}| \xrightarrow{\text{a.s.}} \mathbf{E}[\min\{\varepsilon\|\mathbf{X}\|, 1\} \, |(\mathbf{X}^{(j_1)}| \, |\mathbf{X}^{(j_2)}|] \quad \text{as} \quad n \to \infty,$$

one almost surely has

$$\limsup_{n\to\infty} |(\mathbf{R}_n)_{j_1,j_2}| \le \mathbf{E}[\min\{\varepsilon\|\mathbf{X}\|, 1\} \, |(\mathbf{X}^{(j_1)}| \, |\mathbf{X}^{(j_2)}|].$$

Because the right-hand side converges to zero as $\varepsilon \to 0$, we see that

$$\mathbf{R}_n \xrightarrow{\text{a.s.}} \mathbf{0} \quad \text{as} \quad n \to \infty.$$

It follows from the strong law of large numbers that

$$\frac{1}{n} \sum_{i=1}^n \ell'(\beta_0^T \mathbf{X}_i) \mathbf{X}_i \mathbf{X}_i^T \xrightarrow{\text{a.s.}} \mathbf{E}[\ell'(\beta_0^T \mathbf{X}) \mathbf{X} \mathbf{X}^T] \quad \text{as} \quad n \to \infty,$$

which completes the proof. □

In the following, we often use the inverse $\widehat{M}_n^{-1}$ of $\widehat{M}_n$. Since, under the assumptions of Theorem 6.3, $\mathbf{E}[\ell'(\beta_0^T \mathbf{X}) \mathbf{X} \mathbf{X}^T]$ is invertible, it follows from Lemma 6.5 that the probability that $\widehat{M}_n$ is invertible converges to one as $n \to \infty$. This is sufficient for our purposes as we consider only the asymptotic behaviour.

Theorem 6.6. Assume the situation of Theorem 6.3. Let $\mathbf{a} \in \mathbb{R}^k$, $c_0 \in \mathbb{R}$ and $\alpha \in (0, 1)$. For

$$H_0 : \langle \mathbf{a}, \beta \rangle = c_0 \quad \text{against} \quad H_1 : \langle \mathbf{a}, \beta \rangle \ne c_0 \tag{6.16}$$

the test

$$\varphi((Y_1, \mathbf{X}_1), \ldots, (Y_n, \mathbf{X}_n)) = \begin{cases} \text{cannot reject}, & c_0 \in I_n, \\ \text{reject}, & c_0 \notin I_n, \end{cases}$$

with

$$I_n = \Big(\langle \mathbf{a}, \widehat{\beta}_n \rangle - z_{1-\frac{\alpha}{2}} \sqrt{\mathbf{a}^T \widehat{M}_n^{-1} \mathbf{a}/n}, \langle \mathbf{a}, \widehat{\beta}_n \rangle + z_{1-\frac{\alpha}{2}} \sqrt{\mathbf{a}^T \widehat{M}_n^{-1} \mathbf{a}/n} \Big)$$

has asymptotic level α. Moreover, I_n is an asymptotic confidence interval for $\langle \mathbf{a}, \beta \rangle$ with error level α.

Proof. The statements follow directly from Theorem 5.8 since its assumptions (5.10) and (5.11) are satisfied with $\Gamma = \mathbf{E}[\ell'(\beta_0^T\mathbf{X})\mathbf{X}\mathbf{X}^T]^{-1}$ and $\Gamma_n = \widehat{M}_n^{-1}$ for $n \in \mathbb{N}$ by Theorem 6.3 and Lemma 6.5, respectively. □

Let β have the components $\beta^{(1)}, \ldots, \beta^{(k)} \in \mathbb{R}$. One is often interested in the test problem

$$H_0 : \beta^{(j)} = 0 \quad \text{against} \quad H_1 : \beta^{(j)} \neq 0$$

for some $j \in \{1, \ldots, k\}$. By choosing $\mathbf{a}$ as the vector where the j-th component is one and all other components are zero in Theorem 6.6, we see that we can reject H_0 for a fixed $n \in \mathbb{N}$ at asymptotic level $\alpha \in (0, 1)$ if

$$\frac{\widehat{\beta}_n^{(j)}}{\sqrt{(\widehat{M}_n^{-1})_{j,j}/n}} \notin (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}), \tag{6.17}$$

where $\widehat{\beta}_n^{(j)}$ is the j-th component of $\widehat{\beta}_n$. However, as this is only an asymptotic test, it is not guaranteed that we have the desired level α for a fixed $n \in \mathbb{N}$. The asymptotic p-value is given by

$$2\left(1 - \Phi\left(\frac{|\widehat{\beta}_n^{(j)}|}{\sqrt{(\widehat{M}_n^{-1})_{j,j}/n}}\right)\right),$$

where Φ denotes the distribution function of the standard normal distribution.

Example 6.7. We continue with the damages of O-rings from Example 6.1 and Example 6.2. Using the command `summary.glm` in `R`, we obtain the following output:

```
> summary(Model)

Call:
glm(formula = cbind(damage, 6 - damage) ~ temp, family = binomial,
data = orings)

Coefficients:
             Estimate  Std. Error  z value  Pr(>|z|)
(Intercept)  11.66299     3.29626    3.538  0.000403  ***
temp         -0.21623     0.05318   -4.066  4.78e-05  ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’  0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance:  38.898 on 22 degrees of freedom
Residual deviance:  16.912 on 21 degrees of freedom
AIC: 33.675
```

```
Number of Fisher Scoring iterations:  6
```

The first column of the table contains the estimates, which were already computed in Example 6.2. The standard errors are the denominators of the test statistics in (6.17), while the z-values are these test statistics. Thus, the column with the probabilities provides the asymptotic p-values for the null hypotheses that the intercept and the coefficient of the temperature are zero. Since they are small, we can reject the null hypotheses at all reasonable levels. For the coefficient of the temperature we can conclude that it is negative, which shows that low temperatures increase the probability of a damage at an O-ring.

Appendix A

Results on convergence in distribution

In this appendix, we collect two important results on convergence in distribution, which are employed in the proof of Theorem 5.8.

Theorem A.1 (Slutsky's lemma)**.**

a) Let $(\mathbf{Z}_n)_{n\in\mathbb{N}}$ and $(\widetilde{\mathbf{Z}}_n)_{n\in\mathbb{N}}$ be sequences of random vectors in $\mathbb{R}^m$ and let $\mathbf{Z}$ be a random vector in $\mathbb{R}^m$. If

$$\mathbf{Z}_n - \widetilde{\mathbf{Z}}_n \xrightarrow{\mathbf{P}} \mathbf{0} \quad \text{as} \quad n \to \infty \quad \text{and} \quad \widetilde{\mathbf{Z}}_n \xrightarrow{d} \mathbf{Z} \quad \text{as} \quad n \to \infty,$$

then $\mathbf{Z}_n \xrightarrow{d} \mathbf{Z}$ as $n \to \infty$.

b) Let $(Y_n)_{n\in\mathbb{N}}$ be a sequence of random variables such that $Y_n \xrightarrow{\mathbf{P}} c$ as $n \to \infty$ with $c \in \mathbb{R}$. Let $(\mathbf{Z}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{Z}$ be a random vector in $\mathbb{R}^m$ such that $\mathbf{Z}_n \xrightarrow{d} \mathbf{Z}$ as $n \to \infty$. Then,

$$Y_n\mathbf{Z}_n \xrightarrow{d} c\mathbf{Z} \quad \text{as} \quad n \to \infty.$$

Recall that $\langle \mathbf{a}, \mathbf{b} \rangle$ denotes the inner product of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^m$.

Theorem A.2 (Cramér-Wold device)**.** Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$. Then, one has

$$\mathbf{X}_n \xrightarrow{d} \mathbf{X} \quad \text{as} \quad n \to \infty$$

if and only if

$$\langle \mathbf{a}, \mathbf{X}_n \rangle \xrightarrow{d} \langle \mathbf{a}, \mathbf{X} \rangle \quad \text{as} \quad n \to \infty$$

for all $\mathbf{a} \in \mathbb{R}^m$.

Bibliography

[1] A. J. Hayter (1984): A proof of the conjecture that the Tukey-Kramer multiple comparison procedure is conservative, *Ann. Statist.* **12**, 61–75.

[2] T. S. Ferguson (1996): *A Course in Large Sample Theory*, Chapman & Hall.